

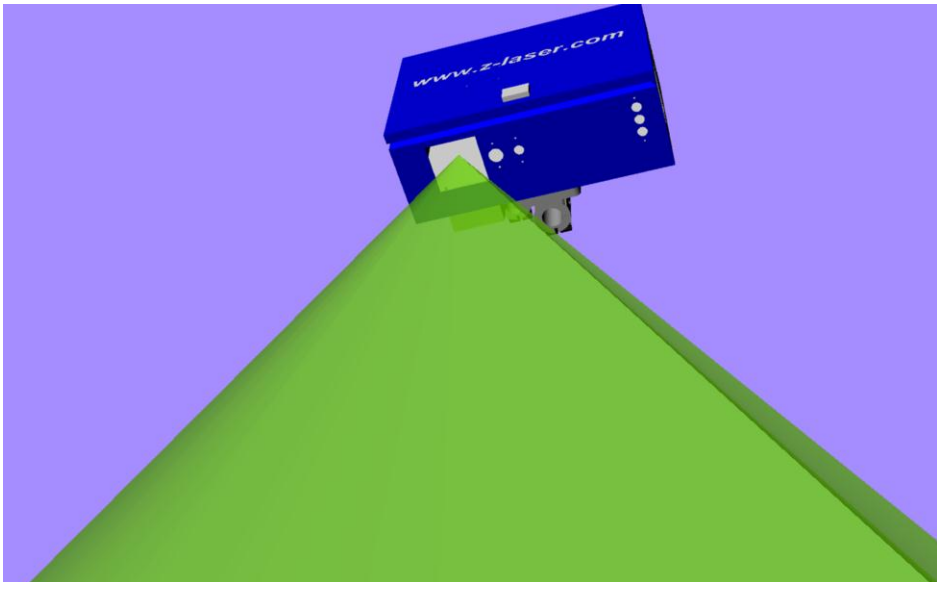
Z-LASER

Z-Laser Optoelektronik GmbH

Merzhauserstr. 9, 79346,
Freiburg, Germany

Laser Projector

3D Calibration and Setup



© *Z-Laser* Optoelektronik GmbH 2012

All rights to text and pictures reserved. Unauthorised publishing prohibited.

Table of Contents

Foreword 4
Rhinoceros 5
Introduction 5
Installation 6
ExportToLPM Module 9
Introduction 9
Installation 10
Configuration 12
LPM 15
Introduction 15
System Requirements 16
Required preparation for the installation 17
LPM Software Installation 18
The Mold 25
3D Calibration Process 26
Introduction 26
Creating the Calibration File in Rhinoceros 27
3D Calibration of the Projector 32
The Sigma Values 38
3D Projection Process 39
Introduction 39
Create the Main/Parent Layer 42
Create the Sub-Layers 43
Extracting Curves from Surfaces or Solids 44
Copying Existing Contours to Sub-Layers 46
Preparing the Objects for Projection 47
Exporting the data with the ExportToLPM module 49

Opening the HPGL 3D file in LPM 52
Appendix 56
Rhino Basics 56
The User Interface 56
Layers 57
Objects 59
Opening 3D Data in Rhino 67
Projector Installation 68
MEM-PIO Driver Installation 69
Table of Figures 74

Foreword

This document is a supplementary manual to the **LP Manual**. This manual contains additional information not found in the main manual regarding the new 3D calibration process for ***Z-Laser*** laser projectors. Also included is the installation, configuration and uses of the 3D modelling software Rhinoceros 4.0 and the plug-in module ExportToLPM. Any information not contained in this manual can be found in the LP Manual on the CD provided with your projector. If you are unable to find something, please contact your supplier or Z-Laser directly.

The LP manual must be read first with regards to safety, installation of the projectors and the various uses and commands of LPM.

Most of the various commands and controls of Rhinoceros 4.0 are not included in this document. Only the importing of the customer data into Rhino and its conversion into data that can be read by LPM 8.1 is detailed here. A variety of tutorials on learning Rhino can be found at http://www.rhino3d.com/tutorials.htm.

Rhinoceros

Introduction

Rhinoceros is a Windows 3D modelling software program.

The following is an excerpt from the Rhinoceros website http://www.rhino3d.com/

*"Rhino can **create**, **edit**, **analyze**, **document**, **render**, **animate**, and **translate** NURBS curves, surfaces, and solids with no limits on complexity, degree, or size. Rhino also supports polygon **meshes** and **point clouds**. Rhino offers:*

***Uninhibited free-form 3-D modeling.** Tools like those found only in products costing 20 to 50 times more. Accurately model any shape you can imagine.*

***Extreme precision**. Design, prototype, engineer, analyze, document, and manufacture anything of any size no matter how small or how large.*

***Unrestricted editing**. Revise freely without concern about how you got there or where you want to go next.*

***2-D drafting, annotation, and illustration**. In addition, flatten 3-D surfaces into 2-D patterns and drive laser, plasma, and water jet cutters.*

***Large project organization**. Work with both large models and large teams.*

***Compatibility**. Exchange files with your other design, drafting, CAM, engineering, analysis, rendering, animation, and illustration software.*

***File read and repair**. Fix extremely challenging **IGES** and **mesh** files.*

***Direct connection**. Support for wide variety of 3-D digitizing arms, 3-D scanners, and 3-D printers.*

***Customization**. Easy-to-use tools let both you and software developers make Rhino fit your special needs.*

***Short learning curve**. Focus on design and visualization without being distracted by the software.*

***Speed**. Even on an ordinary laptop computer Rhino is fast. No special hardware is required.*

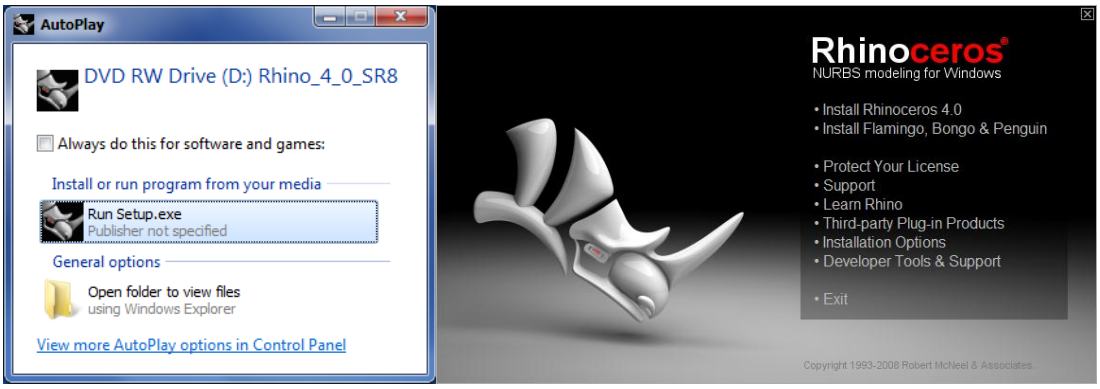
***Affordability**. Ordinary hardware. Short learning curve. Priced like other Windows software. No maintenance fees. Worldwide support is included."*

Rhinoceros is used to import customer files, such as STEP or IGES files, for modification before exporting the file into a format that can be read by our projector control software LPM.

Installation

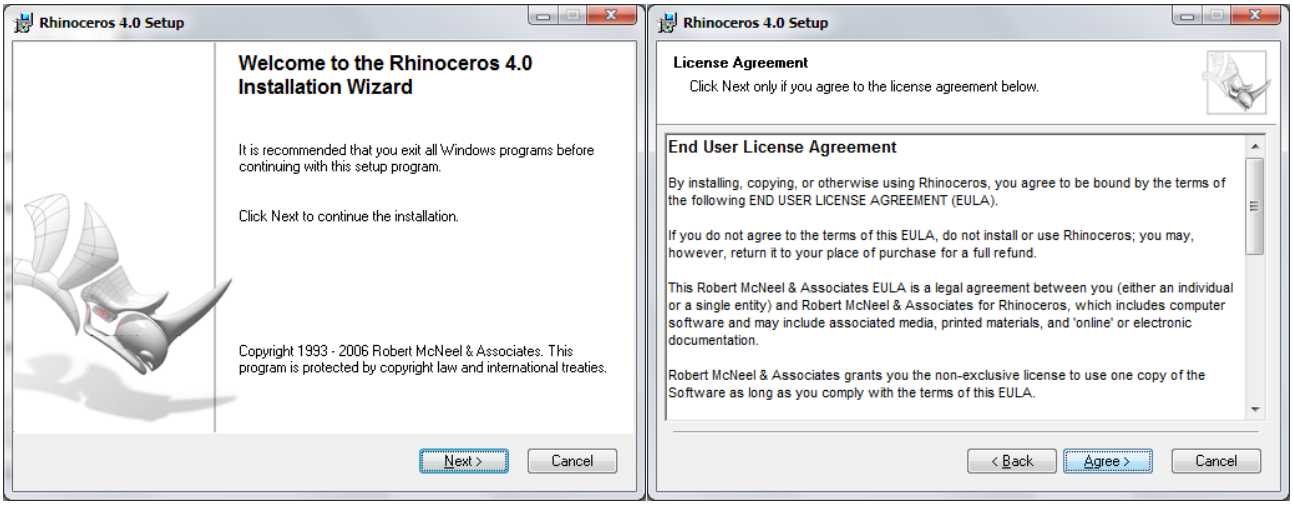
Insert the **Rhino 4.0 SR8** CD into the CD tray and close the tray.

Autoplay should start and ask the user to run the **setup.exe** file. If not, use Windows Explorer to go to your CD and navigate to the **setup.exe** file.



Select the setup.exe file and wait for the homepage to appear. Select **Install Rhinoceros 4.0**.

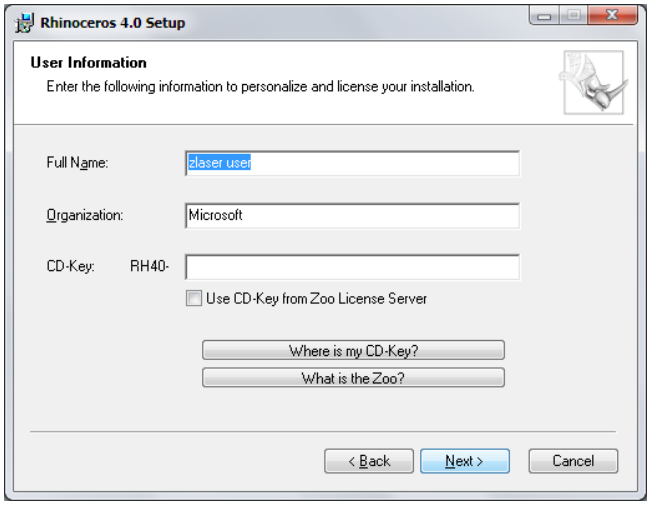
Wait for Windows Installer to prepare the files and then the **Welcome to Rhino 4.0** page will appear. Press **Next**. Select **Agree** at the **License Agreement** page.



The **User Information** window will appear. Enter the **user name** and your **organization name**.

If you have a license for a single Rhino installation (which can be found on the cover page of your manual in the CD case), enter it under **CD-Key**.

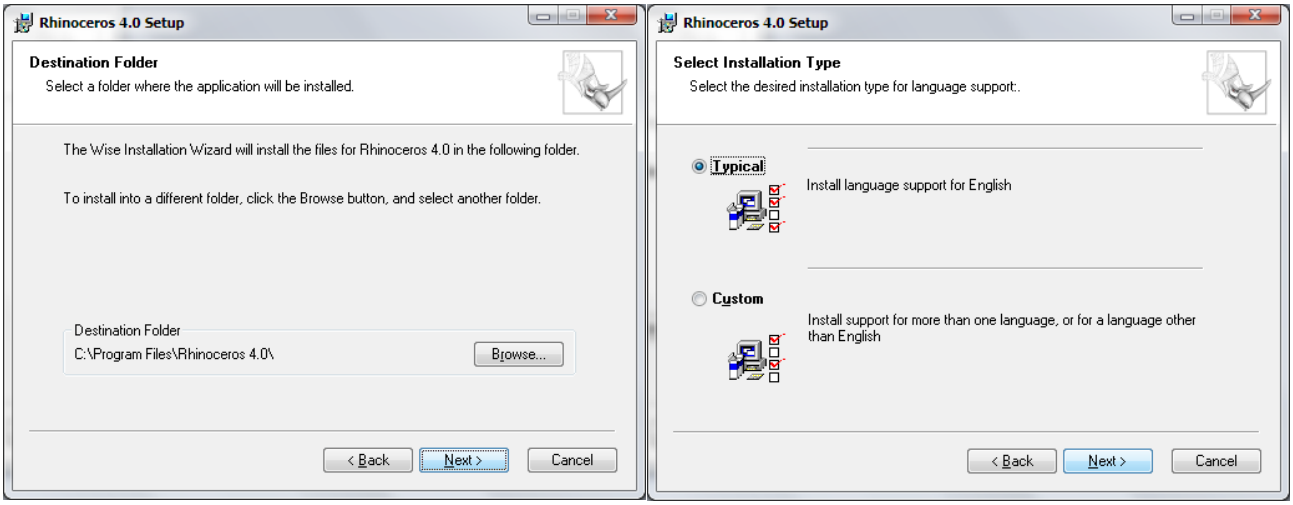
If you will be using the Zoo License Server for multiple Rhino installations, tick the box marked **Use CD-key from Zoo License Server**.



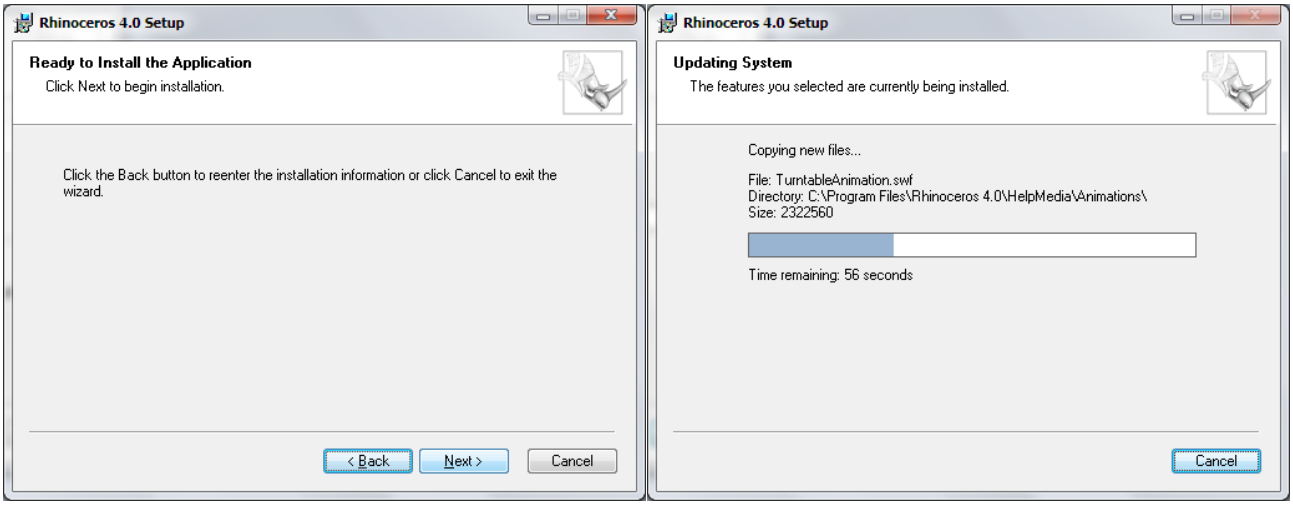
Select **Next**.

Select the **Destination Folder** for Rhino at the next page. If you wish to save it elsewhere, select **Browse** and navigate to the folder of your choice. Check it is correct before selecting **Next**.

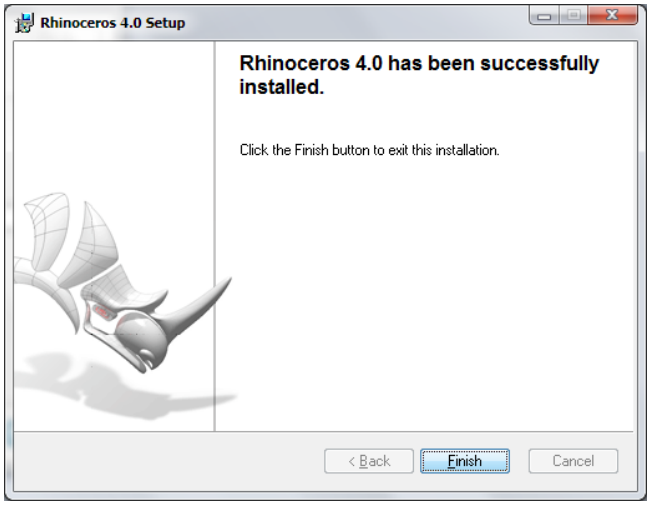
Select a **Typical** installation under **Installation Type** and select **Next**.



Select **Next** at the **Ready to Install** window. The files will now be copied to the destination folder.



Select **Finish** if your installation is successful.



ExportToLPM Module

Introduction

The **ExportToLPM** plug-in module is a script written at ***Z-Laser*** for the extraction and conversion of 3D data modified in Rhinoceros for use with LPM.

It has two main functions:

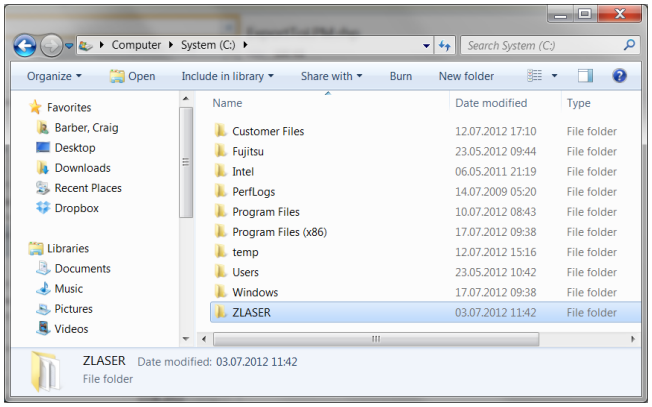
- To export X, Y and Z coordinate information of the calibration points to a text file.
- To export contours, points and text information to a HPGL 3D file.

Both of these files can be read by the projector control software LPM in order to calibrate the projector to the environment and project the customer's files.

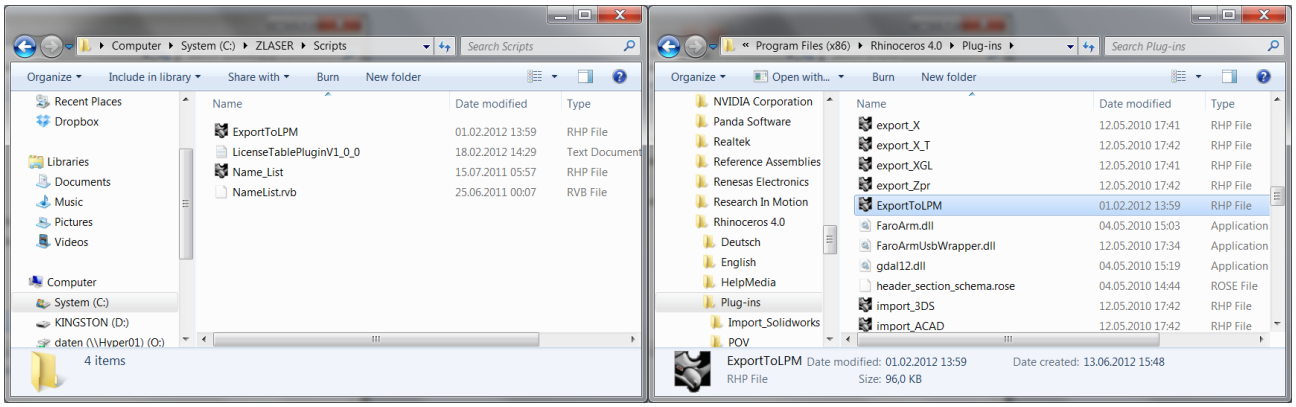
This manual will describe how to install it in Rhinoceros and how to configure it for the user's preferences. The Z-Laser folder containing the module is not supplied with the CD. Please contact your local Z-Laser dealer for this folder.

Installation

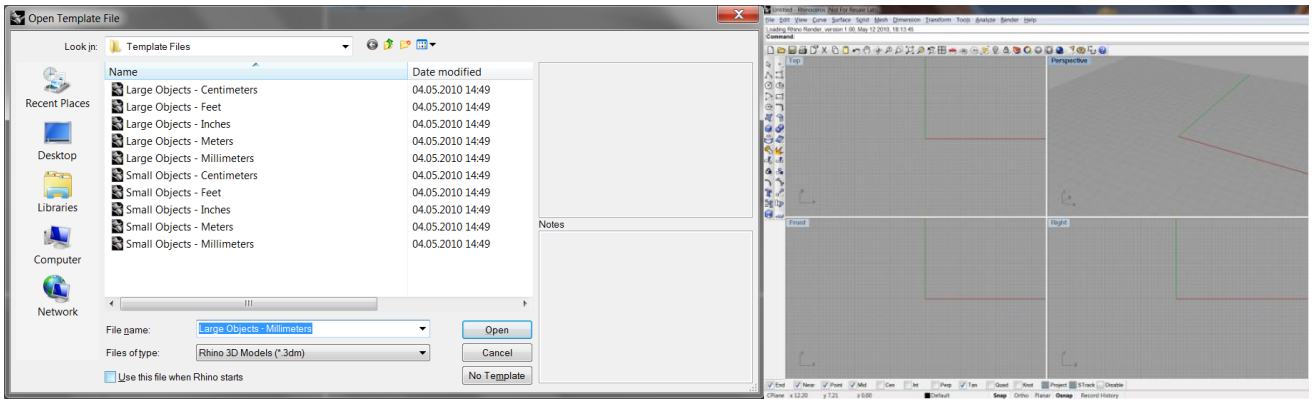
Open Windows Explorer and install the **ZLASER** folder to **C:\ZLASER**.



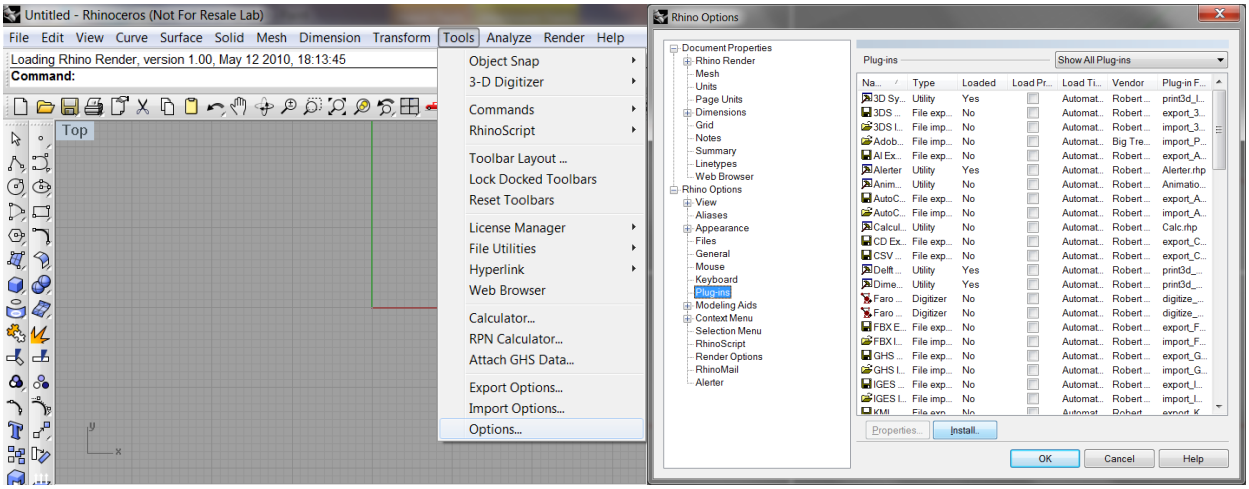
Open **C:\ZLASER\SCRIPTS** and copy the file **ExportToLPM.RHP** to the path **...\Rhinoceros 4.0\Plug-ins**.



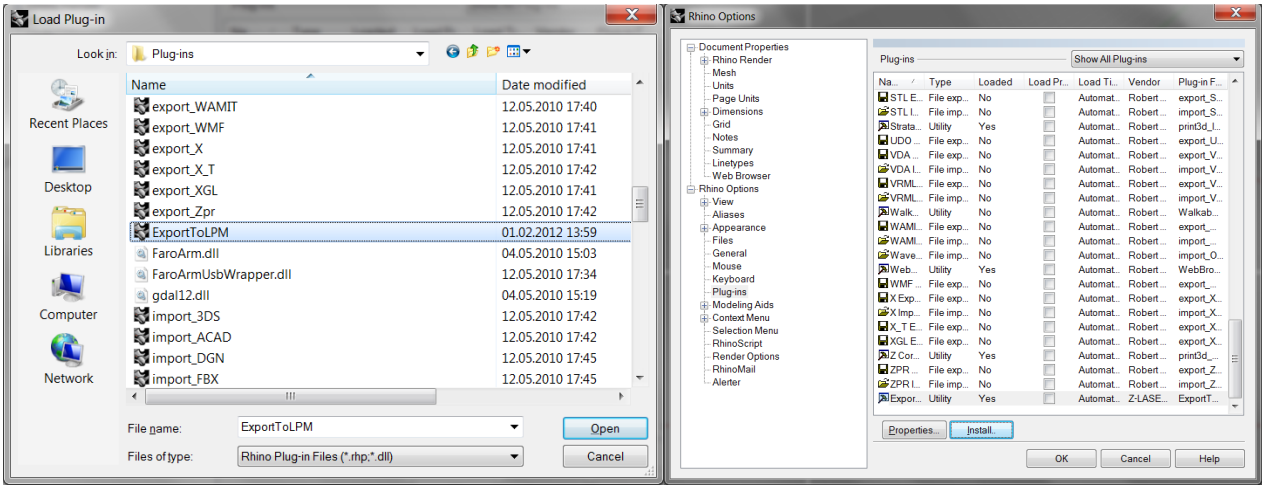
Open **Rhino 4.0**. If Rhino asks you to select a template, use **Large Objects – Millimetres**.



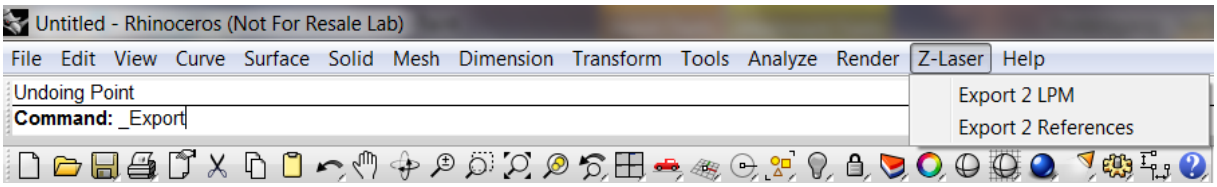
Go to the menu bar and select **Tools** → **Options**. The Rhino Options window will open. Select **Plug-ins** in the left hand window and then **install** in the right hand window.



In the **Load Plug-in** window, navigate to the **...\Rhinoceros 4.0\Plug-ins** folder and select the **ExportToLPM.RHP** file. You will now see it loaded into the Plug-In window at the bottom.



Select **OK** and you will return to the main Rhino screen. You will now find **Z-Laser** on the menu with two options: **Export 2 LPM** and **Export 2 References**.



Configuration

ZLASER.INI is the initialization file for the ExportToLPM.RHP file which is used to convert Rhino 3DS information into HPGL 3D files for LPM. It contains settings for the General, Plotting and Export options.

General Options

```
RefTargetName=T*

DelOrgCurve=0

SuppressDialogs=0
```

RefTargetName is the object name prefix of the calibration points. When the Export 2 Ref function is called, all points starting with this prefix is found and loaded into the table of points to be selected.

DelOrgCurve converts the curves that have been selected when exporting to a *.LAS file to polylines or not. You can either choose to keep the original curves (with information), or delete them and have them replaced with the new polylines (with no information). To delete, select 1. To leave it as it is, select 0.

SuppressDialogs should be set to 0 if you would like warning and error messages to appear.

Plot Options

```
X=1

Y=2

Z=3

AngleTolerance=5

Tolerance=1

crossPoint1=-15,0,0

crossPoint2=15,0,0

crossPoint3=0,-15,0

crossPoint4=0,15,0
```

x,y,z are the 3 settings are used to correlate the projector's axes to the Rhino drawing's axes. X, Y, and Z refer to the projector's axes and 1, 2, and 3 refer to Rhino's X, Y and Z axes.

AngleTolerance sets the maximum angle between the tangents of the curve and the end of line segments of the newly created polyline and it is measured in degrees. The smaller the angle, the closer the new polyline will resemble a natural curve. The greater the angle, the closer the new polyline will resemble a triangle.

Tolerance is the maximum distance allowed between the curve and the line segment's midpoint in millimetres. If the distance is small, then there will be more line segments created in the polyline, and the new polyline will more closely resemble a curve. If the distance is great, then there will be less line segments created in the polyline, and the new polyline will more closely resemble a triangle.

Note: The **AngleTolerance** and the **Tolerance** parameters determine how a curve is projected as a polyline by the projector. Be aware that creating a polyline with very small line segments will cause the frequency rate of the projector to lower and flickering may occur. If the projector is a great distance away from the mold or work-piece and the line segments are too small, then the projector will perceive these line segments as dots and it will not project it for laser safety reasons.

Conversely, if the line segments are too long then the curve will appear blocky.

Figure 1: Distance Tolerance and Maximum Angle

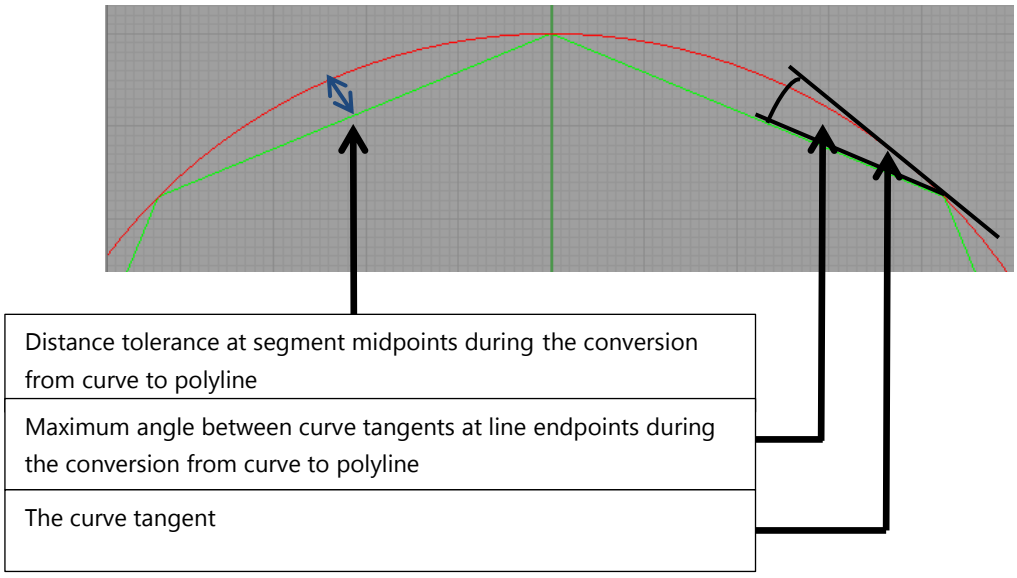
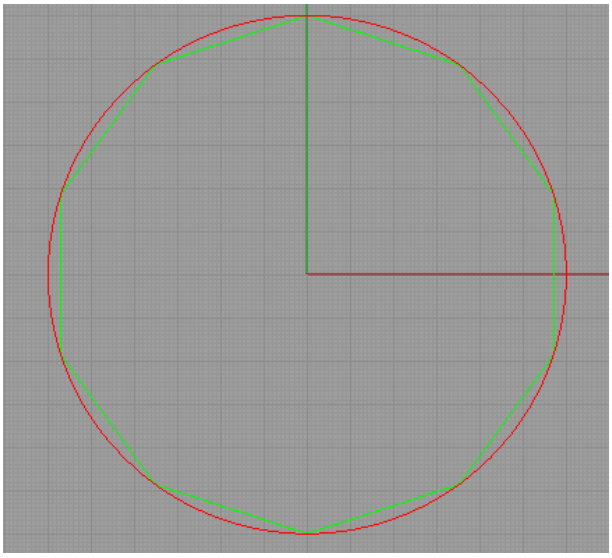
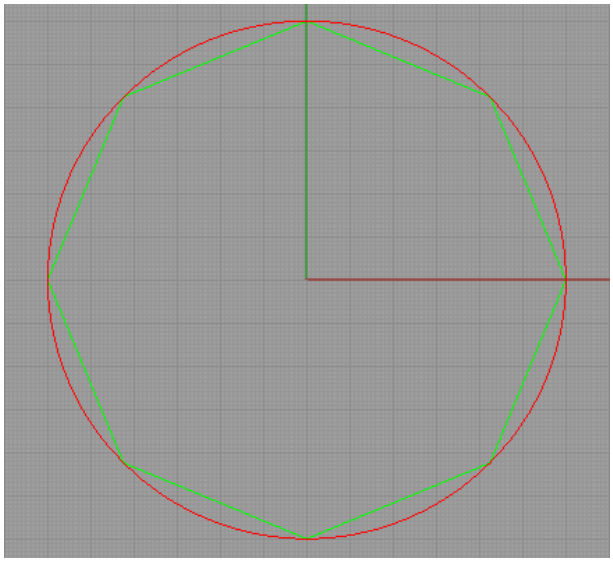


Figure 2: Difference in Angle Tolerance



The green polyline, produced from the red circle, has a small **AngleTolerance** and **Tolerance** and 10 line segments are produced. This image is more likely to cause flickering.



This green polyline has a greater **AngleTolerance** and **Tolerance** and 8 line segments are produced. This image is less likely to cause flickering.

crossPoints 1, 2, 3, and 4 sets the size of the cross projected when displaying a point in Rhino. This is useful for projecting drill holes. A cross is projected for safety reasons.

Export Options

```
AutoPrefix=1

HPGL_Suffix=las

CAL_Suffix=ref

Directory=C:\ZLASER\Export
```

AutoPrefix adds a three digit prefix to the beginning of each new *.REF or *.LAS file so that the last file is not overwritten. If set to 0, there is no added prefix and the previous file of the same name is overwritten.

HPGL_Suffix is the file extension given to all HPGL 3D projection files (*.LAS) used by LPM.

CAL_Suffix is the file extension given to all calibration point files (*.REF) used by LPM.

Directory is the path and folder name to which all files are saved.

LPM

Introduction

The laser projection software LPM (Laser Projection Manager) is tasked to prepare the graphical data for the projector from various import modules and transmit it to the laser projector. At the same time, LPM offers a visualisation of the data on-screen and enables interaction with the position, rotation and height of the projection display. Using the new modifiers, LPM has gained the function to manipulate image data and optimally display it.

On the following pages, we will advise you how to use the software. Should you have any questions, please contact your local Z-Laser dealer.

System Requirements

To control the laser projector via the included LPM (Laser Projector Manager) software, your PC should meet the following minimum requirements:

1. Computing power: Intel Pentium / AMD Processor with 300 MHz
2. Operating system: Windows 7, Windows XP, Windows Vista.
3. Graphics card and monitor with 800 x 600 pixel resolution and 8 bit colour depth
4. CD-ROM / DVD drive
5. 20 MB free hard disk space
6. Keyboard and mouse
7. A free RS232 port or Ethernet network interface card (depending on desired communication)
8. In network systems: TCP / IP protocol installed and running

You will also require a copy of the LPM8.1 setup and the projector must have the latest version of the Linux client installed on it. If unsure which version you have, please contact your local Z-Laser dealer.

Required preparation for the installation

To use the full functionality of the LPM set-up, you should have already prepared the following:

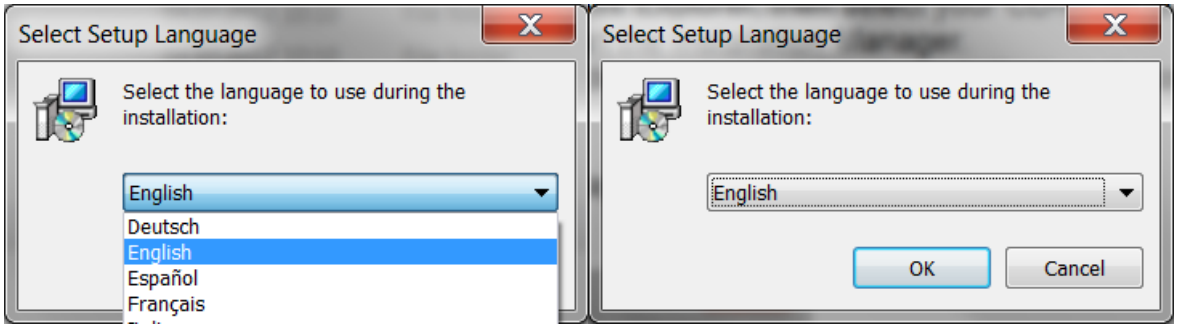
1. Connect the PC and the laser projector together with the communication cable.
2. Apply the supply voltage to the projector and switch the projector on.
3. If the status indicators are flashing green and solid red, then a network connection should be made.
4. If the status indicator is flashing green only, then a serial connection should be made.
5. The auto-run function of your version of Windows should be activated.
6. Ensure you have administrator rights on the PC that LPM is to be installed on.

The set-up program, after successful installation, automatically tries to find the laser projector using a serial connection or over Ethernet. Ensure the search process is not influenced by other equipment.

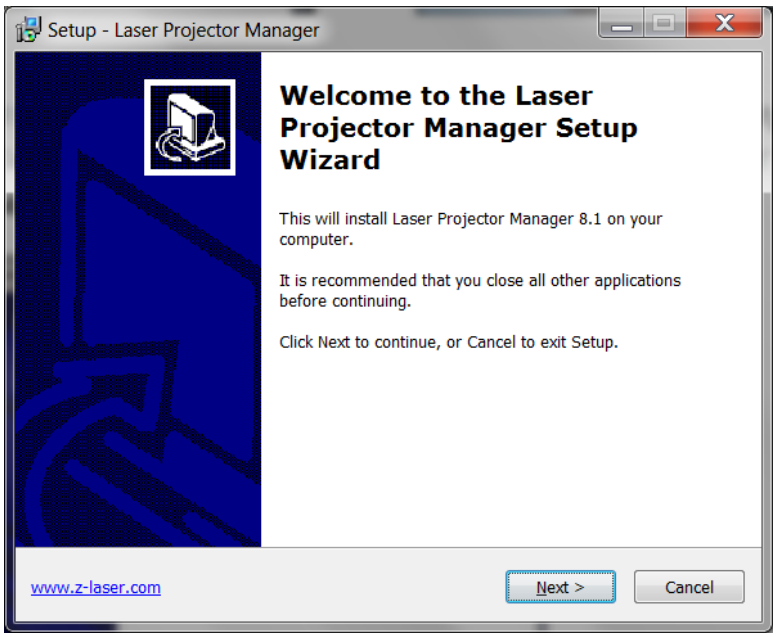
LPM Software Installation

1. Insert the included CD-ROM into your CD-ROM or DVD drive. After closing the drive, the installation process should start automatically if the auto run feature is enabled. If this is not the case, open Windows Explorer, then select your CD-ROM drive icon. Double-click the file **SETUP.EXE** to start the Installation Manager.
   The set-up wizard will guide you step by step through the installation of LPM:

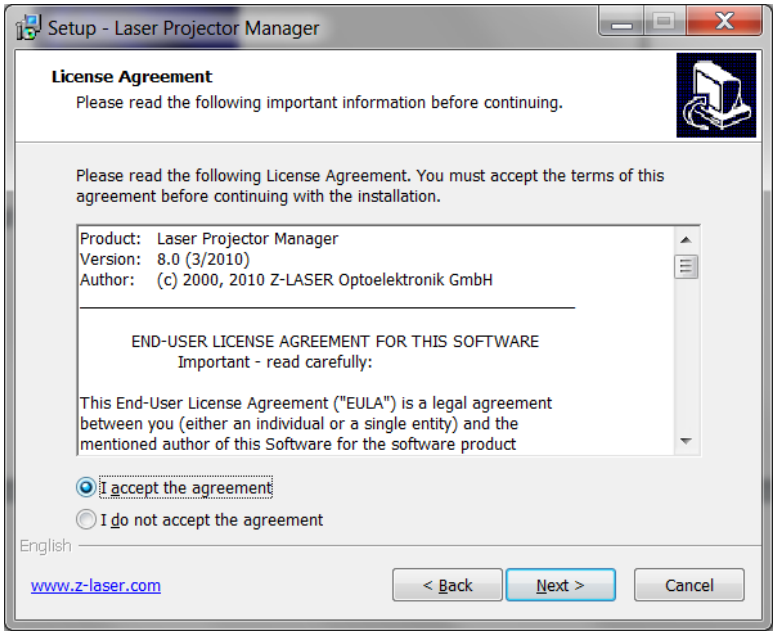
2. Select your language and then click "**OK**".



3. The LPM set-up wizard will start in the language of your choice. Follow the instructions in the wizard and click either "**Continue**" or "**Cancel**."



4. Read the license agreement that appears carefully.

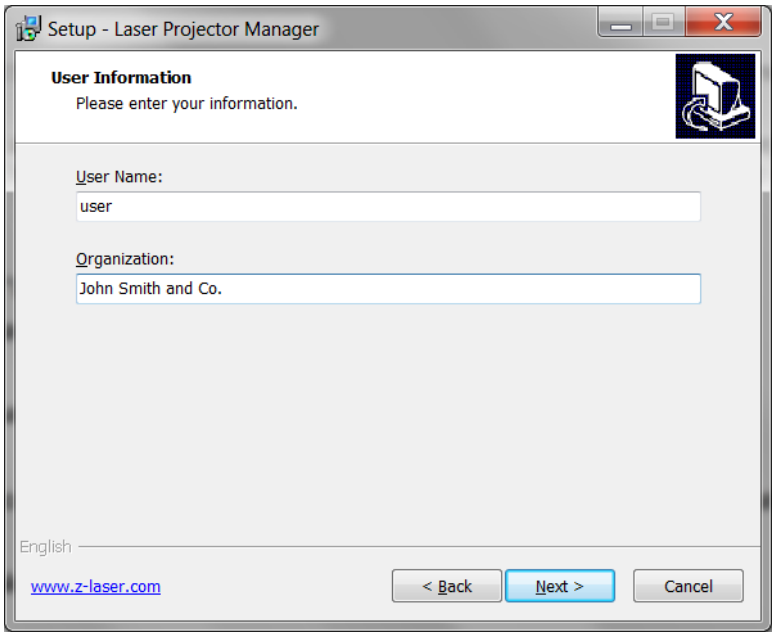


5. If you accept the listed agreements, activate the check box "**I accept the agreement**" and click "**Next**".

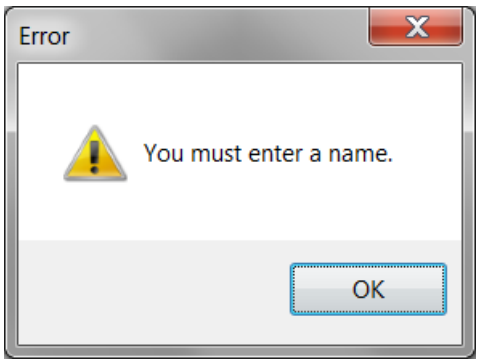
6. If you do not accept the listed agreements, activate the check box "**I do not accept the agreement**". In this case, the "**Next**" button is inactive and you will not be able to install the software.

7. The "**User Information**" window appears next.

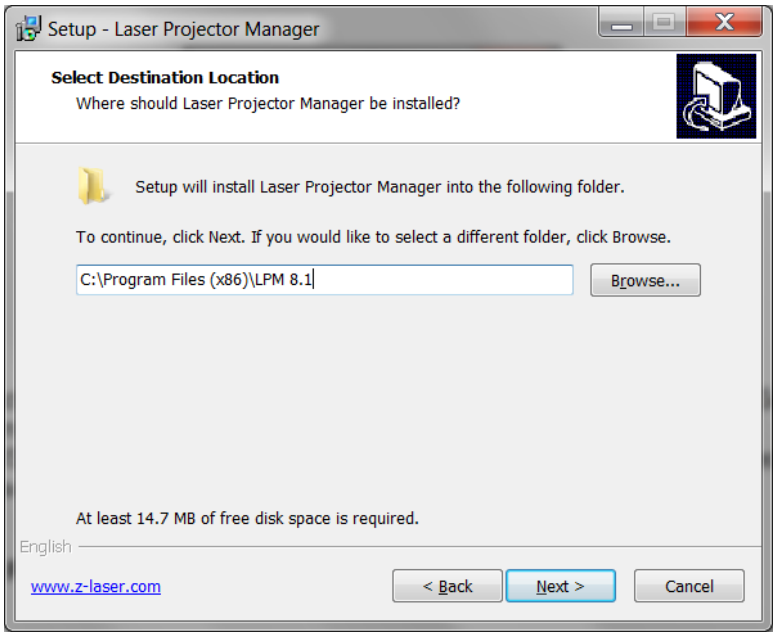
8. Enter your name and company name in the fields provided. Click "**Next**".



9. A dialogue box will appear as a reminder if you forgot to enter the above information.



10. Select the destination folder you would like LPM to be installed in. The Set-up Wizard automatically creates a folder "LPM" under the path: C:\WINDOWS\LPM. If you wish to choose another path, select "**Browse...**". Select "**Next**" when you have made your choice.



11. In the next window, you can select all the import filters for your application needs. To make your selection, simply activate the filter's check box.

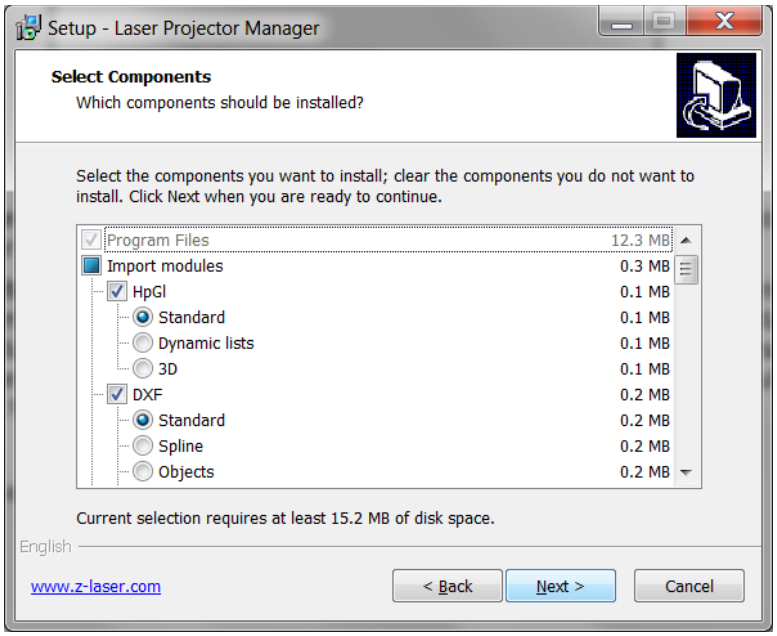
    The import filters for DXF and HPGL are already pre-selected. If you wish to project files created by Rhino, select the **HPGL 3D** import module. You can make only one selection from any set of import filters and extension modules. If you wish to install multiple filters or modules, then the set-up will need to be started again for each additional module. All import filters already installed will remain after the new set-up. Ensure that sufficient space exists for the desired modules.

    Optional selections include:

- Extension Modules → meM-PIO USB → Radio remote control → 866MHz

- Extension Modules → meM-PIO USB → 3 Ports I/O
- Extension Modules → Polling → Standard
- Extension Modules → Barcode Scanner → Barcode Scanner (RS-232)
- Extension Modules → Barcode Scanner → Barcode Scanner USB

Then click "**Next**".



12. Apply your settings as follows:

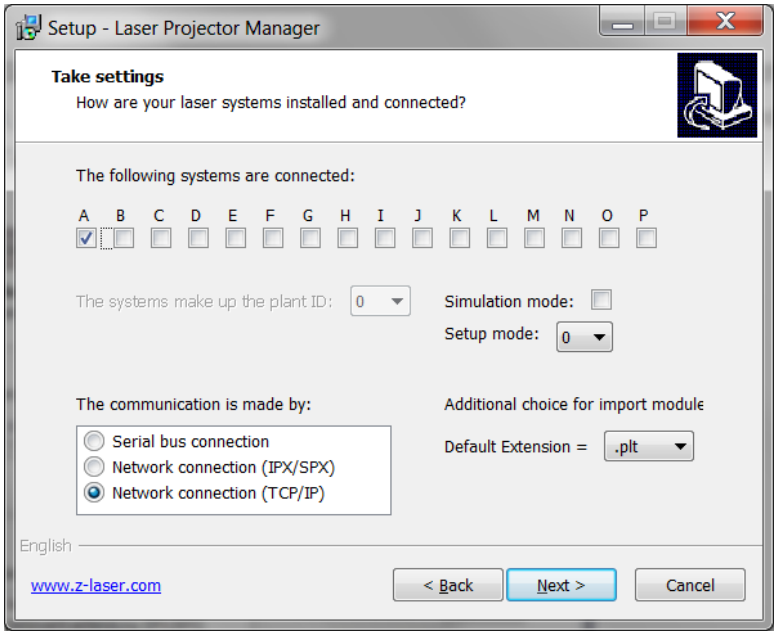
    Enter the number of projectors to be connected by selecting the check box of the corresponding character. If you have only one projector, then select the letter "A". For two projectors, select letters "A" and "B", for three, select letters "A", "B" and "C".

    You can connect up to 16 projectors.

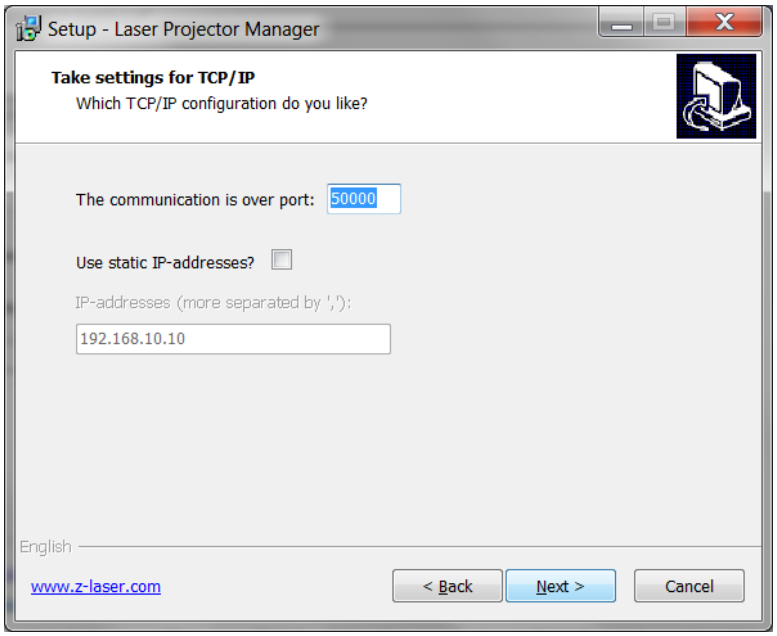
    Set the communication type to TCP/IP protocol or serial. IPX/SPX protocol is not supported in 3D mode.

    Set Setup Mode to 2 to activate 3D mode.

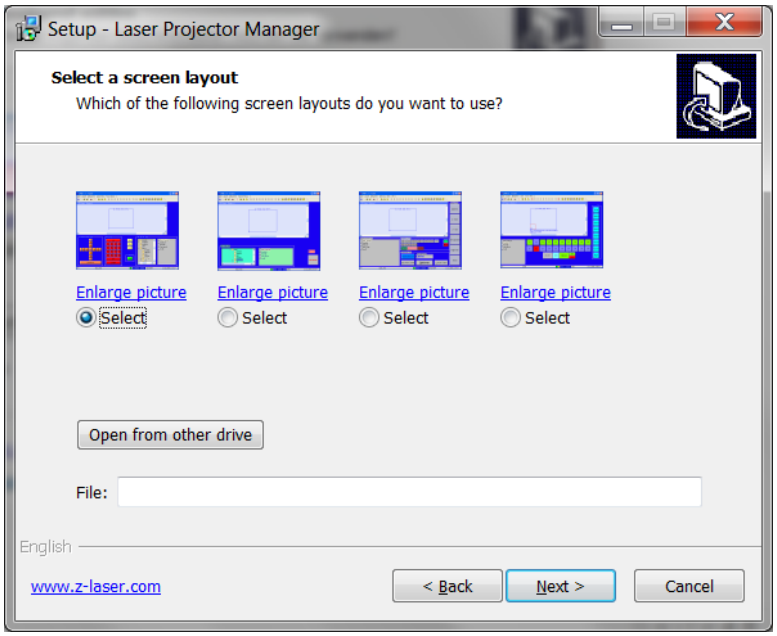
    Finally, select an import filter that you would like to open your files with from the "**Default Extension =**" drop-down menu. HPGL 3D files end in *.LAS extensions. When you are finished, select "**Next**".



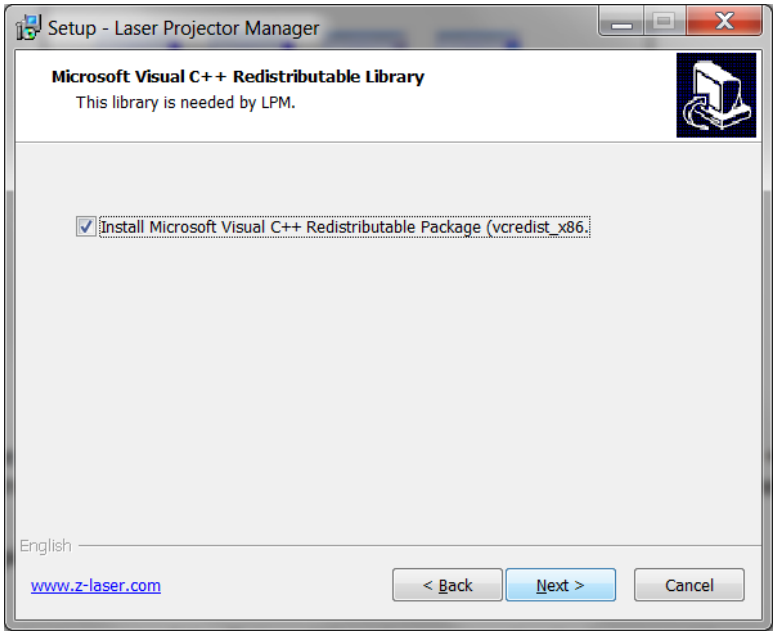
13. If you selected network communication over TCP/IP protocol, an additional window will open which will allow you to adjust the settings of the communication port. The port is pre-initialised as 50000. Allowed values are 50000 – 60000. If you use the same IP address or if there is no DHCP server available, then activate the "**Use static IP address?**" check box and enter your IP address. Afterwards, select "**Next**".



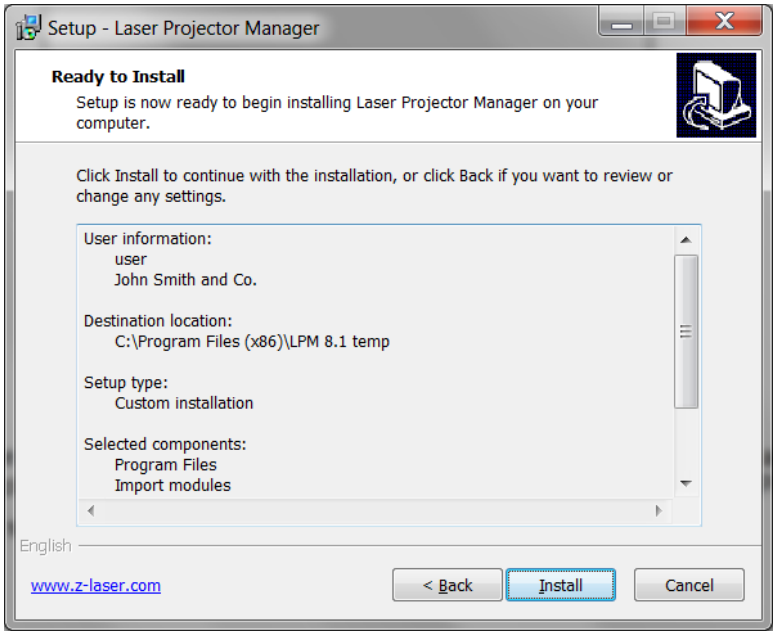
14. In the following menu, you can select the screen layout of the LPM interface. You can enlarge the thumbnails of the user interfaces on display to make your decision. If you already have a specially configured user interface of LPM, you can load it by selecting "**Search the hard disk**". Then click "**Next**".



15. In the next step, you will be prompted to install the Microsoft Visual C + + library. If this library already exists on your computer, remove the check mark from the check box. Click "**Next**".



16. In the last screen of the Set-up Wizard, you can check all your settings again. Click on "**Back**" to change your settings. Click "**Install**" to install on your PC to LPM.

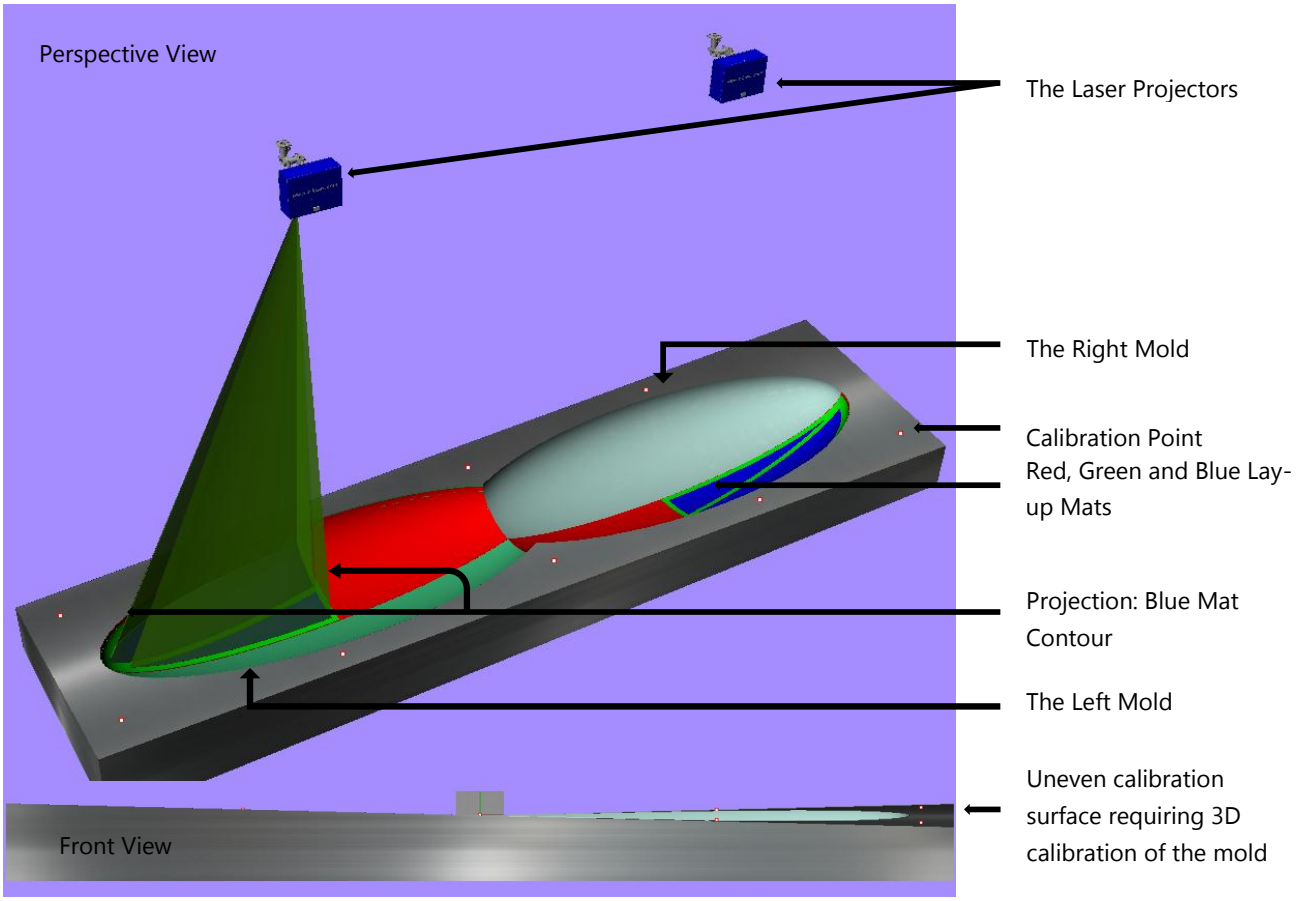


17. The software is now installed and ready for use. You can either start LPM by pressing Ctrl, Alt and L simultaneously on your keyboard, via "**Start** → **Programs** → **LPM** → **LPM**" or simply via the automatically created shortcut on your desktop. Please note that LPM is copy protected. After installation, your installed modules are free of charge for 30 days. After this period, the modules will stop working and LPM will need to be supplied a registration code from Z-LASER. For information about the registration process, see the chapter "**Software Registration**" in the main user manual.

The Mold

This manual will refer to a mold, which was created in Rhino, as a way to guide the end user through the various processes required to project 3D data. A screenshot of the mold can be seen below.

Figure 3: The Example Mold



The outer structure is approximately 8000 x 2000 x 500 mm in size. Two shells, based on an ellipsoid measuring 4000 x 750 x 1000 mm, bisect each other and form the left mold and right mold.

The top level of the outer structure is uneven. Two diagonal corners are at a height of -100 mm while the other two diagonal corners are at +100mm. The calibration points are placed on this level therefore 3D calibration of the projector is necessary. A total of 10 glass reflectors are placed at various points on the top level of the outer structure.

Each projector has a 60° degree opening angle and can cover only one mold. There is an overlap in the centre where both projectors can project. Each projector has 6 glass reflectors within its maximum field borders.

Four lay-up mats are placed inside each mold. The red mat is placed first, then the green mat followed by two smaller blue mats. Each mat has a thickness of 3 mm.

3D Calibration Process

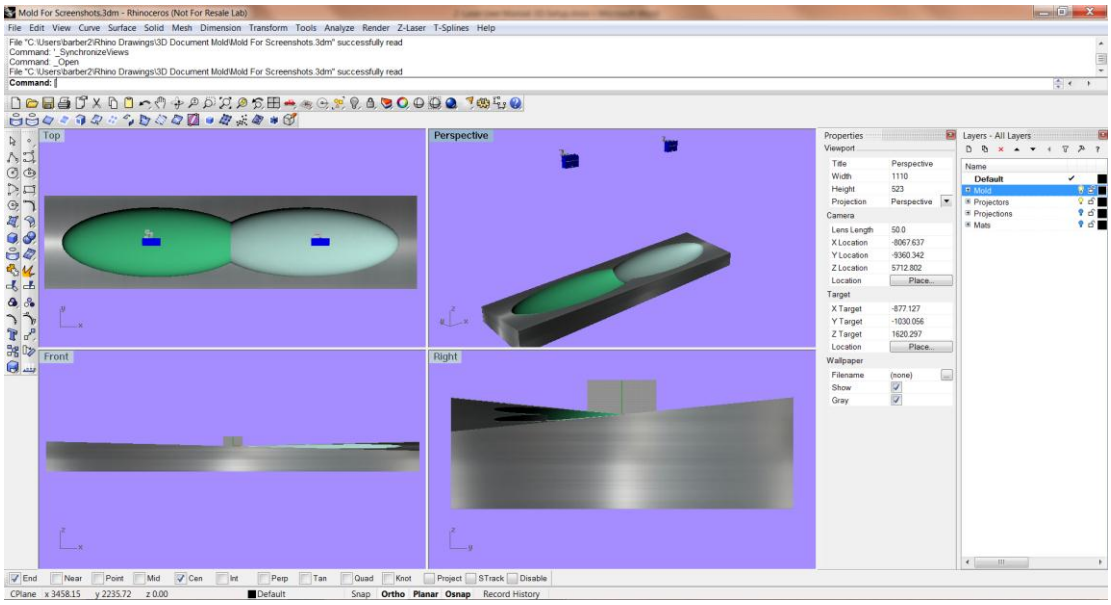
Introduction

A ***Z-Laser*** projector must be calibrated to its working environment in order for it to project accurately. A minimum of 6 points, including their X, Y and Z coordinates within this environment, are required for the 3D calibration process. This allows the projector to learn about its position with regards to the mold or tool on which it must project.

The 3D modelling software Rhinoceros allows the user to import a wide range of different files which contain the aforementioned points in relation to the mold or tool. The ExportToLPM module then allows the user to export the coordinates of these points to a text file. This file is then imported into LPM and it is used to calibrate the projector to the environment.

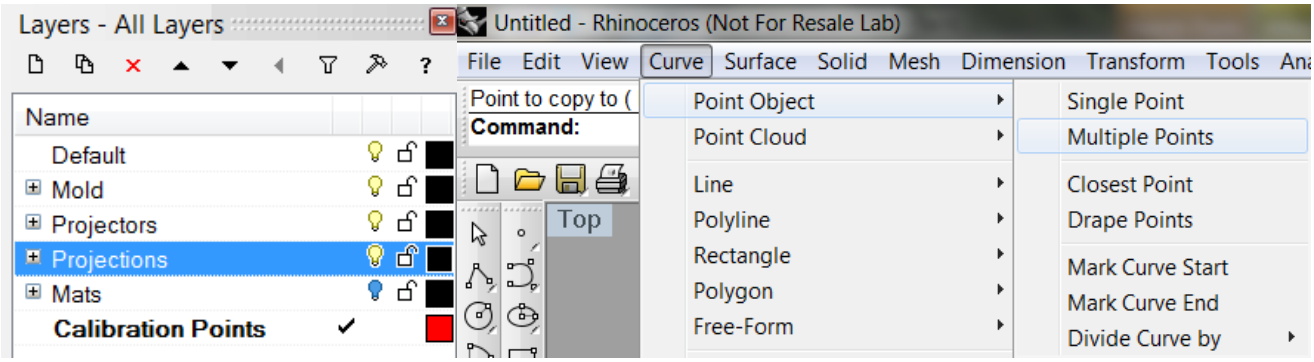
Creating the Calibration File in Rhinoceros

Open your 3D file in Rhino. Ensure the **Main1**, **Main2**, **Standard** and **Curve From Object** toolbars are docked to the top and side of your user interface. The Object Properties and Layer dialog boxes should also be visible.



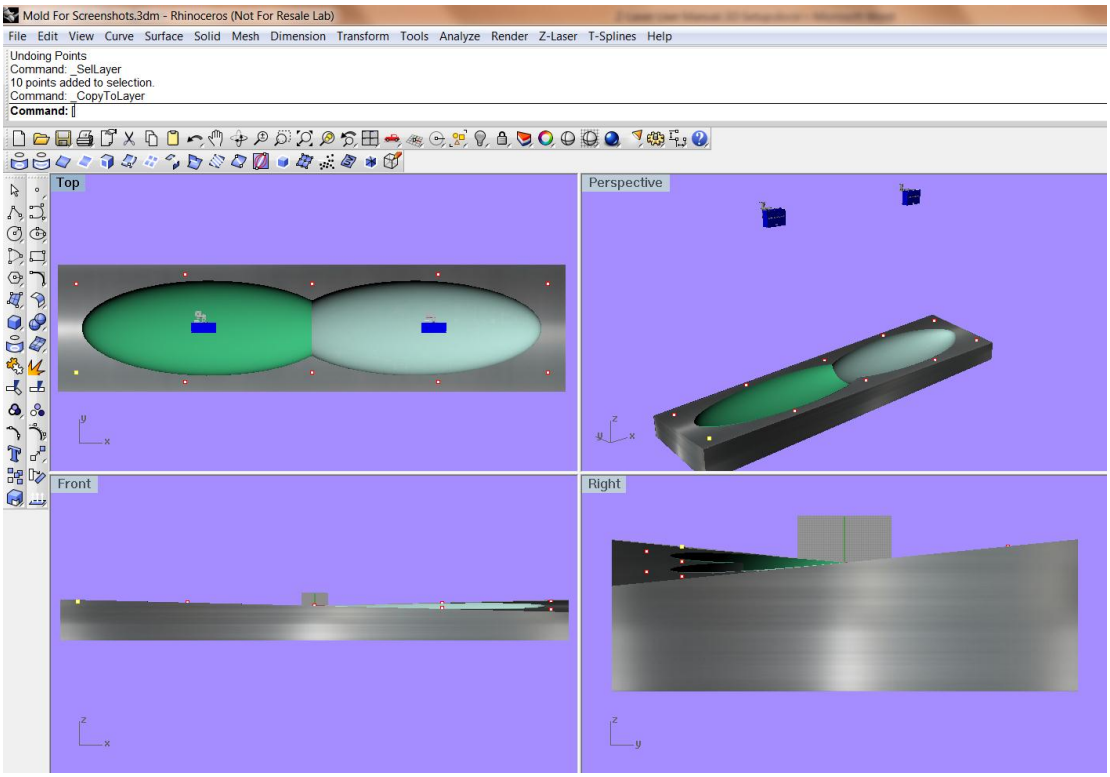
In order for the projectors to be calibrated to the mold, each projector requires a minimum of 6 reflective points placed on the mold. These points must be visible to the projector and the X, Y and Z coordinates of each point must be known. Once these points have been placed on the mold, the points must be placed into the 3D file.

Create a layer called **Calibration Points**. Set the colour to red and make it the active layer by placing a tick next to the colour. Next, select **Curve** → **Point Object** → **Multiple Points** from the menu bar or type **_Points** and press enter.

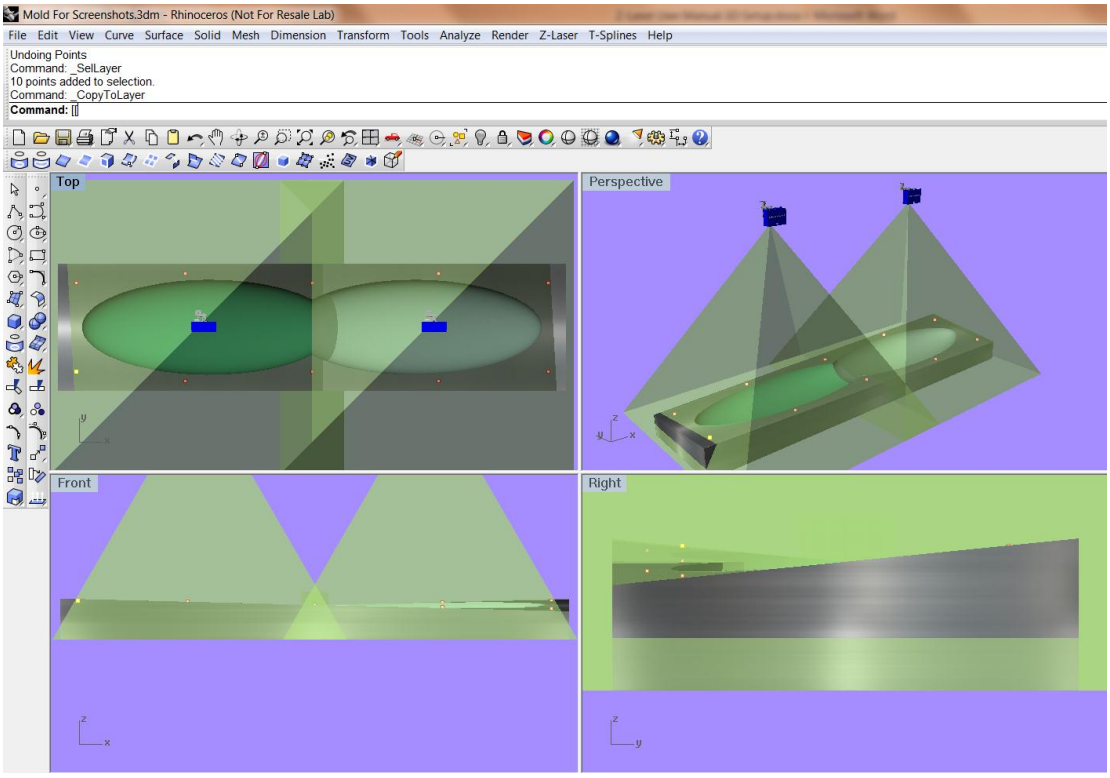


Place the point using an object snap to place it manually or move the mouse into the top viewport and simply enter the X, Y and Z coordinate in the command prompt. Separate each co-ordinate with a comma and press enter to place the point.

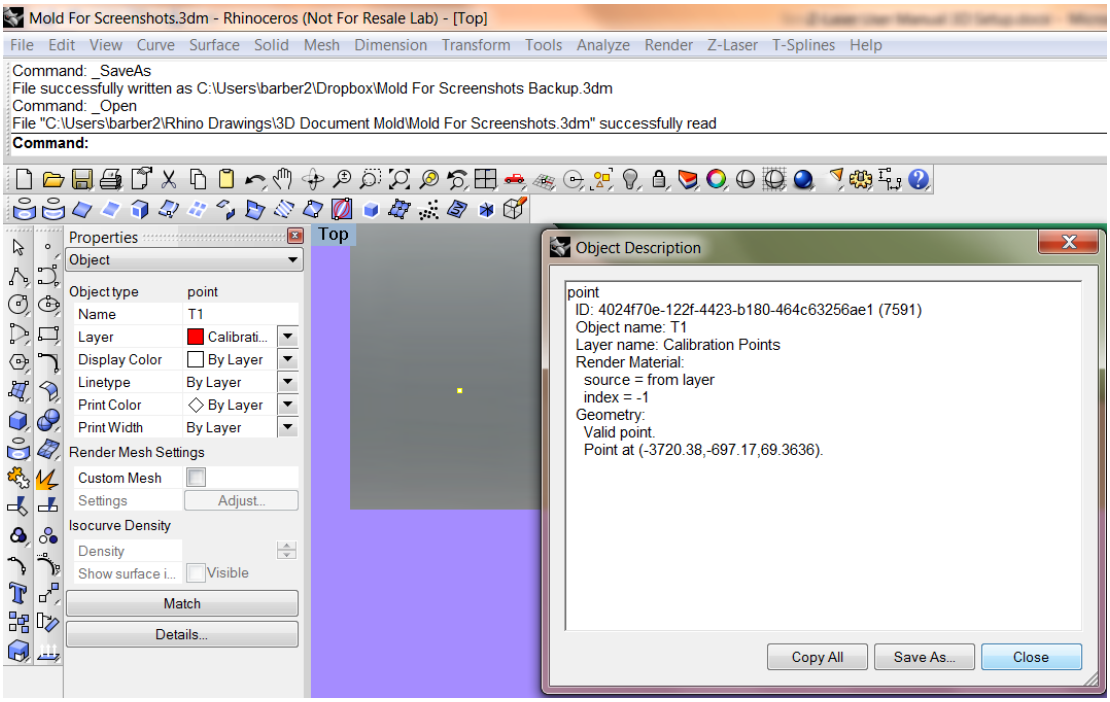
Continue placing all 12 points in their positions. When finished, press **Enter** to stop placing points.



Ensure all points are inside the maximum field border of the two projectors otherwise the points will not be found during calibration.



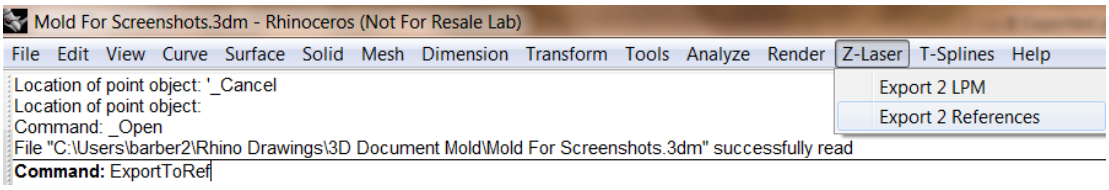
The ExportToLPM module exports all points that start with the letter T by default. To change this prefix, modify the setting **RefTargetName** in the ZLASER.INI file. Select one of the calibration points. In the Object Properties dialog box, change the **Name** to **T1**. Select the Details button in the Properties pane to see details of the point, including object and layer name and coordinates.



Continue to rename the other 12 points until you have 12 points labelled T1 → T12.

At this point, the calibration point file is now ready to be created. You can do this by:

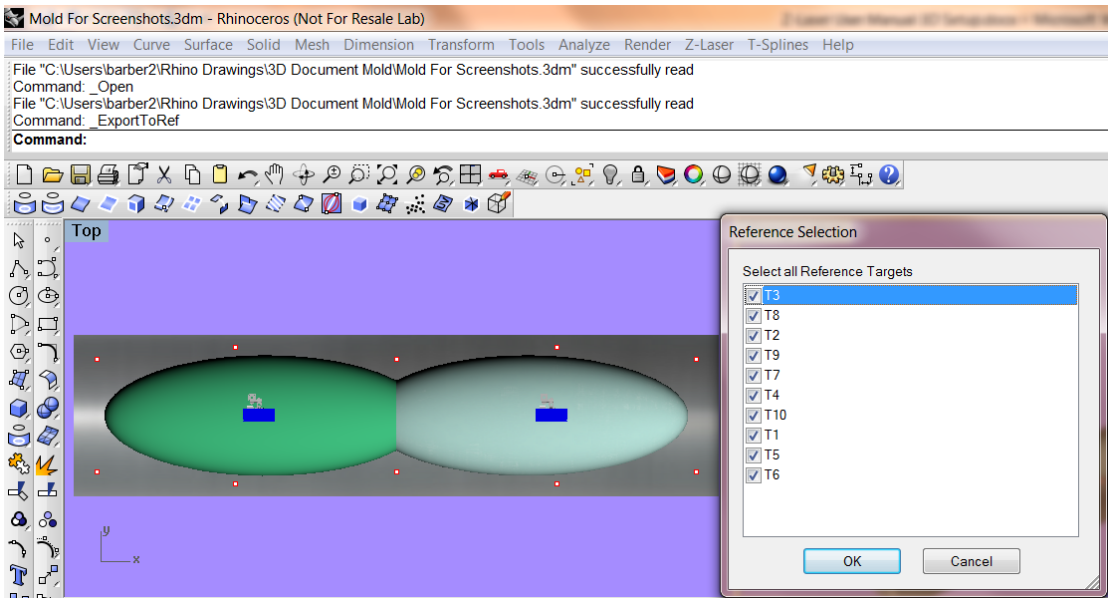
- Typing **ExportToRef** in the Command field at the top of the screen.
- Selecting **Z-Laser** → **Export 2 Ref** in the menu bar.



After selecting **Export 2 Ref**, a window will appear with a table listing all the points starting with T which have been found in the Rhino file.

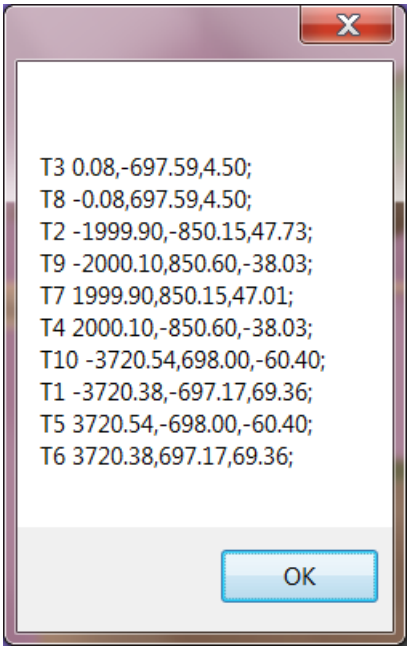
- Select only the points you wish to use by placing a tick in the box next to that point.
- Make sure you have selected at least 6 points
- Make sure that if you select a point from the Rhino file that you know which point it relates to on the real mold or work-piece.

Select **OK** after you have made your selection.

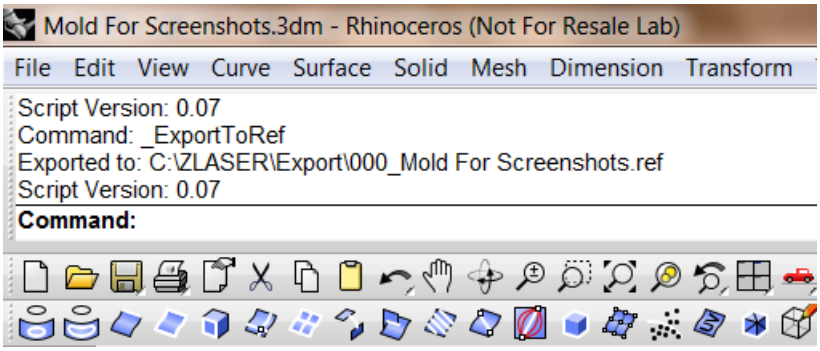


Another window should now appear showing the points which have been selected, the name of the points and the X, Y and Z co-ordinates of that point. You can confirm these co-ordinates by selecting the point in Rhino and press the **Details** button in the **Properties** pane.

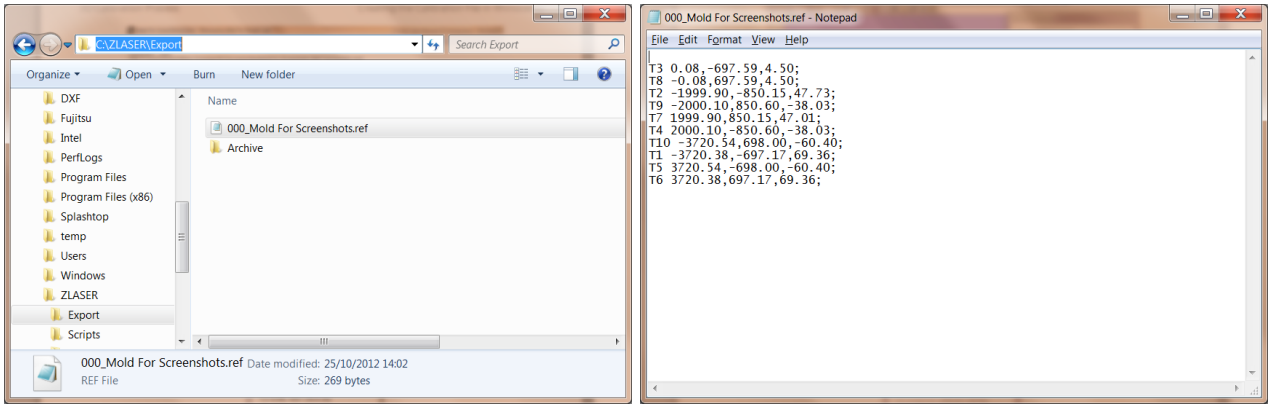
Select **OK** to continue.



The path and file name will be displayed in the information field. If it is not displayed, use the up arrow found at the right hand side of the field.



Navigate to the file and open it with Notepad. Check the information matches the details displayed in the window shown in point 9. Close Notepad. If you made changes, do not save them when closing Notepad.



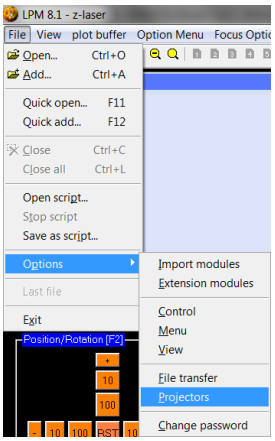
Alternatively, if you know the calibration point numbering and X, Y and Z coordinates, you could open Notepad and create the text file yourself. Simply list the calibration point name, add a space and list the X, Y and Z coordinates separated by a comma, all followed by a semicolon. All decimal places are preceded by a point. When saving the files, select “Save as”, select the file type as “All Files” and add the filename followed by a point and the suffix REF. e.g. Points.REF. See the Notepad screenshot above as an example.

The file is now ready for uploading to LPM 8.1.

3D Calibration of the Projector

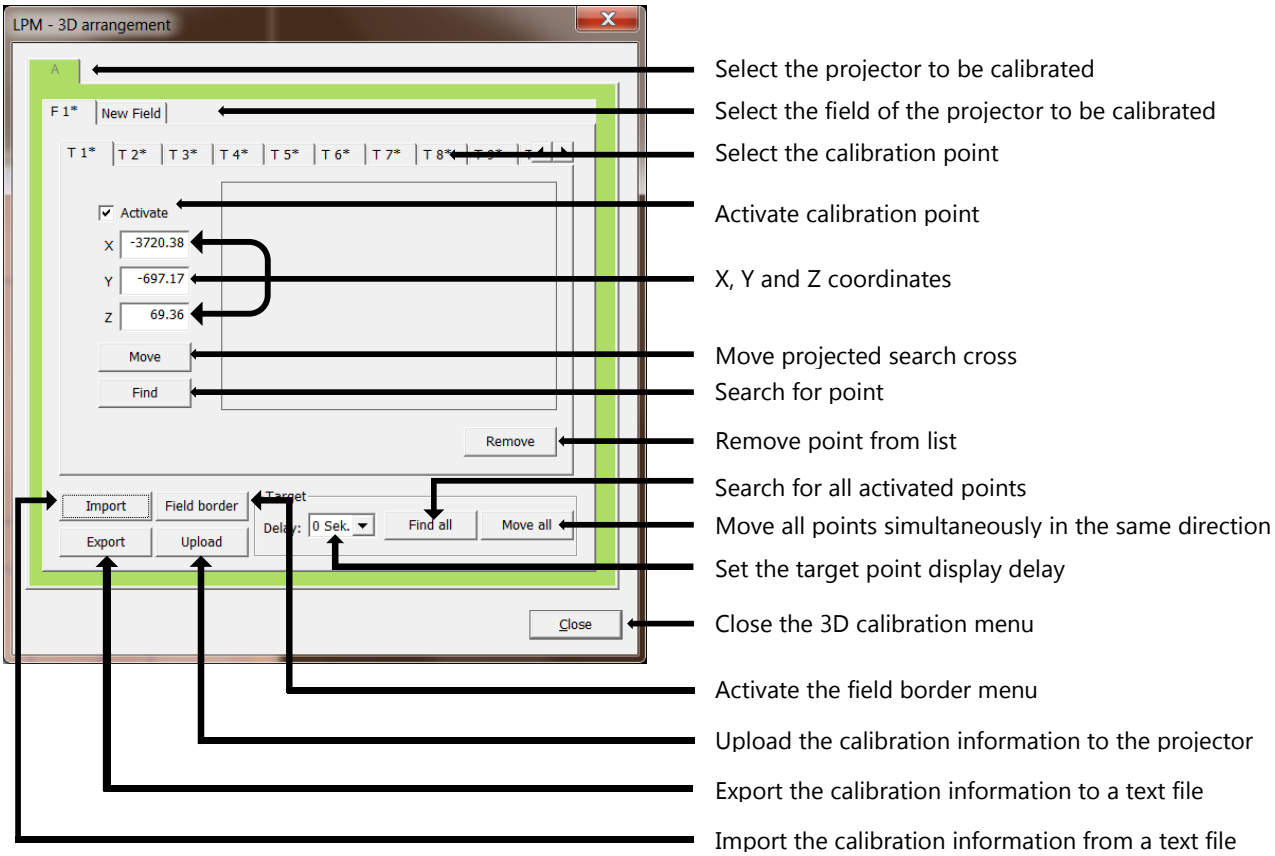
Open LPM. Wait for all projectors to connect with the software. Each projector will be displayed by a blue field border in the graphical interface of LPM.

To start the 3D calibration, go to **File** → **Options** → **Projectors**. Enter the password cps when prompted to do so.



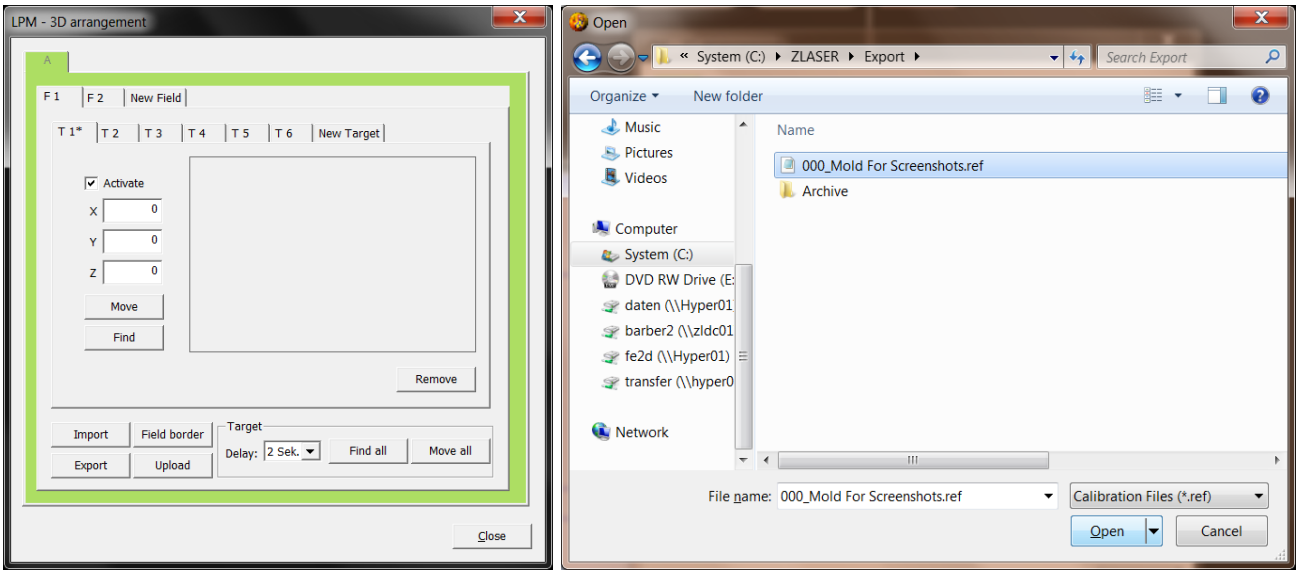
The 3D calibration window appears. See the figure below for the function of each button.

Figure 4: The 3d Calibration Menu

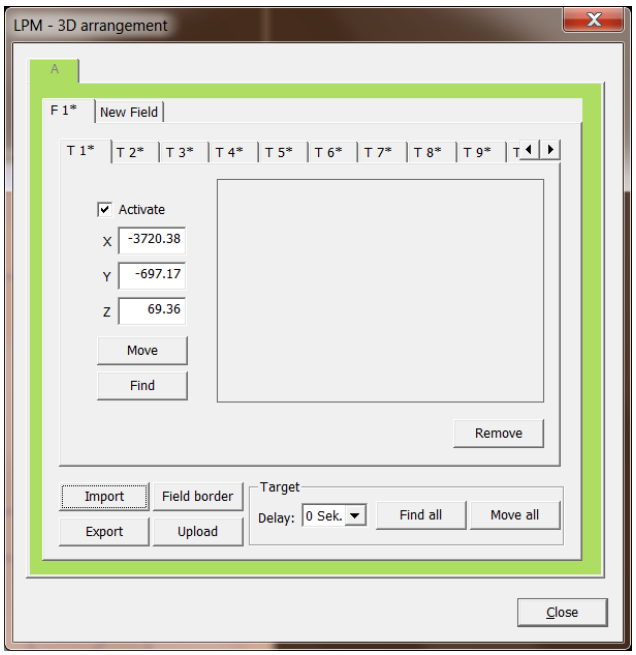


Upon entering the 3D Calibration menu, the projector will display a cross with a number next to it. The number relates to the calibration point that is currently selected. Select the projector name and the field of the projector you wish to calibrate using the tabs. In this example, Projector A, Field 1 is selected.

Press **Import** to select the calibration file. Navigate to the folder containing the file, select it and press **Open**.



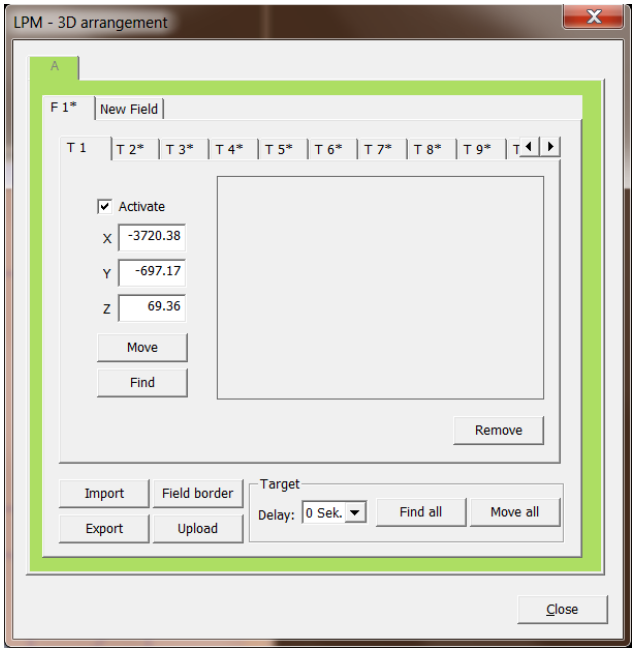
The coordinates for points T1 to T10 have now been uploaded and they can be seen in the X, Y and Z field for each point.



Select the T1 tab. Notice that there is an asterisk (*) after the name of the point. This indicates that the point has not yet been scanned. By selecting **Move**, the projected cross can be controlled by the

mouse or keyboard arrows. Move the point over the glass reflector positioned at T1. Left click the mouse to stop moving the point.

Press **Find** to search for the point. The projector will now project a moving line and scan the glass reflector in two axes. If the point is found, the projected cross should be seen over the T1. The asterisk will also disappear next to the name of the point.

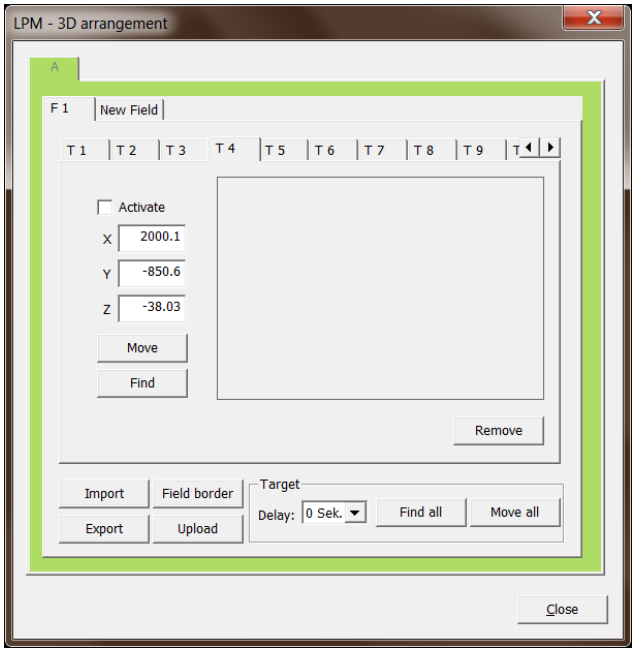


If the point is not found, the asterisk will appear next to the label T1. Ensure that the glass reflector is not hidden or dirty and that the moving line passes directly over the glass reflector.

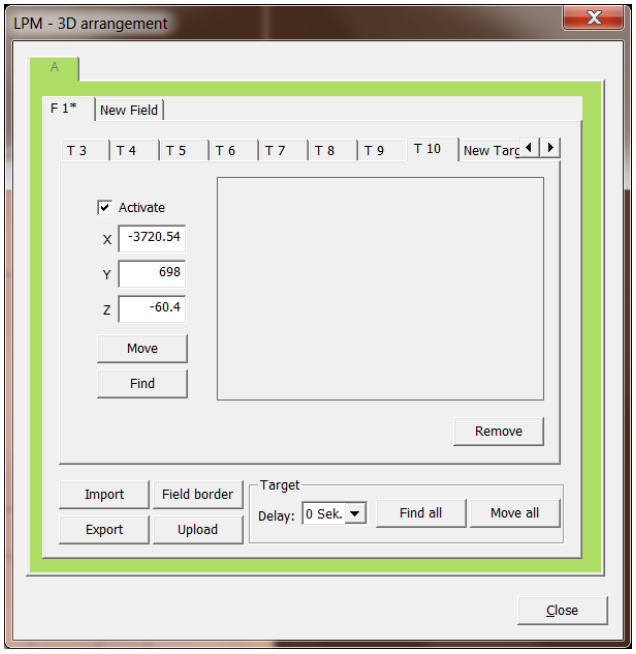




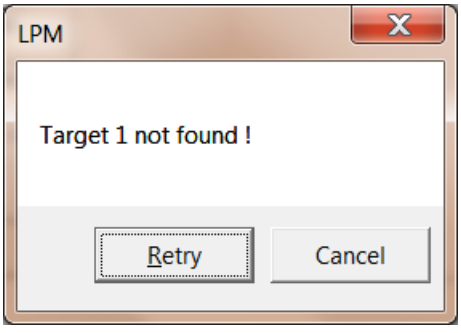
In this example, Point T4 – T7 are outside the maximum field border of Projector A and they will not be found. Simply remove the tick from the tick box for all points not in use by this projector.



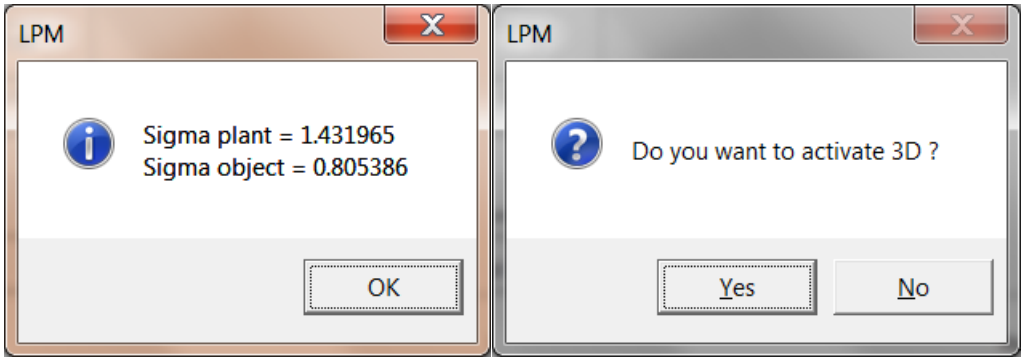
Select T2 and repeat until all 6 points have been found.



Alternatively, move the crosses over all the points and select **Find All**. All points will be scanned. Once a point is scanned, it will display the position of the point for a time determined in the field **Delay**. If a point is not found, the following error message will appear.

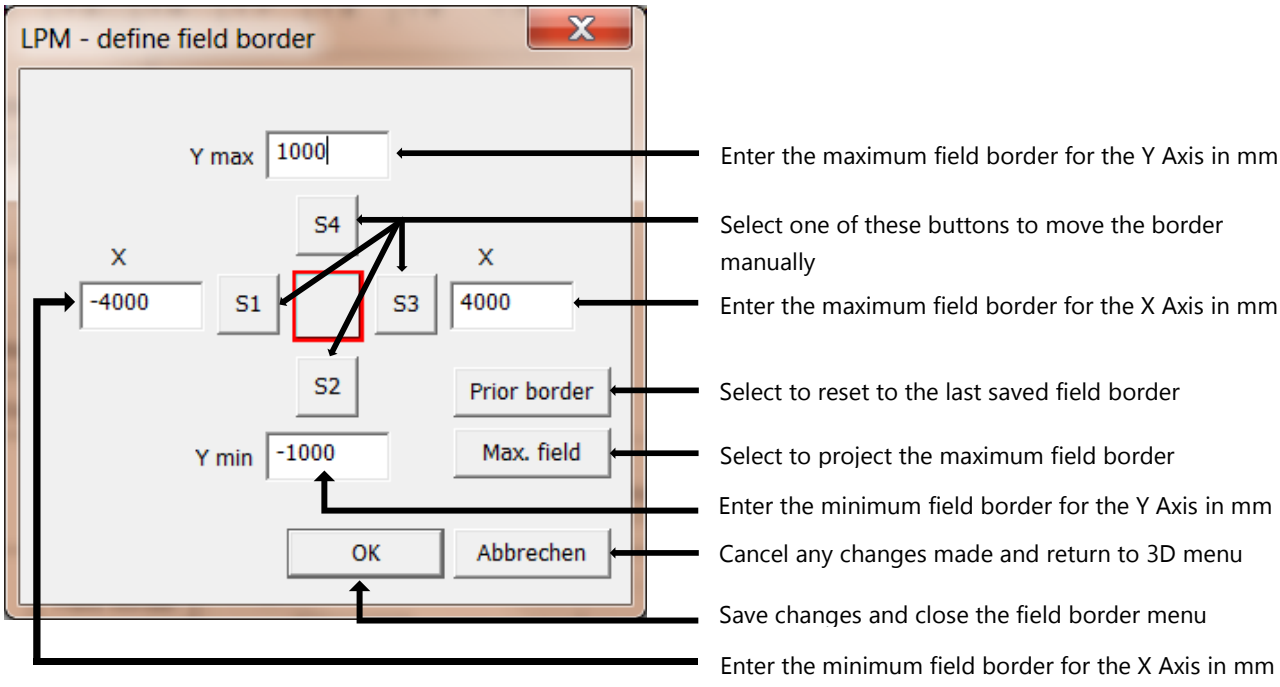


Once all points have been found, select **Upload** to send the information to the projector. The projector will return two values. The Sigma plant and the Sigma Object indicate the accuracy of the projector and maximum deviation of the real world position as compared to the same position of the file. The sigma values should be less than 1. Any values over 1 mean that all data should be rechecked. Select **OK** and another pop-up window asks if you want to upload the data to the projector. Select **Yes**.



Next, select **Field Border**.

Figure 5: The 3D Field Border Menu

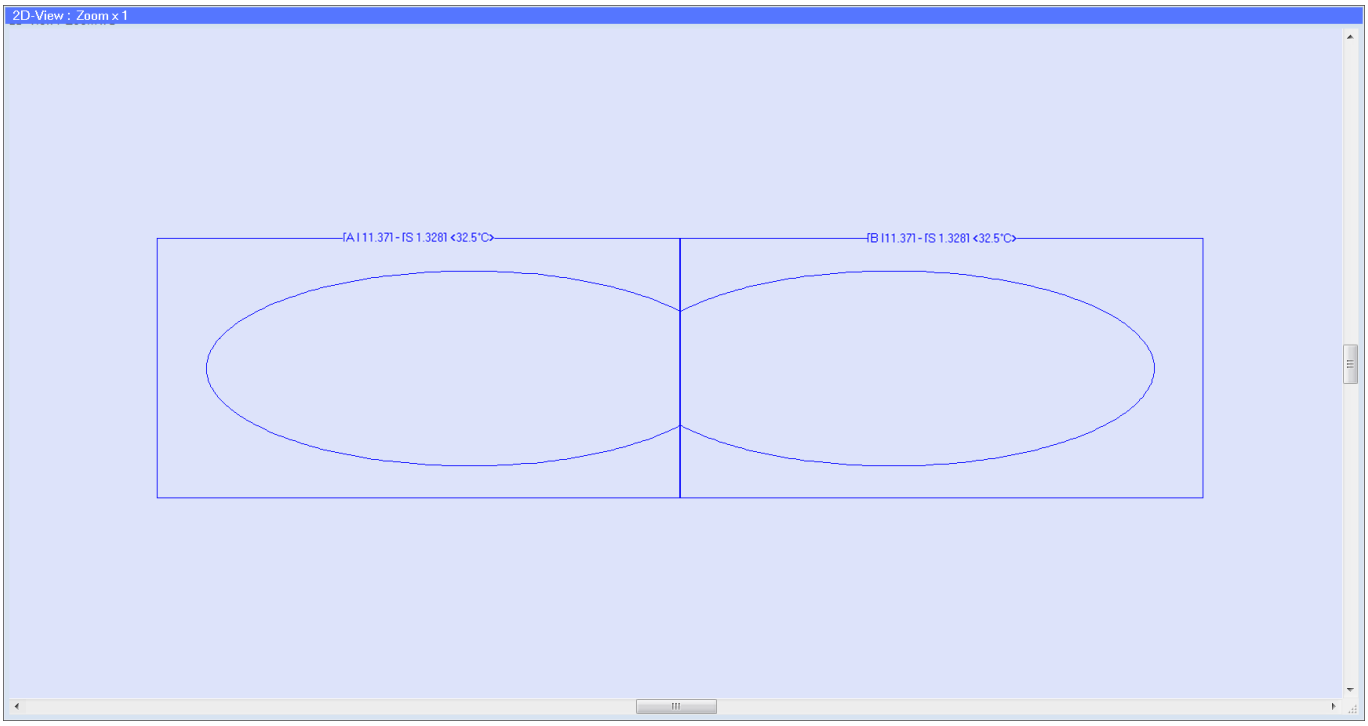


The projector should now display a rectangle with 4 arrows shown in the inside and centre of each of border. If you have previously altered the field border and you are returning to alter it, select **Prior Field Border** for the last saved field border to appear.

Use the **S1...4** buttons to move the field borders manually with the use of a mouse or the keyboard arrows. Alternatively, you can enter the dimensions of the field border in the text fields provided.

Ensure you do not cross the maximum field border edges. This will be indicated by the field border lines deforming or by the arrows decreasing in size and eventually disappearing.

Press **OK** when finished and select **Upload** again for the data to be sent to the projector. The projector will display the sigma plant and object again. Select **OK** and activate **3D** again.



Your projector is now calibrated. If you have more than one projector, select the tab for the name of the projector and repeat the above procedure until the projector is calibrated. Ensure that there is at least one shared field border between the projectors. When finished, select **Close**.

The field border will now be displayed in 3D mode in the graphical interface window. The six points will be shown in relation to their X and Y coordinate and the sigma object will be displayed at the top of the blue field border between the client version number and the temperature.

Repeat this until all projectors are calibrated.

The Sigma Values

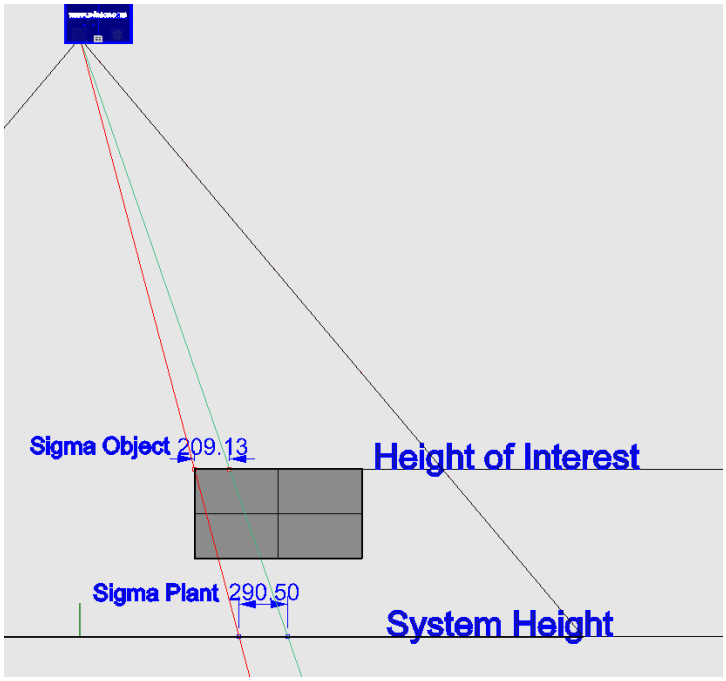
During the production of each Z-Laser projector, it undergoes a process called Teaching. The projector is placed perpendicularly to a grid of reflective points at a distance known as the **system height**. It is then positioned so that the lode point is directly in front of the exact center of the wall. The reflective points have a known X and Y coordinate which is uploaded to the projector. The projector then scans these points and notes the position of the mirror galvanometers (X and Y) at the time when the reflected light is returned. These positions are saved in order to create a calibrated reference grid that is the basis for all future calculations regarding 2D or 3D calibration in a new working environment.

The sigma values are the theoretical results of a calculation based on the calibrated reference grid, the accuracy of the scanned calibration points and the coordinates given, the system height and the object of interest height.

Sigma plant is the deviation between the scanned position and the theoretical (3D model) position of the calibration points at the system height.

Sigma object is the transformation of the sigma object value to the height of the object of interest. If the object is placed between the projector and the system height, then the Sigma Object will always be smaller than the Sigma Plant. If the object is placed further away than the system height, then the Sigma Object will be greater than the Sigma Plant.

Below is a representation of the idea behind the Sigma calculations. The red line depicts the scanned position of a reflector placed on the top left corner of an object. The green line depicts the theoretical position of the reflector. This is calculated by the projector according to the coordinates of the point uploaded to it and the projectors estimation of where the point should be found based on the reference grid at the system height. The distance between the two points at the system height contributes to the value of the Sigma Plant., as all calibration points are required to be measured to give the Sigma Plant.



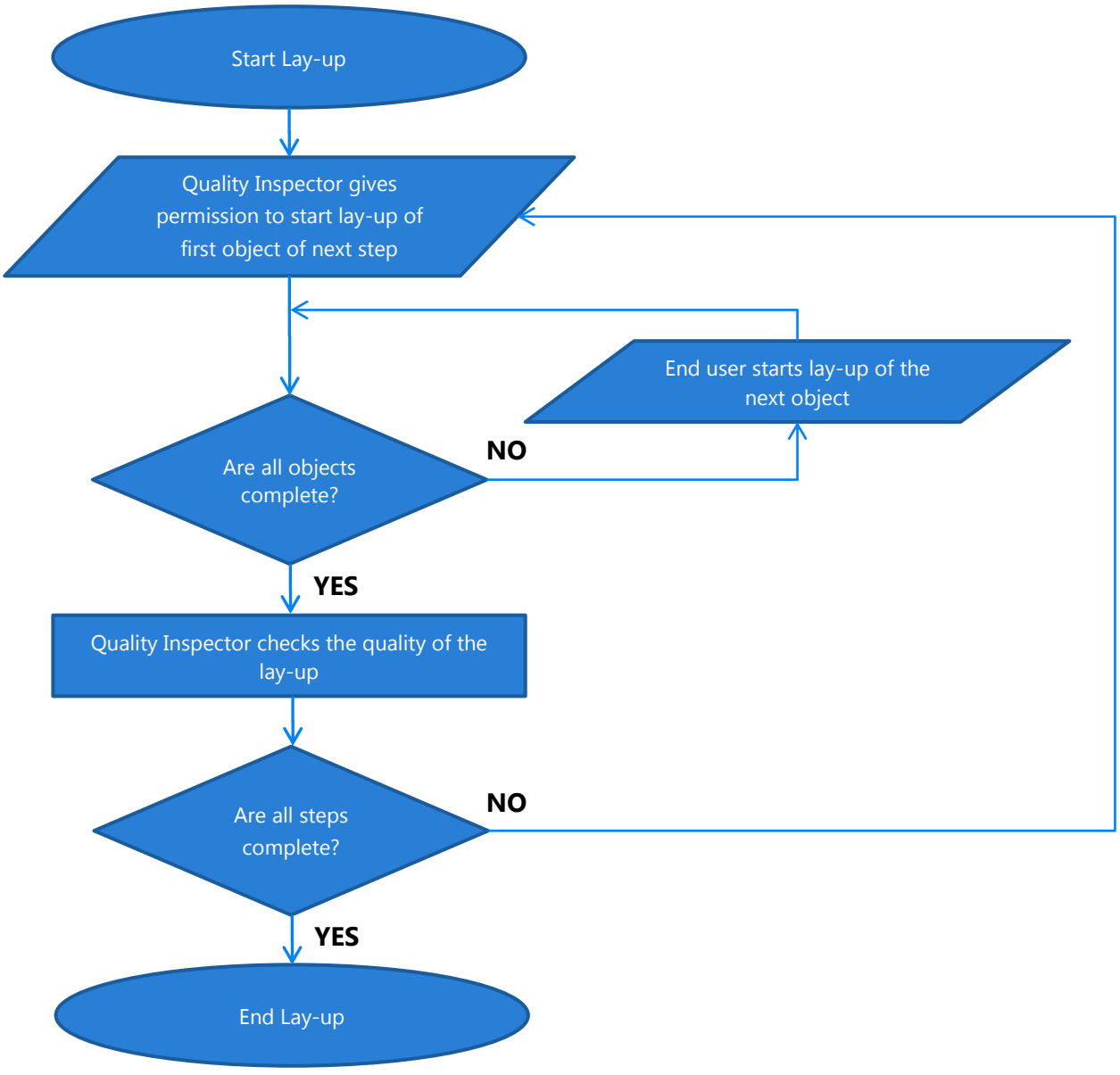
The Sigma Object takes the Sigma Plant value and transforms it to the height of interest. As the figure indicates, when the height of interest is between the system height and the projector, then the Sigma Object is smaller than the Sigma Plant.

3D Projection Process

Introduction

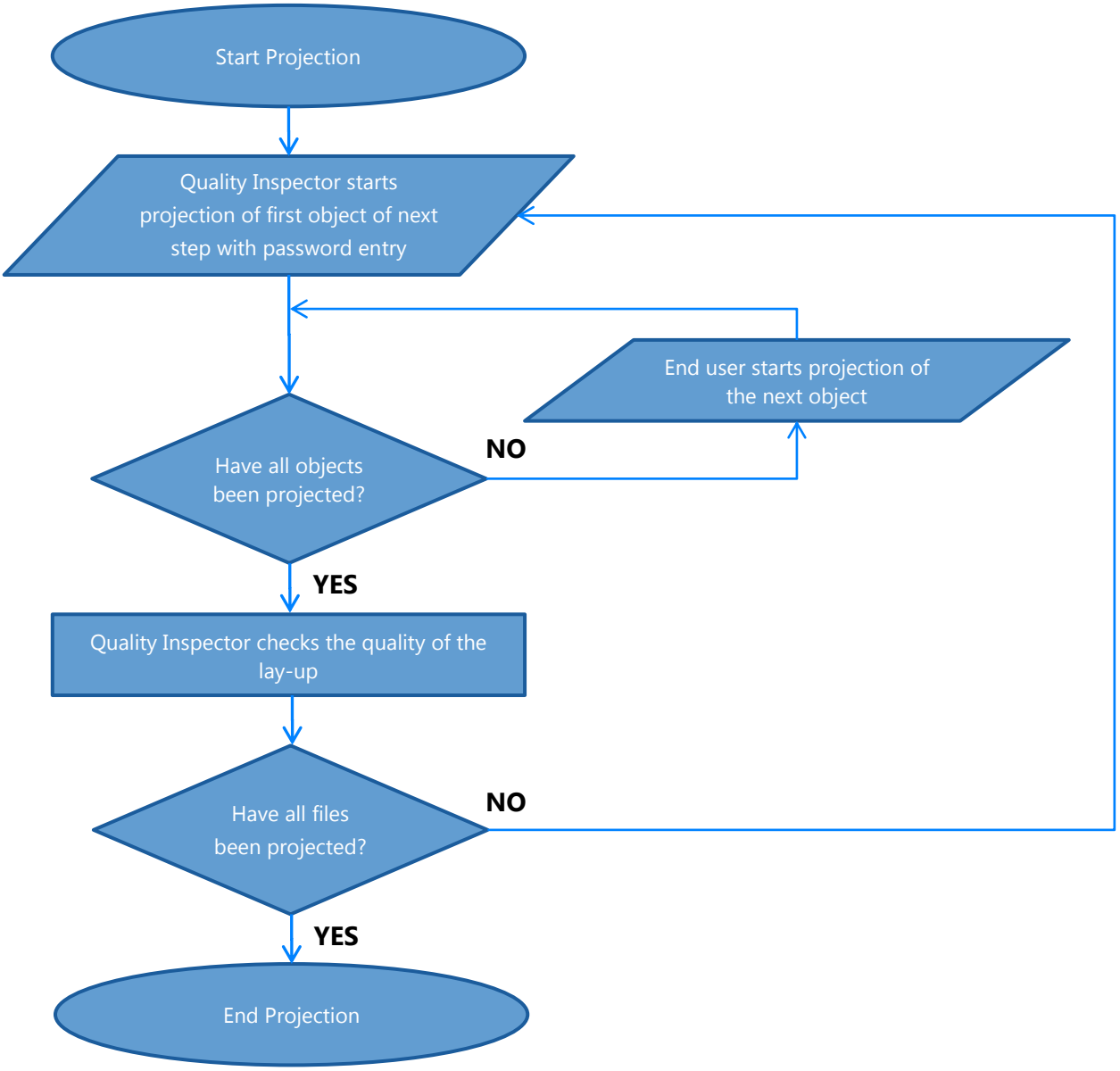
Each customer will have a different working environment and work flow from one another. Generally, the process is divided into steps and objects. Each step contains a set of objects that must be placed before continuing onto the next step. In the case of a rotor blade mold, a step may contain a set of fibre and balsa mats that have to be placed correctly before the quality inspector gives permission to proceed to the next step. In the composite material industry, each step may contain a separate mold, where upon completion of the placement of the objects, requires the mold to be changed. The basic work flow is as follows:

Figure 6: The Step/Object Work Flow



Using Rhino, we will extract the information required for projection and copy it to a set of layers that will replicate this step/object process. The step will be represented by a main/parent layer while the objects will be copied to a sub-layer of the main/parent layer. When the information is exported to a HPGL 3D file, each step will be represented by a separate file whereas each object will be represented by a different pen colour within that file. The projection work flow will look as follows:

Figure 7: The Projection Step/Object Work Flow



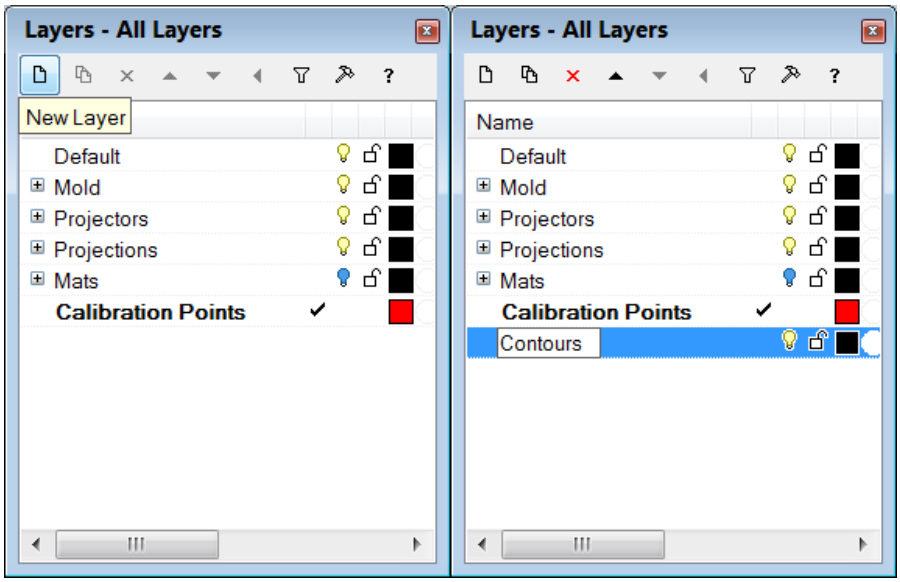
In order to achieve this, the following must happen within Rhino:

1. Create the main/parent layers for each step.
2. Create the sub-layers within the main layers for each object
3. Extract the information from the original data and copy to the correct sub-layer

4. Give each object a name to avoid confusion when exporting
5. Export the first step layer giving each object a pen colour
6. Repeat until all objects that are to be projected have been exported

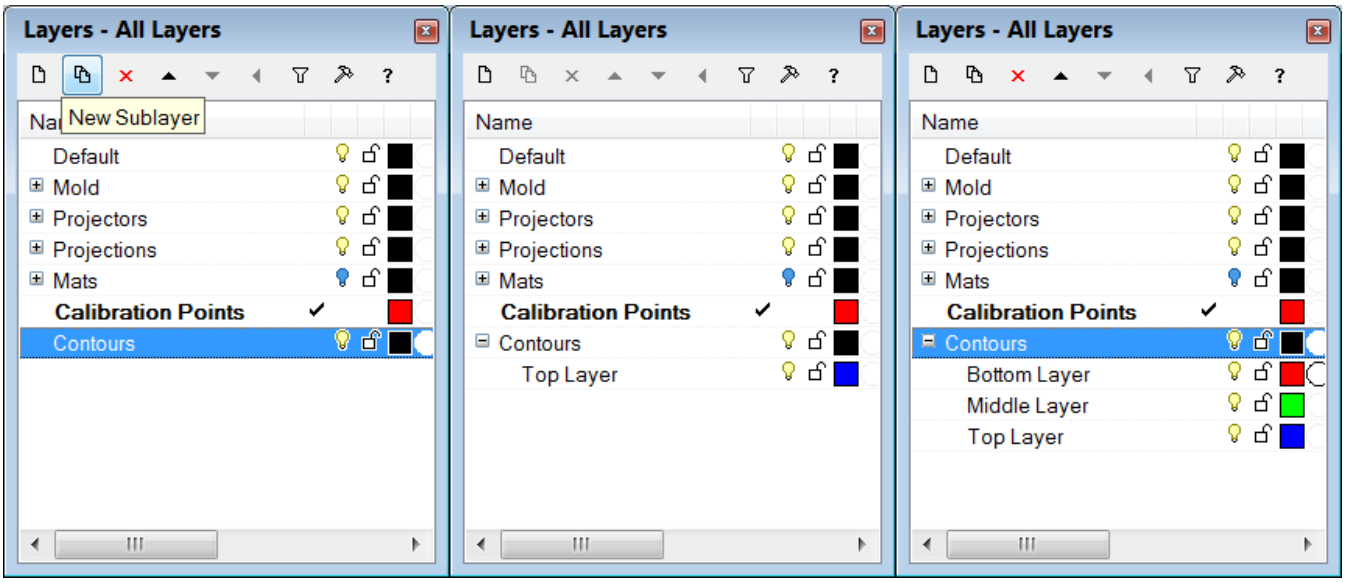
Create the Main/Parent Layer

Open the **Layer Dialog Window**. On the toolbar, select **New Layer**. Rename the layer. In the example, all the data will be exported to one file, separated as different pens to be projected. Therefore only one main layer is created and it is named Contours. Move the layer until it sits at the bottom of the list.



Create the Sub-Layers

Select the layer that will be the main/parent layer. Select the **Sub-Layer** button on the toolbar. Rename the layer and change the colour of the layer. Repeat until all sub-layers have been created. Re-order the sub-layers as you would like them.

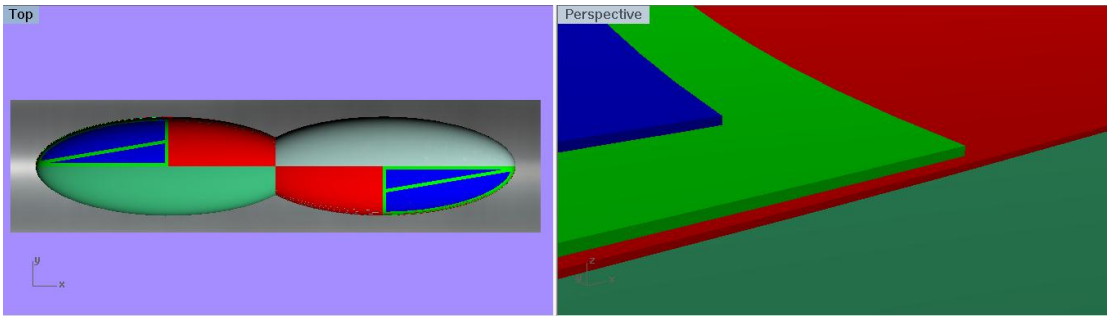


In the example, there will be three sub-layers for the red bottom mats, the green middle mats and the blue top mats.

Extracting Curves from Surfaces or Solids

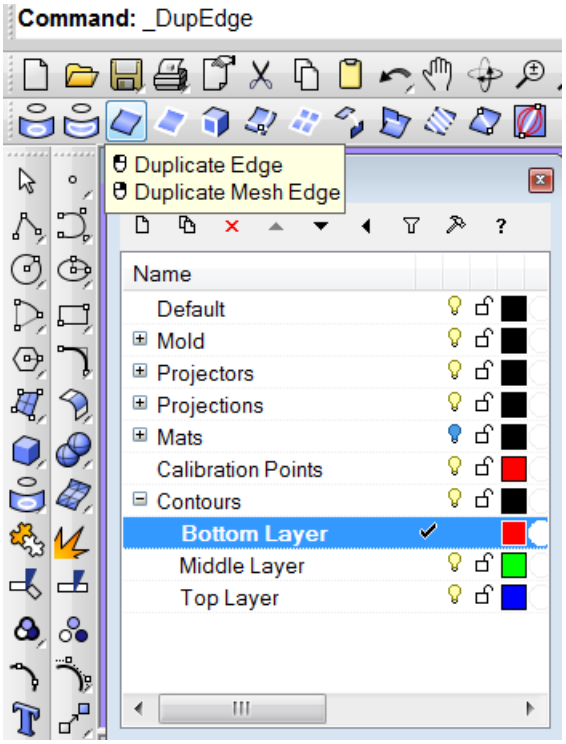
The tutorial example below has 8 pieces of matting that need to be placed into the mold. These mats have a 3mm thickness and they are surfaces with no naked edges (solids).

The 2 red mats are should be placed first followed by the 2 green mats and then by the 4 blue mats. The thickness of the mats can be seen in the perspective view below.



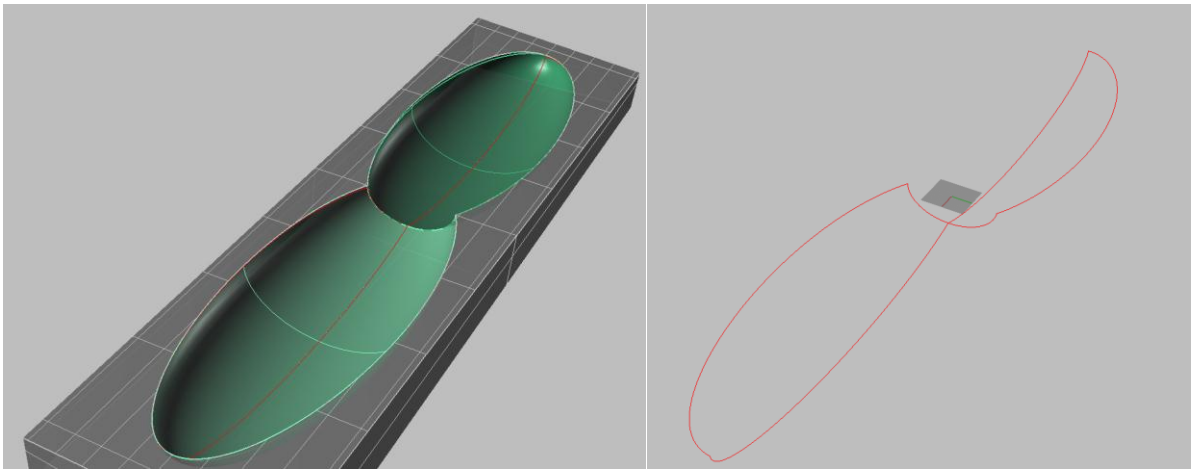
It is important at this point to decide whether you would like to project only the bottom or top edges, or project both on separate layers. Projecting both at the same time will slow the projector down, causing flickering or produce a double line which could confuse the end user. For this example, I will be selecting the bottom edge of all mats.

Make the correct sub-layer active (Bottom Layer) by placing a tick mark in the Active Layer Selection column and then type **_DupEdge** or left mouse click the third button from the left (**Duplicate Edge** button).

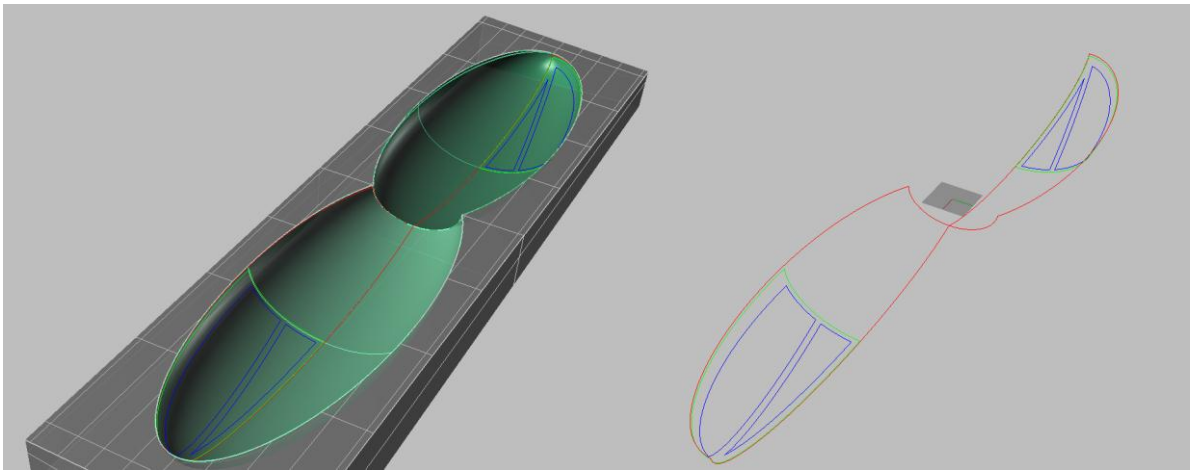


Select the bottom edge of both red mats. Make sure you select the correct edges by using the zoom window button.

Once you have selected all the correct edges, press enter. If you are in Render View, switch to Wireframe or Ghosted in order to see the new curves. You can also switch off all other layers to make sure you have extracted these edges correctly.



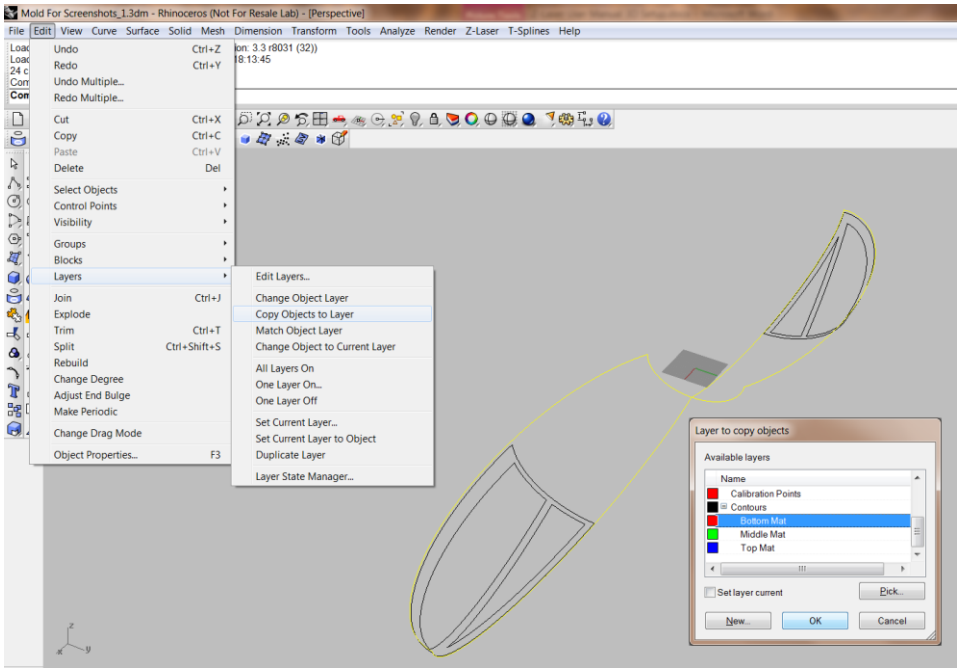
Continue to extract all the curves from the solids to the correct sub-layers.



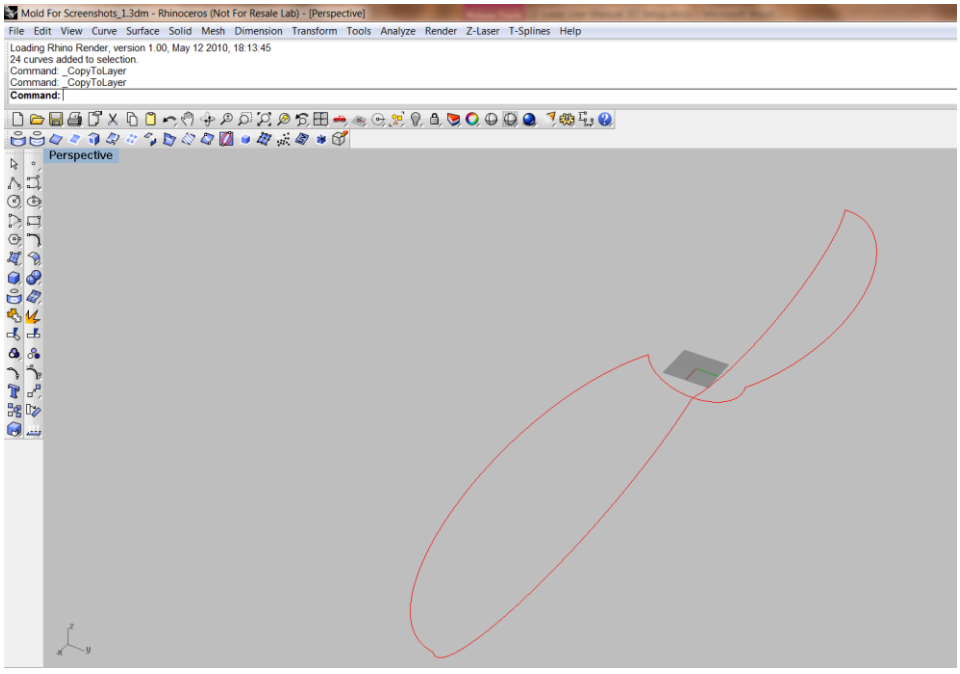
Copying Existing Contours to Sub-Layers

If your 3D data only has existing curves that are to be projected, it is good practice to copy these curves to the correct sub-layer in order to preserve the original data for future use.

Select the curves you want to copy and then go to **Edit → Layers → Copy Object to Layers** or type **_CopyToLayer**. A dialog box will appear. Select the sub-.layer you want to copy the curves to and select **OK**.



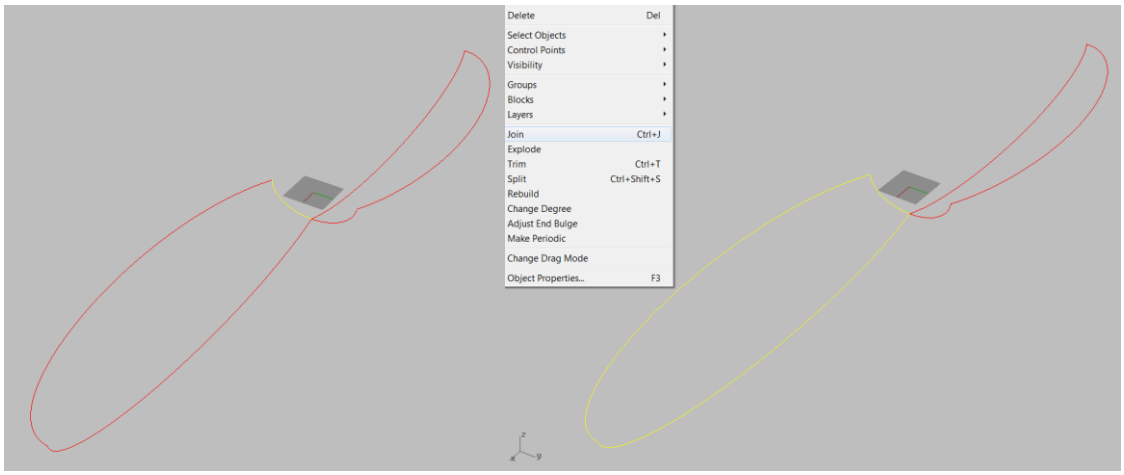
The curves will now be preserved in the original layer with a copy made to the sub-layer used for projection.



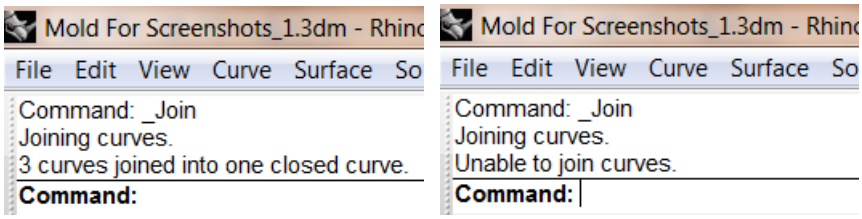
Preparing the Objects for Projection

All sub-layers should now have objects (points or curves) and they should now be prepared for exporting. It is considered good practice to join all connected curves into a single polyline and name them for future reference.

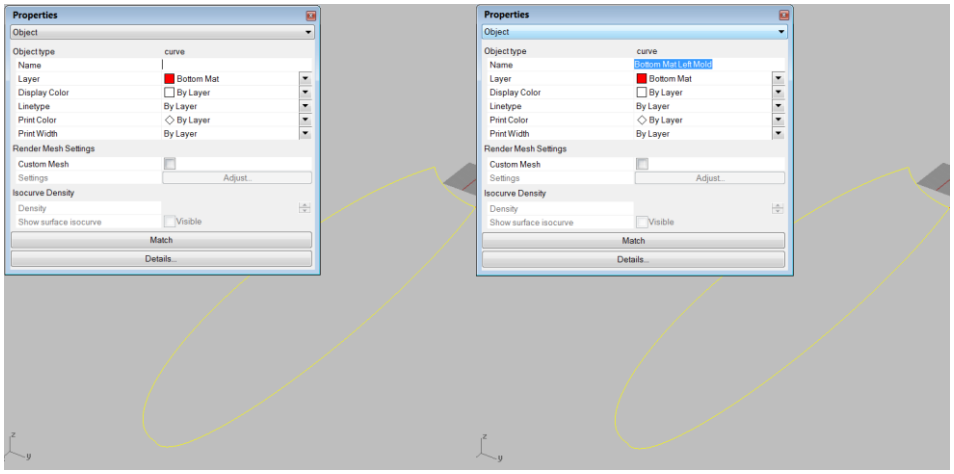
Select the curves you would like to join. Type _Join or go to **Edit → Join** and the curves will join.



Check the command history window to see if the join was successful. If not, it may be that the curve cannot be joined due to an incorrect curve being copied earlier (top edge of a polysurface instead of the bottom edge).



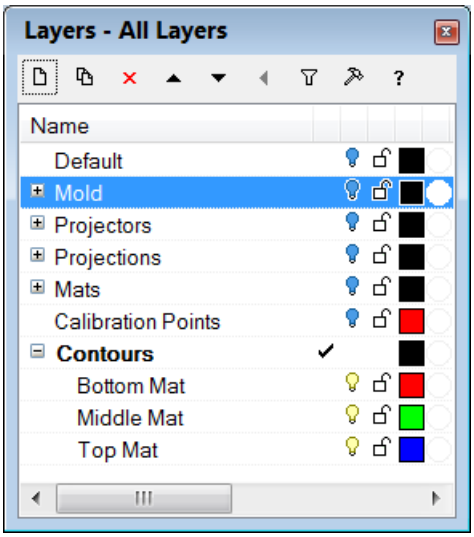
Select the joined curve and open the Object Properties window. Give the object an appropriate name and press enter.



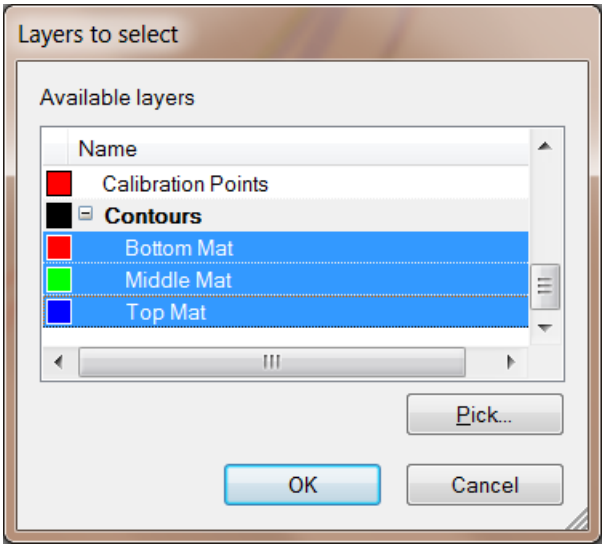
Repeat for all objects.

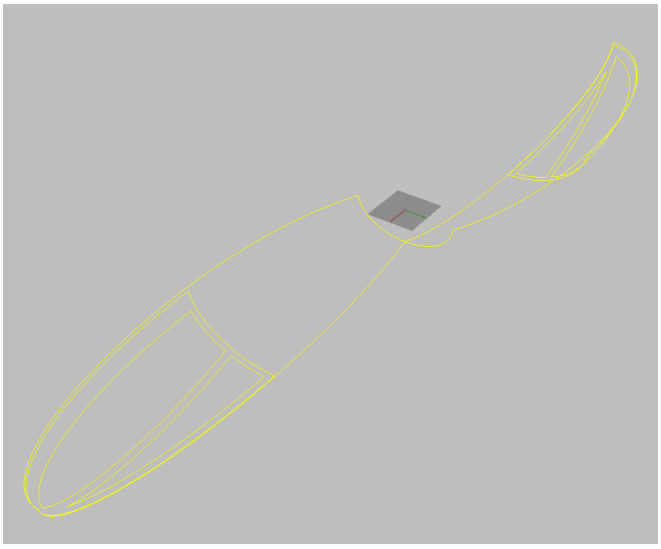
Exporting the data with the ExportToLPM module

The data is now ready to be converted to a HPGL 3D file. Hide all the layers except for the main layer which is to be exported.

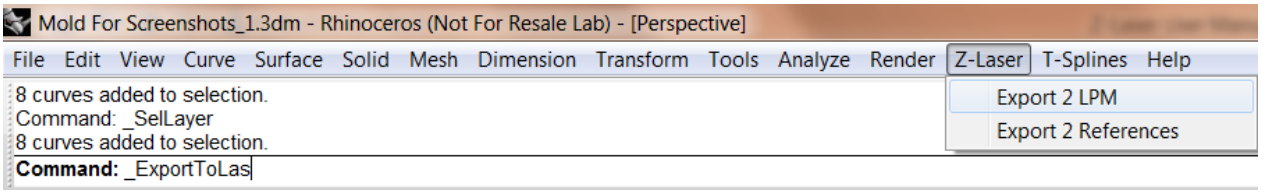


Press the Escape button to deselect all objects. Type **_SelLayer** or go to **Edit → Select Objects → By Layer…** The Select Layer window will appear. Select all sub-layers from the main/parent layer by holding the Alt key and left clicking on all the sub-layers. When finished, press **OK**.

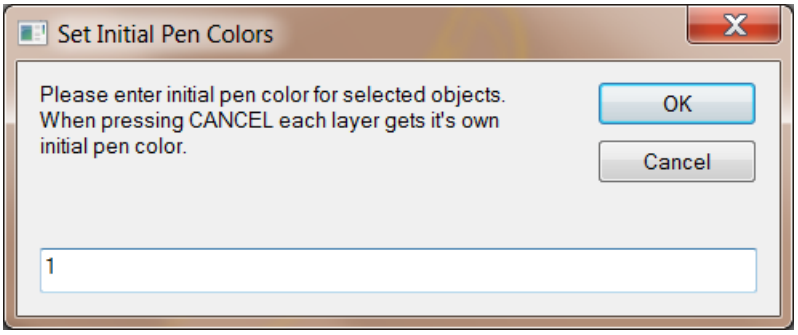




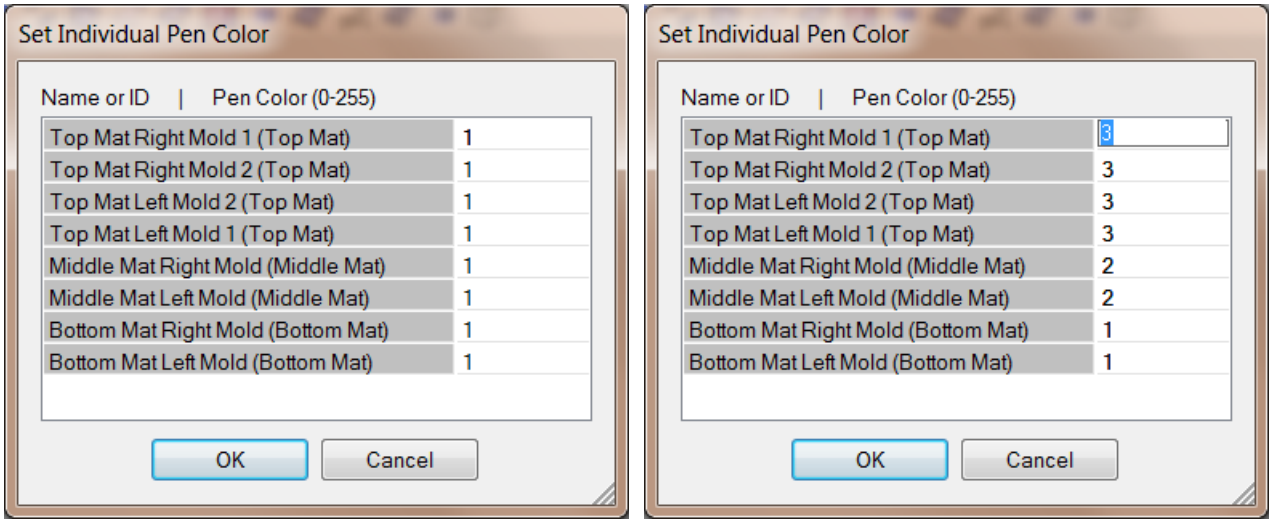
Make sure all the objects you want to export are selected. Type **_ExportToLas** or go **to Z-Laser → Export 2 LPM** and select it.



The module will ask the user to select a starting pen colour. The pen colour you select depends on your set-up in LPM, but Pen 1 is usually the starting colour. Type a number and press **OK**.

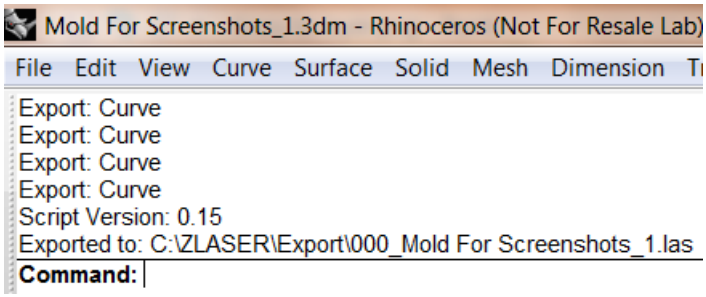


All the objects that were selected for export now appear in a new window. Each object is displayed as the object name followed by the layer name in brackets and the pen colour. All objects will have the starting colour assigned to it by default. If you wish to change the colour of the object, select the colour field for the object and type the correct pen colour in its place. Select **OK** when you want to continue.

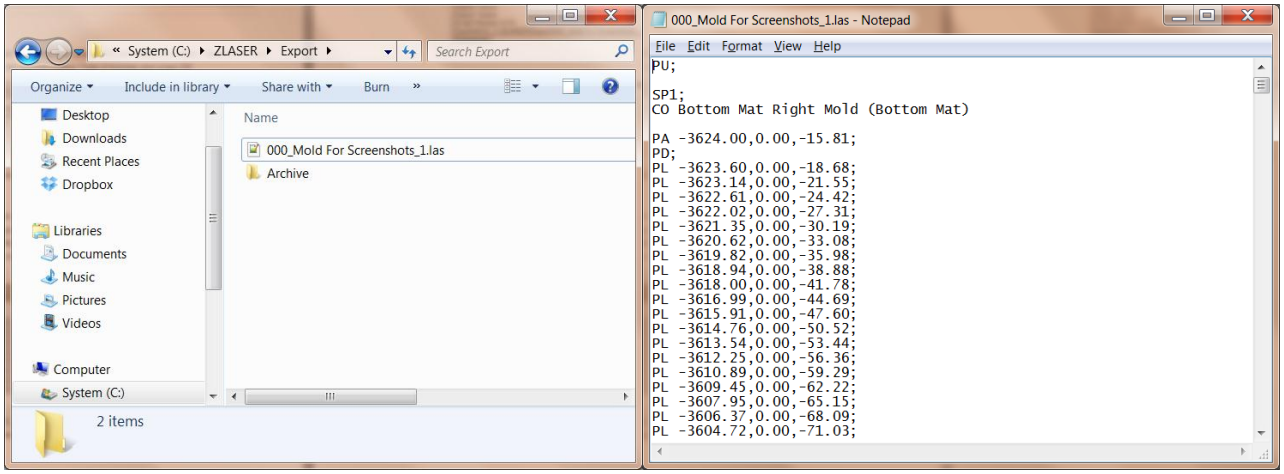


In the case of the manual example, the bottom layer will be projected on Pen 1, the middle on Pen 2 and the top layer on Pen 3.

The ExportToLPM module will then export all the objects to a newly created file. The name, extension and path of the file will be determined by the settings in the ZLASER.INI file. If the export of the file is successful, you will see the path and file name of the new file in the command history window.



Open Windows Explorer and navigate to the folder and open the file in a text editor such as Notepad or Notepad++. Do **not** use WordPad.

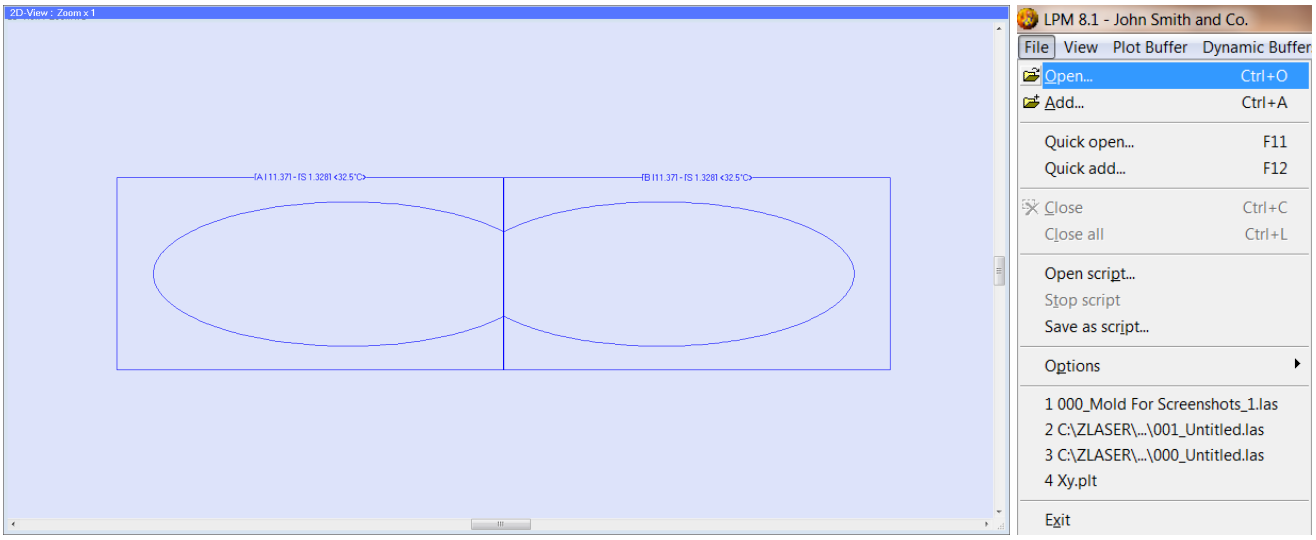


The file will consist of multiple polylines that make up the various objects (**PL** command), the pen colours of the object (**SP** command) and the name and layer name of the object (**CO** command).

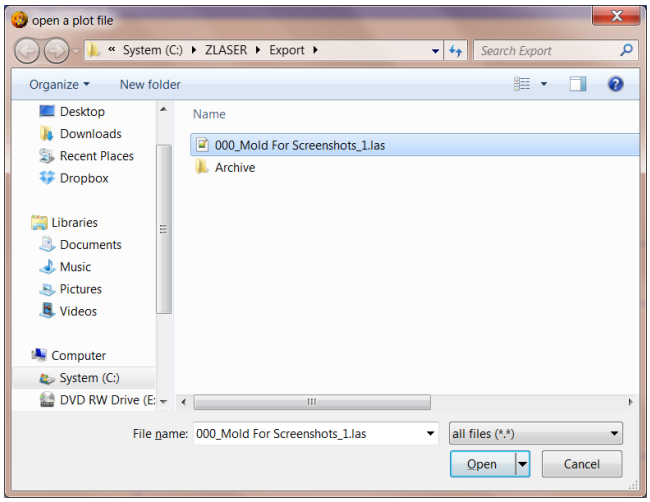
Close the file without saving.

Opening the HPGL 3D file in LPM

Open LPM 8.1 and wait for the blue field borders of the projectors to appear. Go to **File → Open...** on the menu bar or use the keys **Ctrl+O** to open a new file.



In the **Open a Plot File** window that appears, navigate to the file you exported from Rhino, select it and press **Open**.

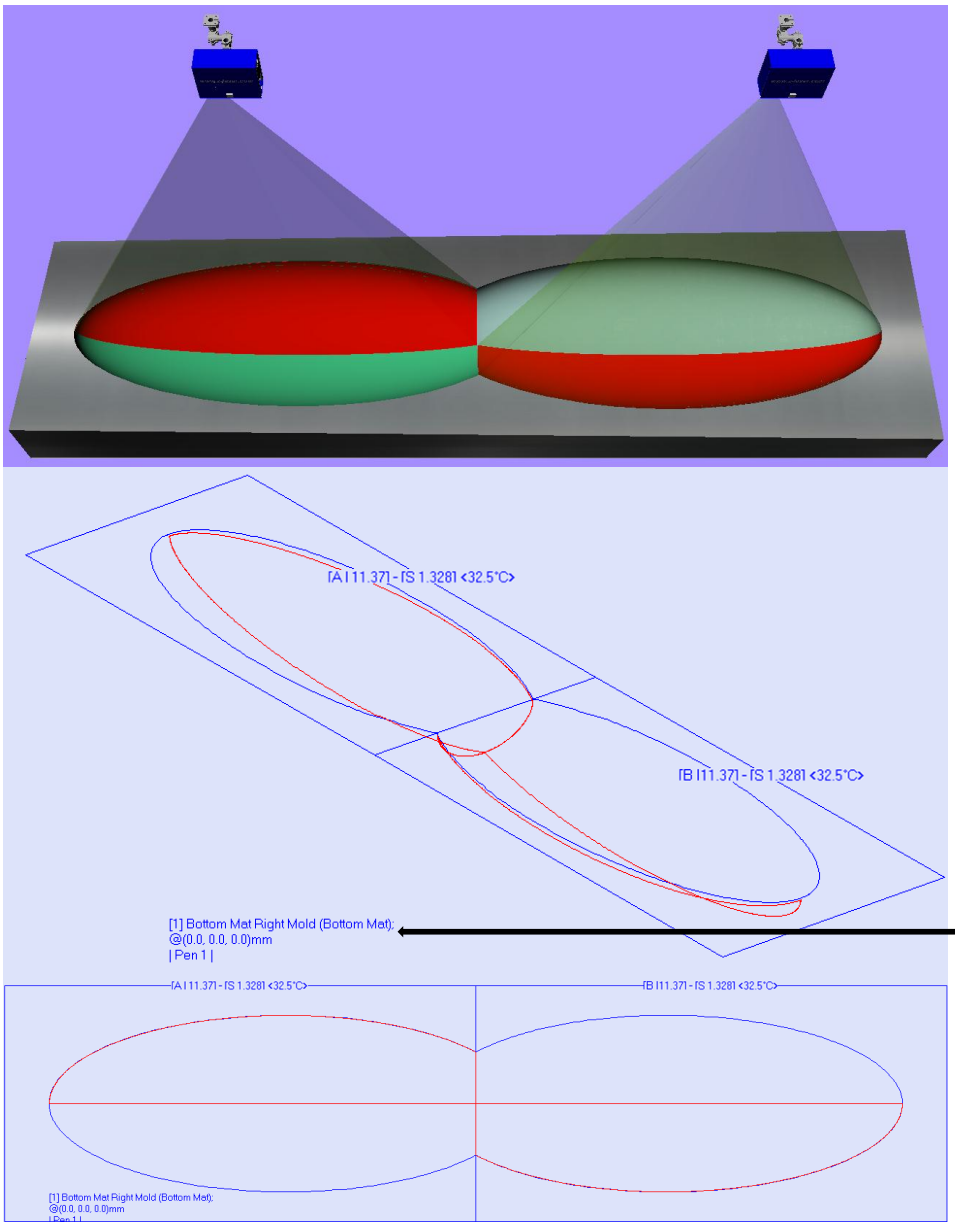


The file will load into LPM and it will send the data to the projector. If the file contains 2D data, the graphical interface will show the data in a top view. If the file contains 3D data, the graphical interface will show data in a perspective view. By selecting one of the graphical view icons on the toolbar, you can change to a view that suits the operator.



Pen 1 should now project. If you exported the file correctly, this should be projecting the first object to be placed. Pen 1 is shown as a red colour in the graphical interface by default. In the manual example, the following is seen:

Figure 8: Projecting the First Object



Projecting the red bottom mats in both shells

Perspective View of the bottom contours of the mat

Object and Layer Name
Coordinates
Pen Colour

Top View

To change the pen colour, go to **File → Option Menu → Direct Command** in the menu bar and type **/GS 2** to turn on Pen 2 and all other pens off.

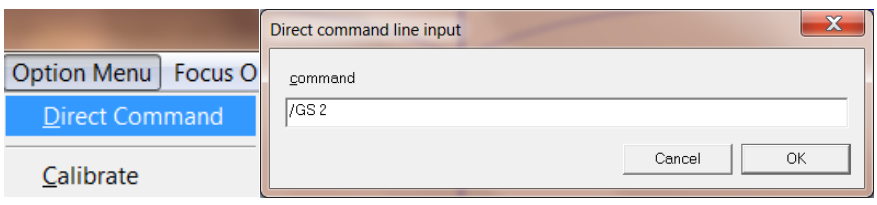
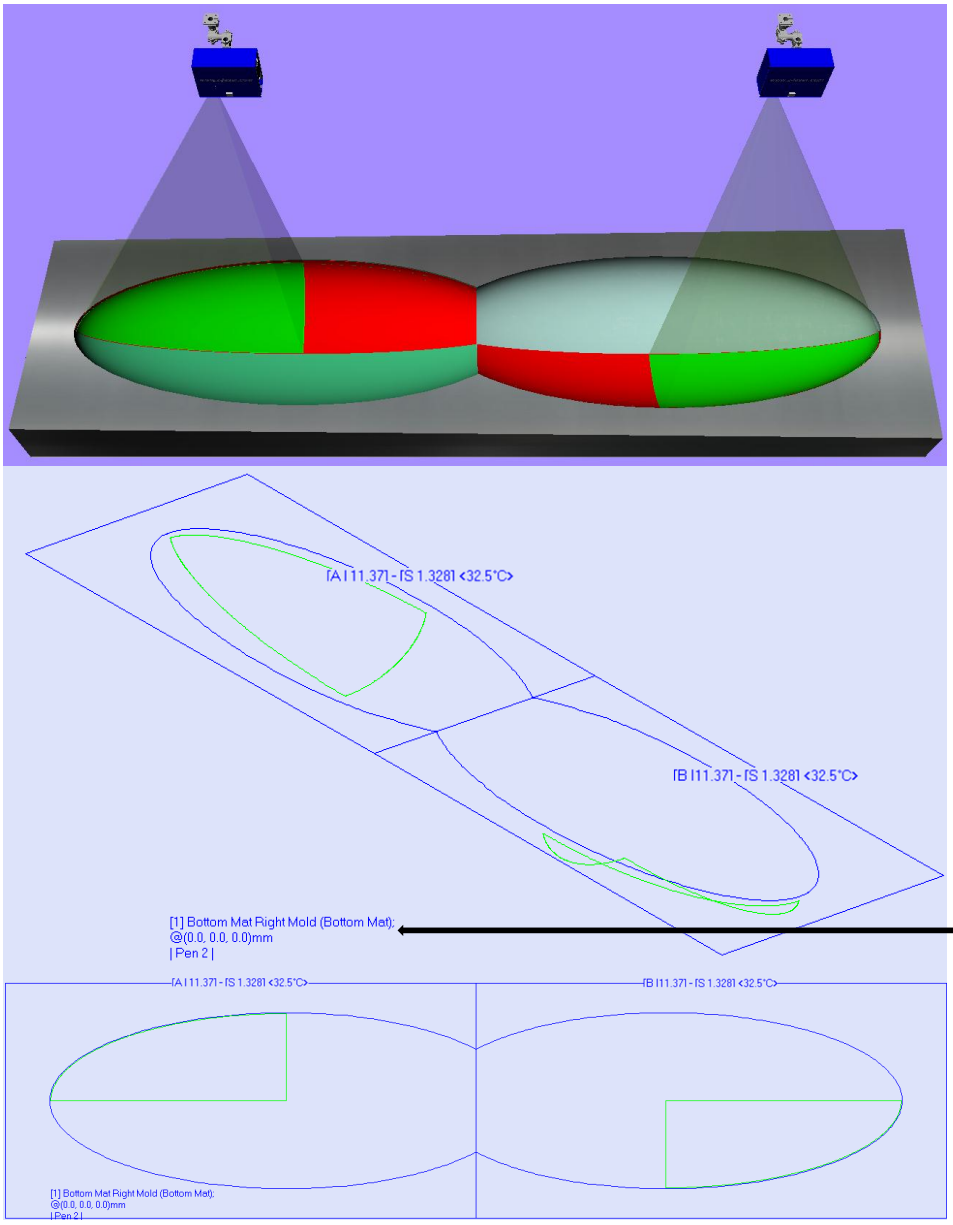


Figure 9: Projecting the Second Object



Projecting the green middle mats in both shells

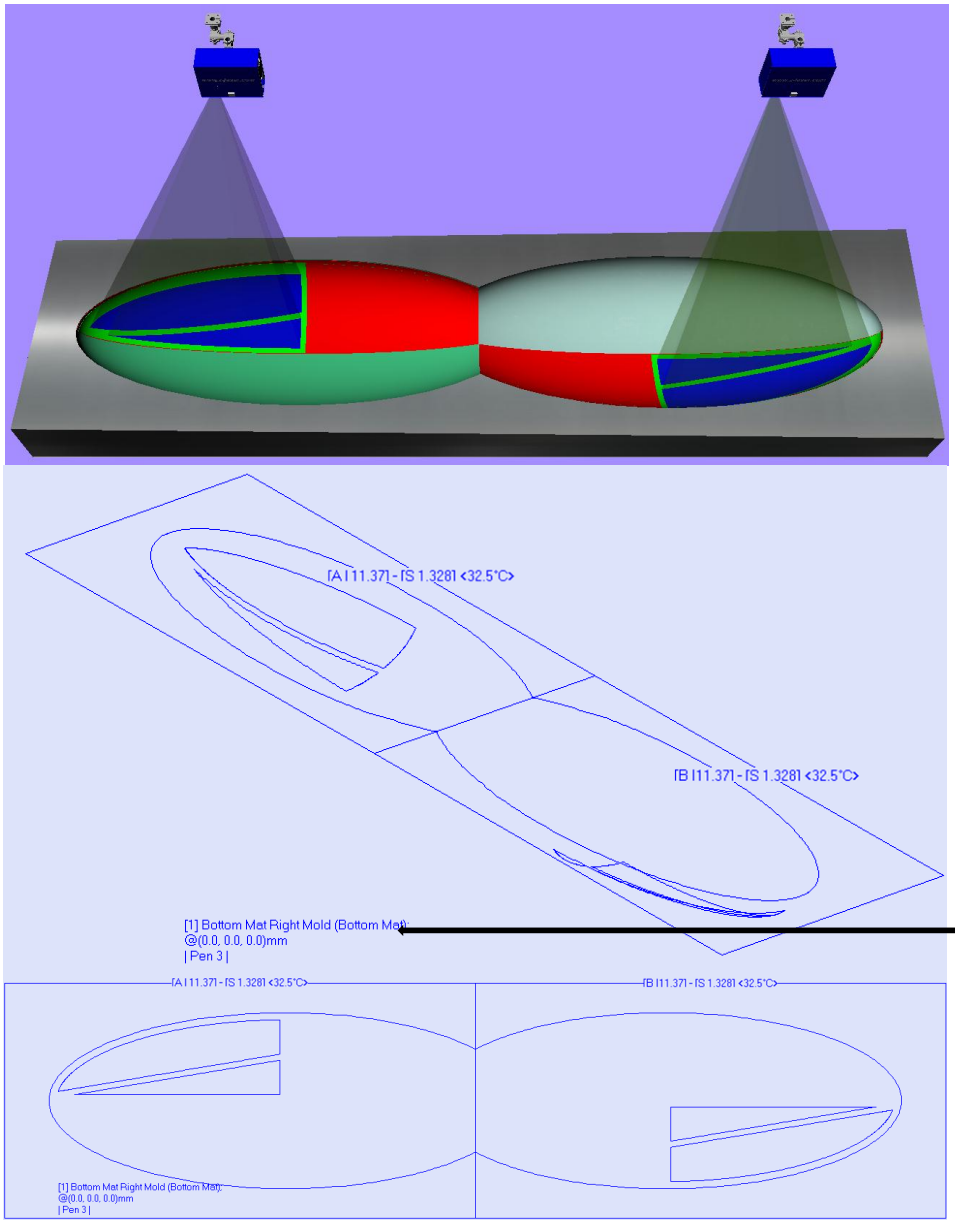
Perspective View of the bottom contours of the mat

Object and Layer Name
Coordinates
Pen Colour

Top View

Change the pen colour again with the command **/GS 3**.

Figure 10: Projecting the Third Object



Projecting the blue top mats in both shells

Perspective View of the bottom contours of the mat

Object and Layer Name
Coordinates
Pen Colour

Top View

Continue to project the next objects until the last object is reached. At that point, proceed to the next step by opening the next file.

Help on using LPM can be found in the **LP Manual English** document as well as a list of commands and user interface configuration.

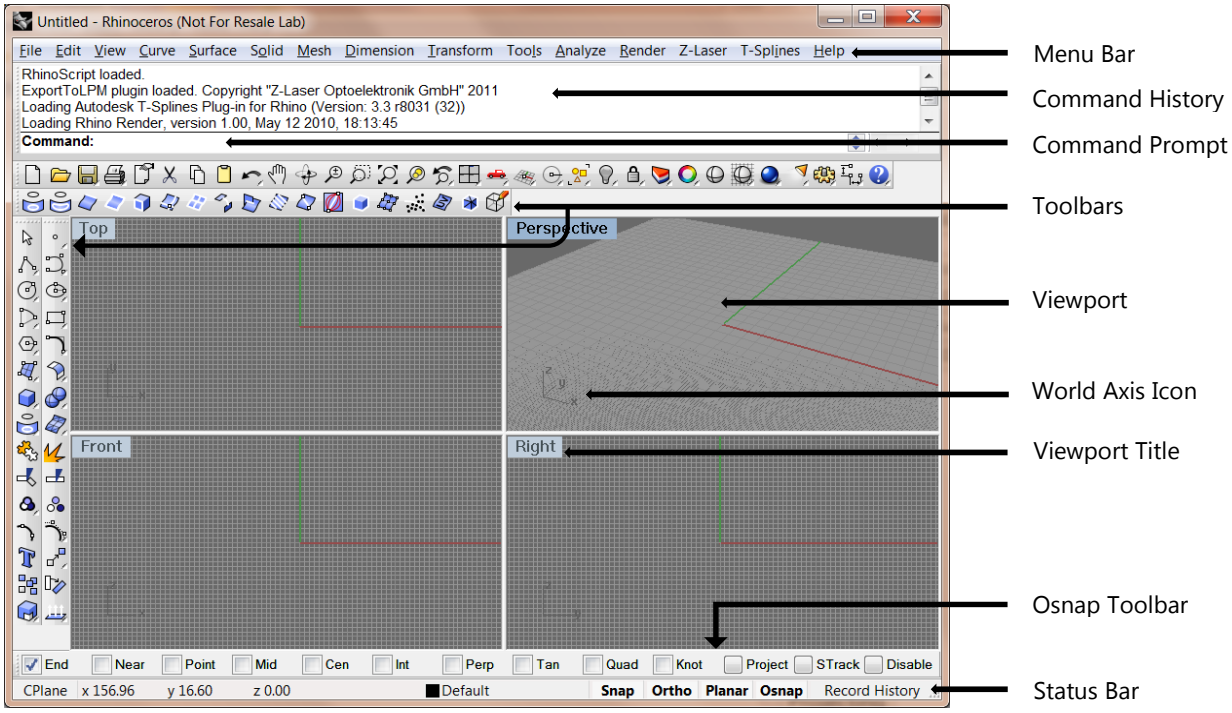
Appendix

Rhino Basics

The User Interface

The user interface is divided into six areas that supply the user with information or prompt the user for input.

Figure 11: The Rhino User Interface



The **menu bar** contains give the user access to commands, options and the help topics.

The **command history** allows the user to scroll through all previous used commands.

The **command prompt** allows the user enter commands using text and displays options.

The **toolbars** give the user quick access to commands and options. Hovering the mouse overa toolbar button will reveal tooltips to its uses.

The **viewports** displays the open model with different views of the model.

The **viewport title** allows the user to change camera views, full screen selection, etc by right mouse clicking on the title.

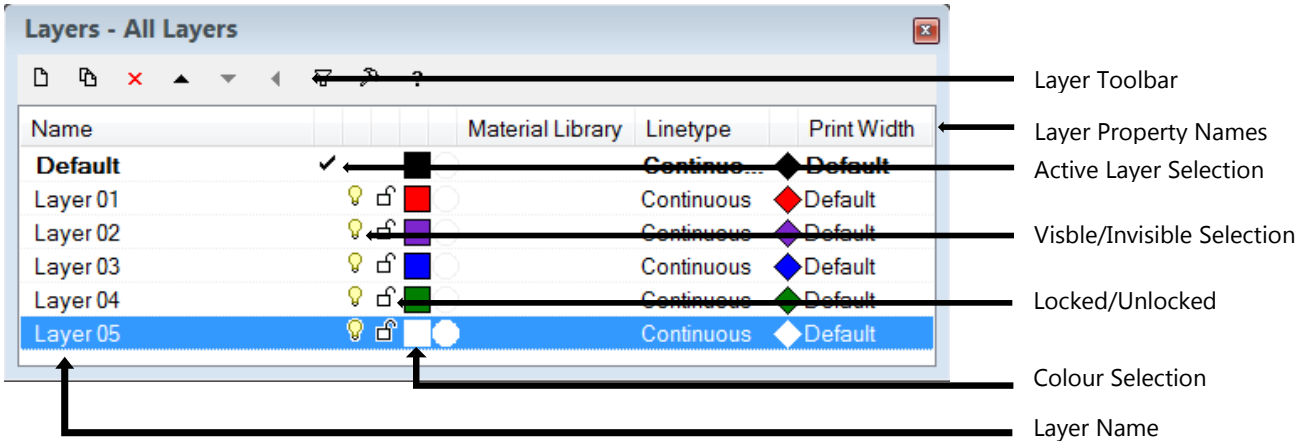
The **osnap toolbar** (short for object snap) shows a selection of snaps which can be toggled to allow the user to snap onto a specific object location when picking a point.

Layers

Layers are a way of organizing objects for the user to manipulate them all at once or keep track of them in some way. When objects are on a layer, you can turn them all off at once, change their wireframe display colour, and select them all with one selection.

The Layer Dialog Window

Figure 12: The Layer Dialog Window



The layer dialog window can be activated by going to **Edit → Layers → Edit Layers...** in the menu bar or typing **_Layer** in the command prompt.

The **layer toolbar** gives the user access to commands which directly affect layers:

- New Layer
- New Sub-Layer
- Delete
- Move Up
- Move Down
- Move Up One Parent
- Filters
- Tools
- Help

The **active layer selection** column displays the current active layer. Simply tick this column next to the layer you would like to make active.

The **visible/invisible** selection column allows the user to hide or unhide the layer. Select the light bulb icon to toggle.

The **locked/unlocked** selection column allows the user to lock or unlock the layer with regards to editing. Select the lock icon to toggle.

The **colour selection** column allows the user to change the colour of the layer. Select the colour to open a colour selection window.

The layer name column displays the name of the layer. Double clicking on the name will allow the user to make a name change (only if a command has not been started).

Creating a new layer

Select the new layer button on the layer toolbar. The layer will appear in the layer dialog window. Give the layer a new name and press enter. Highlight the layer and right mouse click it to bring up a drop-down menu with layer editing options.

Creating a new sub-layer

Sub-layers are a great way grouping similar objects in sub-categories. An example would be different objects which are displayed one at a time within a single step in the lay-up process.

To create a new sub-layer, select the layer which is to become the parent layer and select the sub-layer icon in the layer toolbar. Rename and change the properties as above.

Copying an object to a layer

It is best not to edit the original information that was imported into Rhino. Instead, copying the objects which are important for calibration or projection to a new layer allows the end user to edit the data safely without destroying the original information. It also allows selection of the objects to be easier.

Select the object or objects you wish to duplicate. Either go to **Edit → Layers → Copy Objects to Layer** in the menu bar or type **_CopyToLayer** in the command prompt. The Copy to Layer menu window will appear. Select the layer you wish to copy to and select OK.

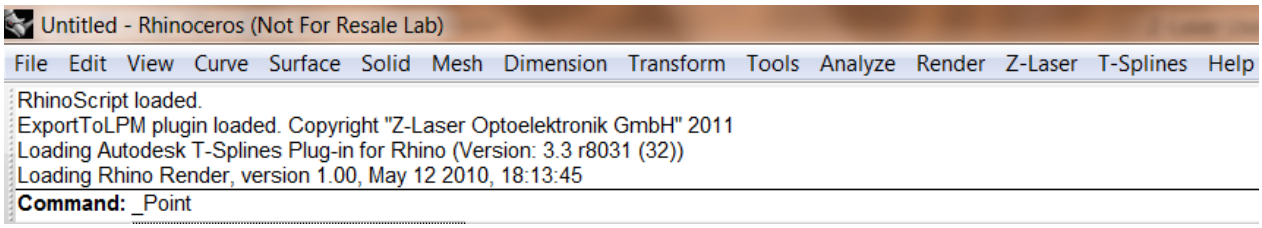
Objects

Points

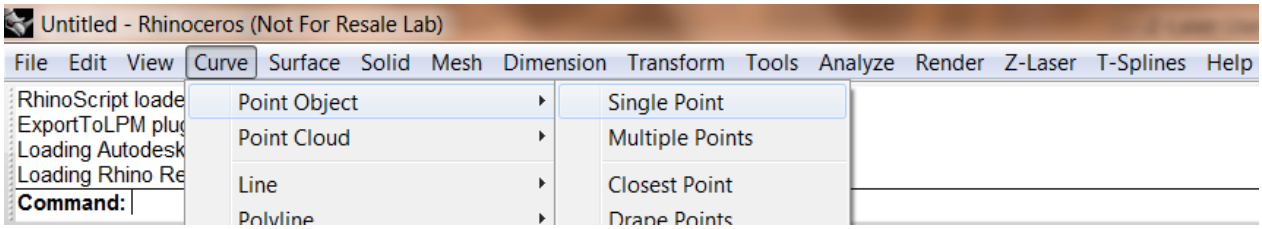
Points are the simplest objects in Rhino. A point is a one dimensional object with no height or width. It is used to mark a point in 3D space with X, Y and Z co-ordinates. It should not be confused with a control point which are used to control bends in a curve.

To start placing a single point in Rhino, you can use one of three options:

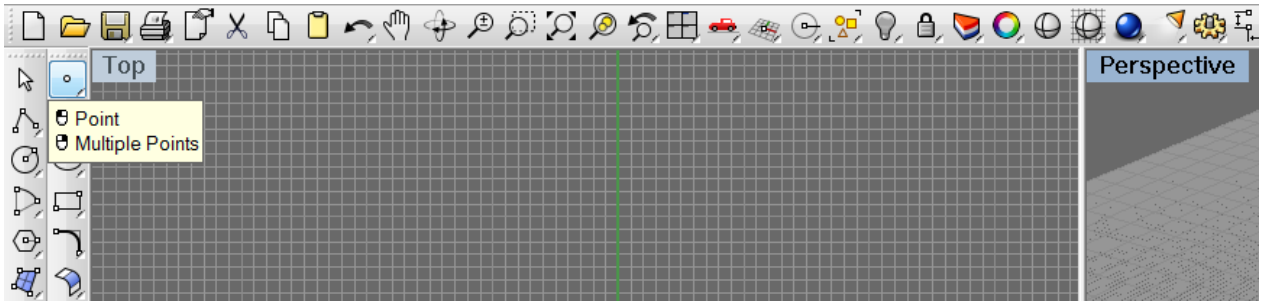
1. Type the command **_Point**.



2. Select **Curve → Point Object → Single Point** in the menu bar.



3. Using the left mouse button, click the **Point** button from the Main2 toolbar.



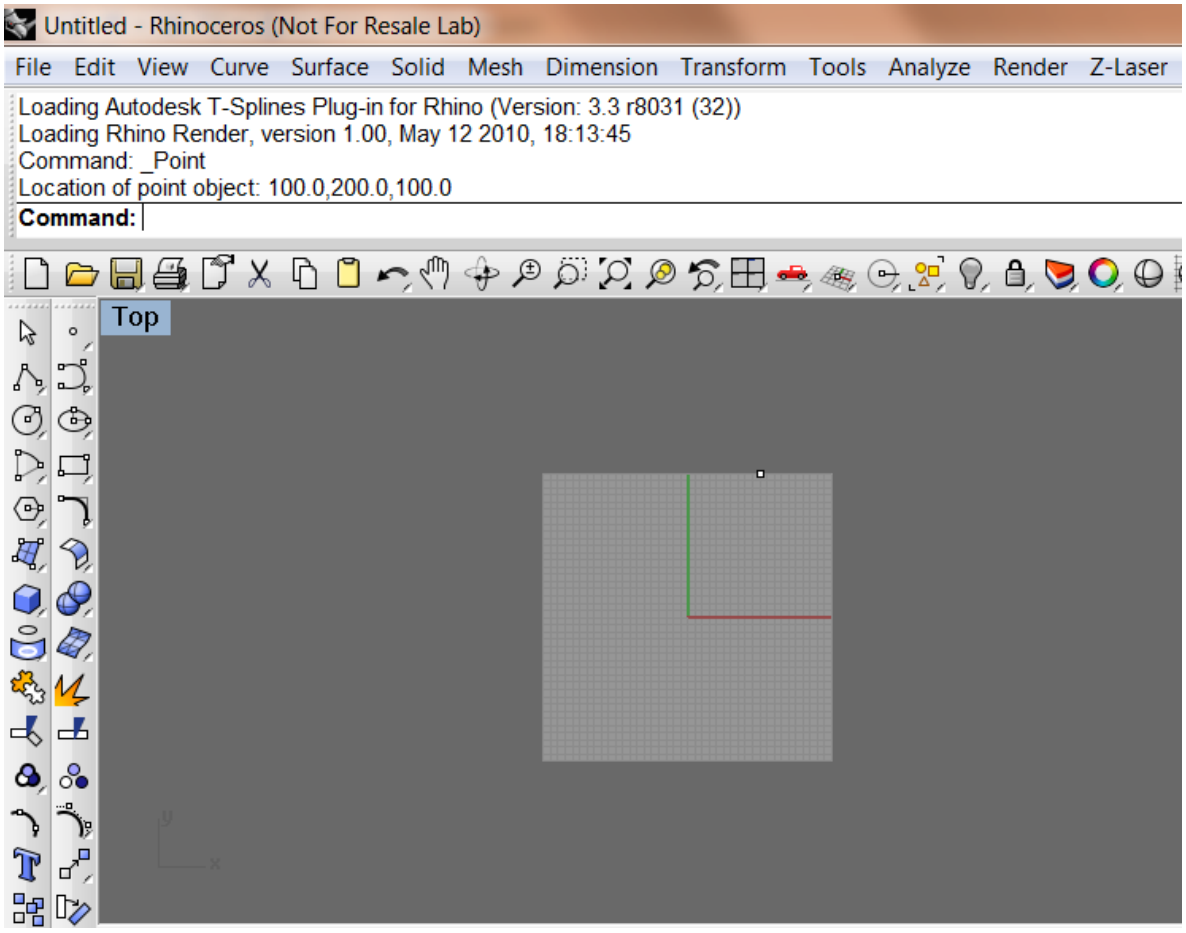
Next you should enter the coordinates of this point. Place the mouse cursor in the **Top Viewport**. Enter three numbers designating the X, Y and Z axis coordinate separated by a comma.

Example: Place the point at X 100.0, Y 200.0 and Z 100.0

Type in the command prompt

100.0,200.0,100.0

Placing the mouse cursor in a different viewport will read the coordinates in a X, Y and Z order different to the order above. It may be read as Y, Z and X or Z, X and Y.



You can also drop the point to a location on an object using an **Object Snap** such as **Endpoint**, **Mid**, **Center**,etc.

To list all available options with regards to points, do one of the following:

1. Type **_Point** in the command prompt and wait for a drop down list to appear with the other options underneath.
2. Select **Curve** → **Point Object** → from the menu bar.
3. Using the right mouse button, click and hold on the **Point** button in the Main2 toolbar.

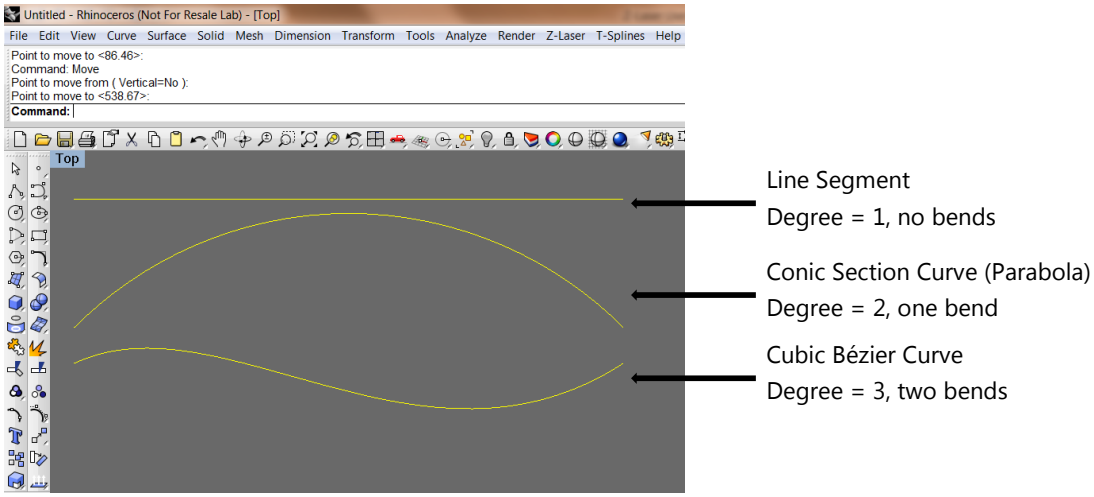
Curves

A curve is basically an object which is similar to a line but it is not required to be straight. All curves have a direction and control points that determine the shape of the curve. These control points cannot be separated from their objects. The amount of control points on any one object determine the degree to which the object can bend.

A line is a special case of curve, namely a curve with zero curvature. A line segment is a line or part of a line that is bound by two distinct end points and contains every point on the line between its end points. It has a degree of 1 which means it has no bends.

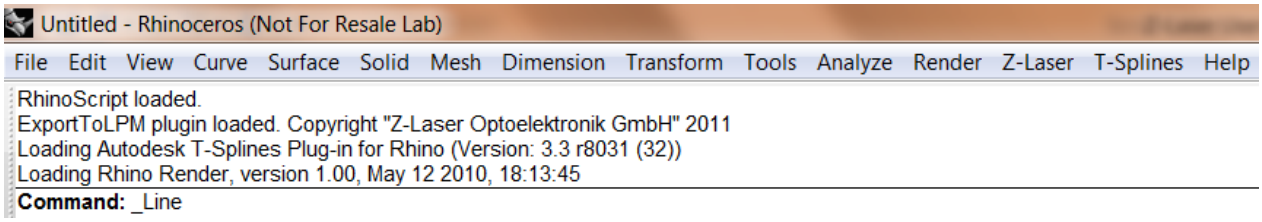
With the exception of a line segment, all curves have bends. Conic section curves such as an arc, parabola or a circle have one bend and a degree of 2 whereas a cubic Bézier curve has at least 2 bends with a minimum degree of 3.

Figure 13: Types of Curves

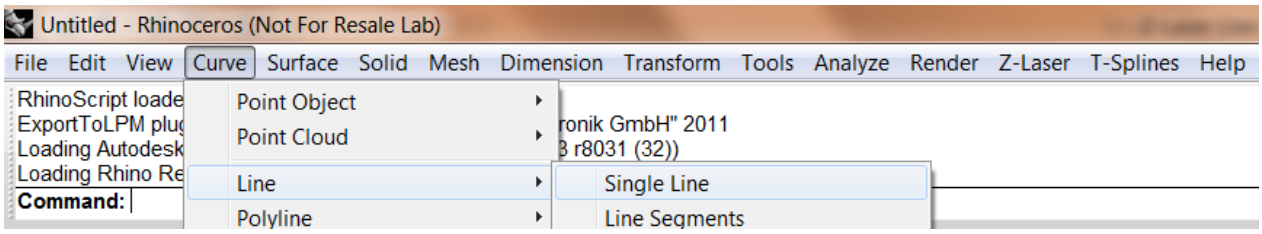


To start placing a line in Rhino, you can use one of the following three options:

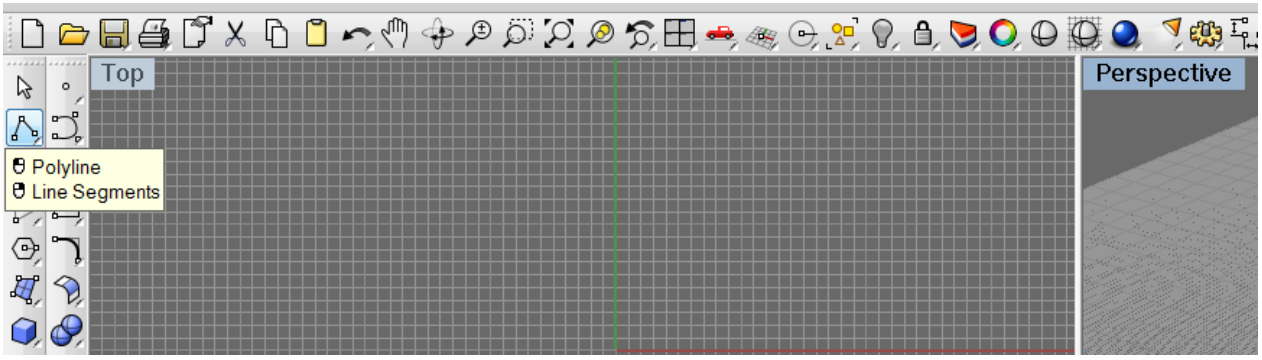
1. Type **_Line** in the command prompt (and select one of the drop-down options).



2. Select **Curve → Line → Single Line** (or one of the other options) from the menu bar.



3. Using the right mouse button, click the Polyline button in the Main1 toolbar (or right click and hold for other options).



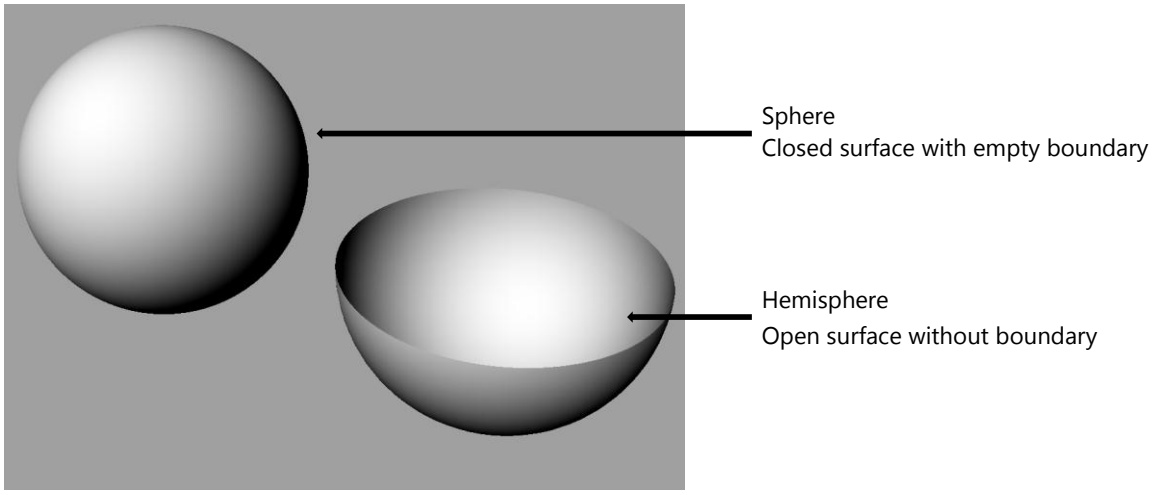
Both starting and end points of the line have a X, Y and Z coordinate which can be given the same as a single point or by dropping the point on an object using an ObjectSnap.

Conic section curves or Bézier curves are placed in much the same way as line segments except that additional control points must be entered or placed into the curves. For further explanations on placing a curve in a Rhino drawing, please refer to the Help Topics included in the program.

Surfaces

A surface is a two dimensional topological manifold. It consists of, generally speaking, width and length but no thickness but any point on the surface may have a three dimensional coordinate (X, Y and Z). There are two types of surfaces: open and closed. A surface without boundary is referred to as an open surface whereas a surface with empty boundary is referred to as a closed surface. Rhino also refers to a closed surface as a solid.

Figure 14: Difference between a Solid and a Surface



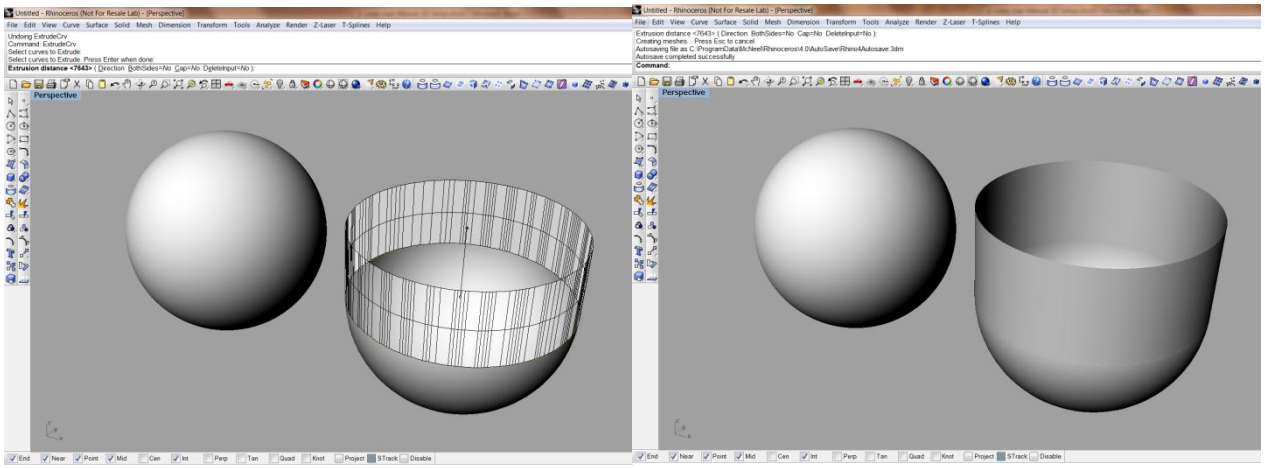
A surface can be created in Rhino in a variety of ways but the three most common are:

1. To extrude a single curve
2. To create a flat surface using a minimum of 3 points
3. To create a surface using multiple curves

For all other options, please refer to the Help Topics included in the program.

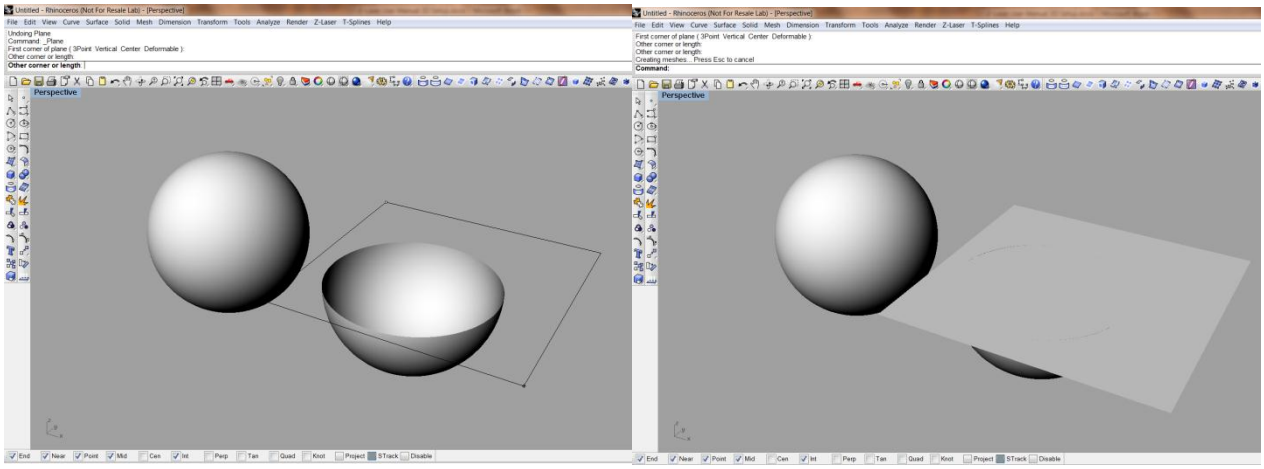
To extrude a single curve

Either go to **Surface → Extrude Curve** in the menu bar or type **ExtrudeCrv** and make a selection based on the path you wish the surface to follow. You should then select the curve and press enter. There will be various settings based on the way you would like the surface to be made, such as the direction of the extrusion, whether or not you wish the surface to be capped, deletion of the original curve and so on. Make your selections by left clicking on the selection and entering the new value and pressing enter. When you have extruded the curve to your liking, press enter for the surface to be created.



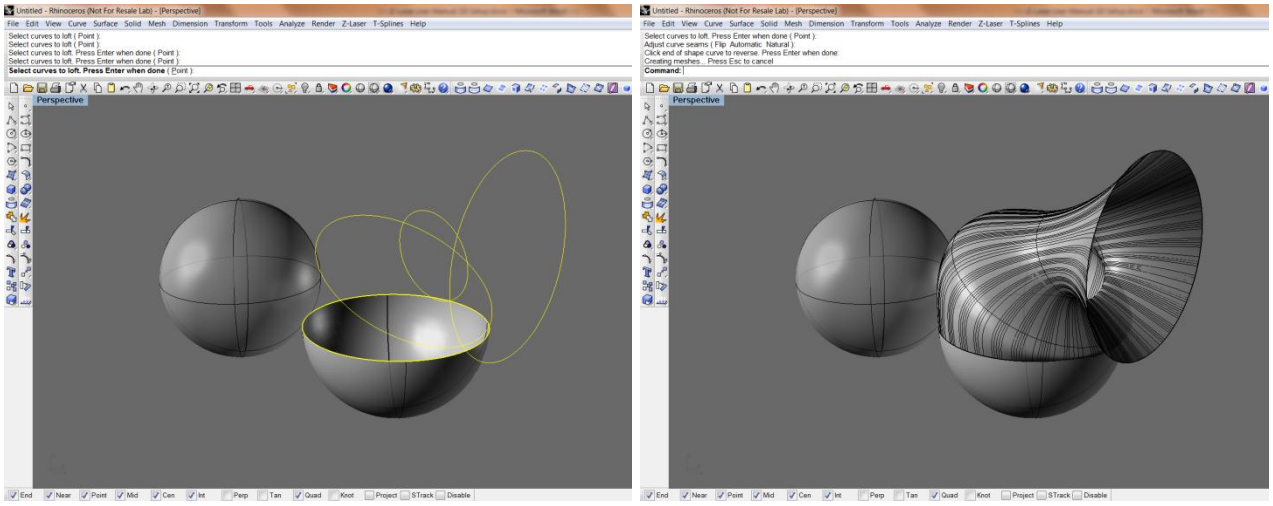
Create a plane with 3 points

Either go to **Surface → Plane → Three Points** in the menu bar or type **_3Point**. Place your starting corner, one of the other corners and then the last corner. Change any surface settings you may require in the command prompt before selecting the last corner. Selecting the last corner creates the surface.



To loft multiple curves

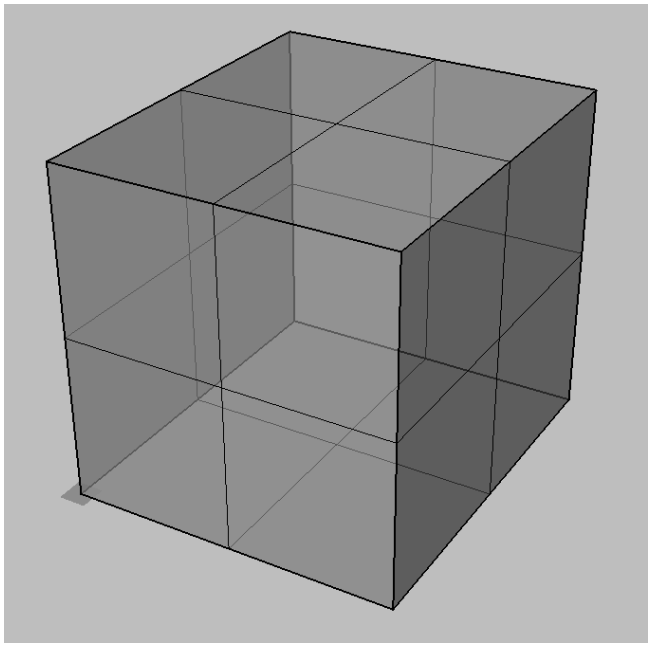
Either go to **Surface → Loft** in the menu bar or type **_Loft**. Select the curves that will define the surface shape and press enter. Adjust the seams of the curves to match the correct direction and press enter. The Loft options menu will appear. Set the settings you wish the loft to use and press enter for the surface to be created.



Solids/Polysurfaces

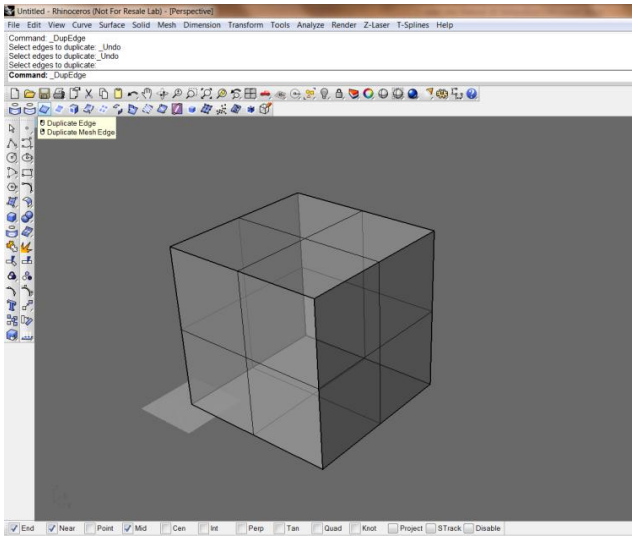
A solid is defined by Rhino as any surface or polysurface that is closed and has no naked edges. It is said that a solid object can be constructed from primatives, such as spheres, cuboids and cones, by means of allowable operations. These are typically Boolean Operations of sets, like union, difference and intersection.

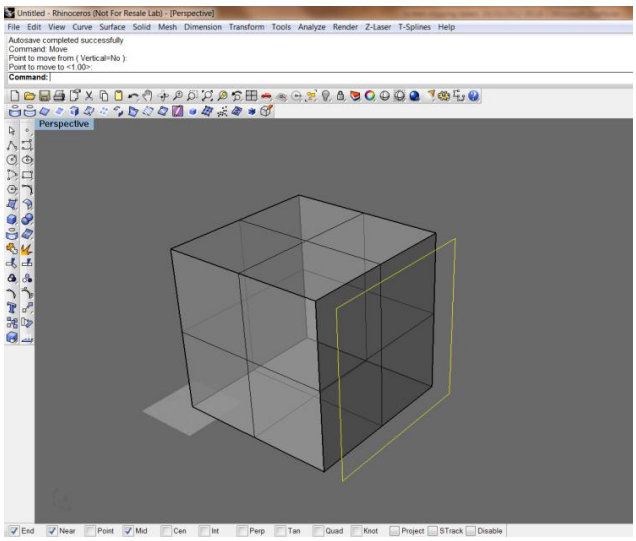
This document will not deal with creating solid or polysurface objects as LPM can only project lines, polylines or curves. Solid objects are normally created by the CAD designers and LPM only requires the edge of these solids in order to project it. It will, however, deal with extracting information from the solid for the purposes of projection.



To extract edges from solids

Either type **_DupEdge** in the command prompt or left mouse click the **Duplicate Edge** button in the **Curve From Object** toolbar. Select any edge from a surface or solid in order to extract that curve to the current active layer.

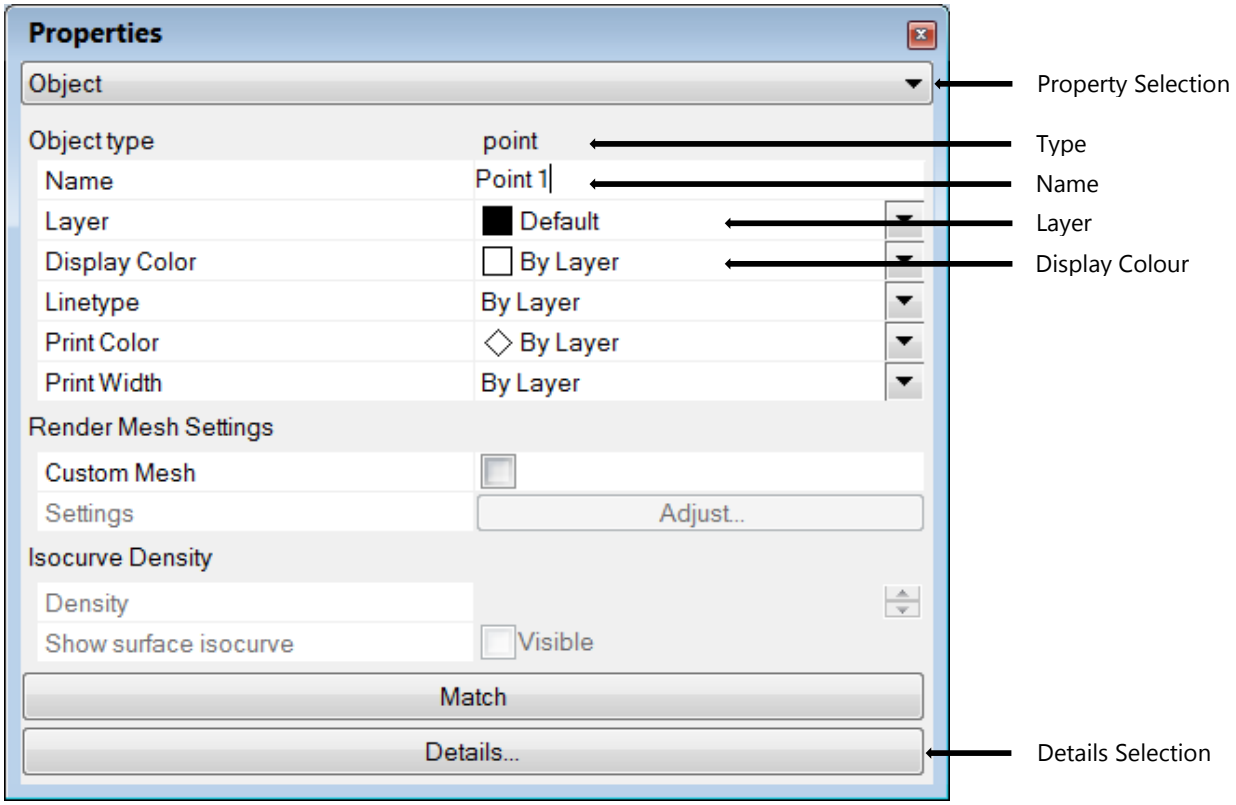




Object Properties

The object properties dialog window allows the user to manage the material, light, text and dimensions of the selected object and view its properties. Press F3 to display the window.

Figure 15: The Object Properties Dialog Window



The object **property** drop-down selection allows the user to change between object, material and texture mapping properties.

The object **type** display the type of object currently selected. Examples are points, curves, surface, polysurfaces and meshes.

The object **name** displays the name of the currently selected object. It is good practice to name all objects which will later be exported to LPM as it will make it easier to these objects to be selected in lists.

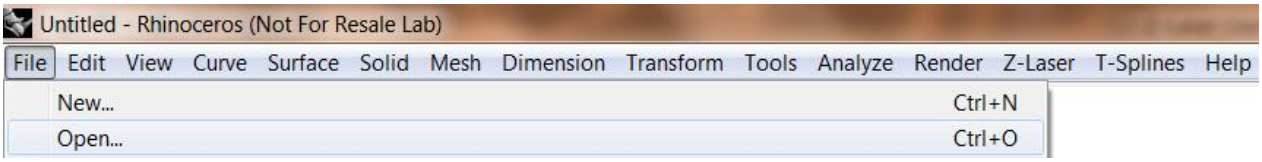
The **layer** of the currently selected object can be changed by selecting the layer from the drop down list to move (not copy) the object to that layer.

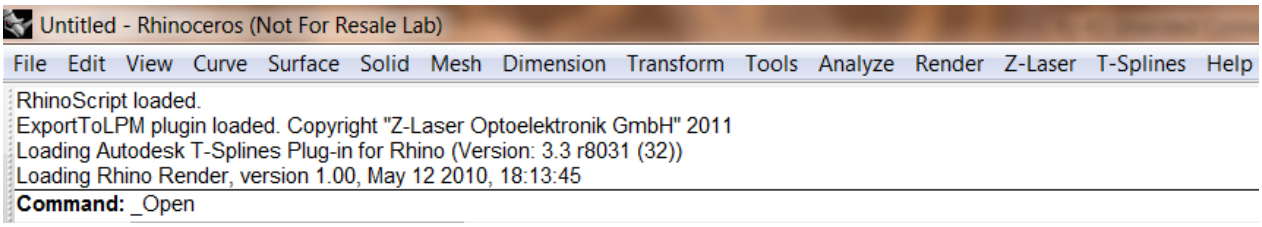
The display **colour** can be changed by selecting the type of colour from the drop down list.

The **details** selection button will open a display window with all information regarding to the currently selected object or objects.

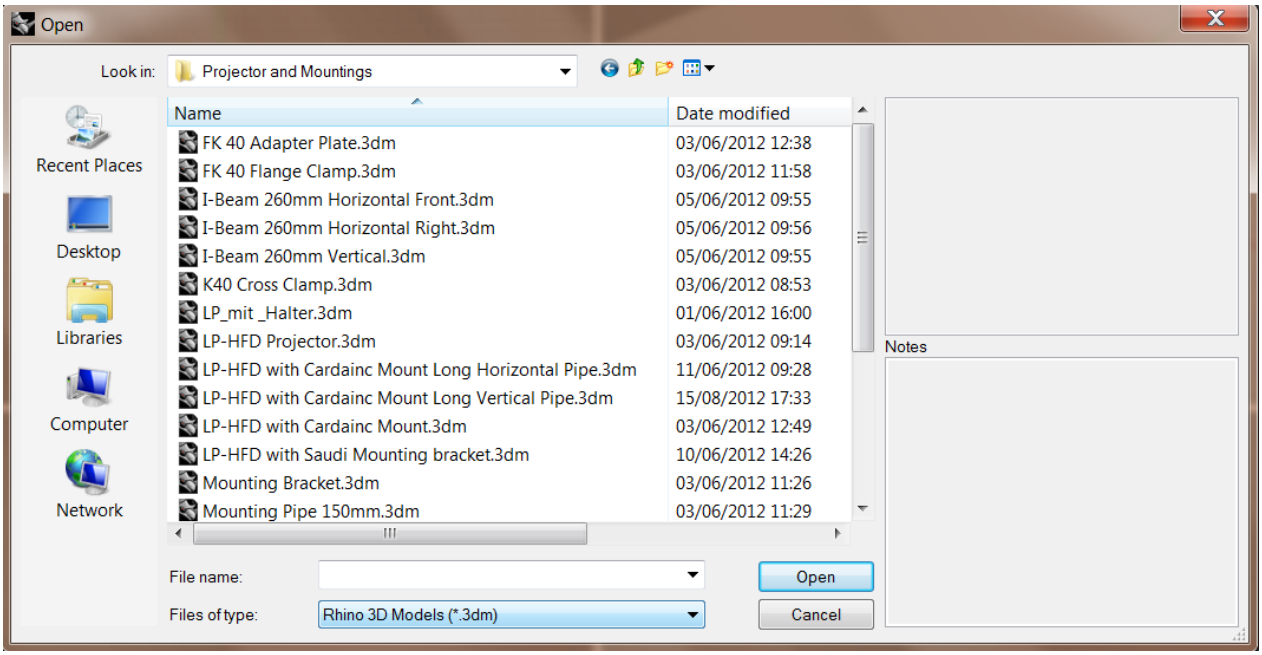
Opening 3D Data in Rhino

Start Rhinoceros 4.0. Type the command **_Open** or go to the menu bar and select **File → Open**.





Navigate to the folder where your file in stored. Left click the arrow on the field marked **Files of Type**. Select the type of file you wish to open and select the Open button. Depending on the size and type of your file, it may take a few minutes to import the file into Rhino.



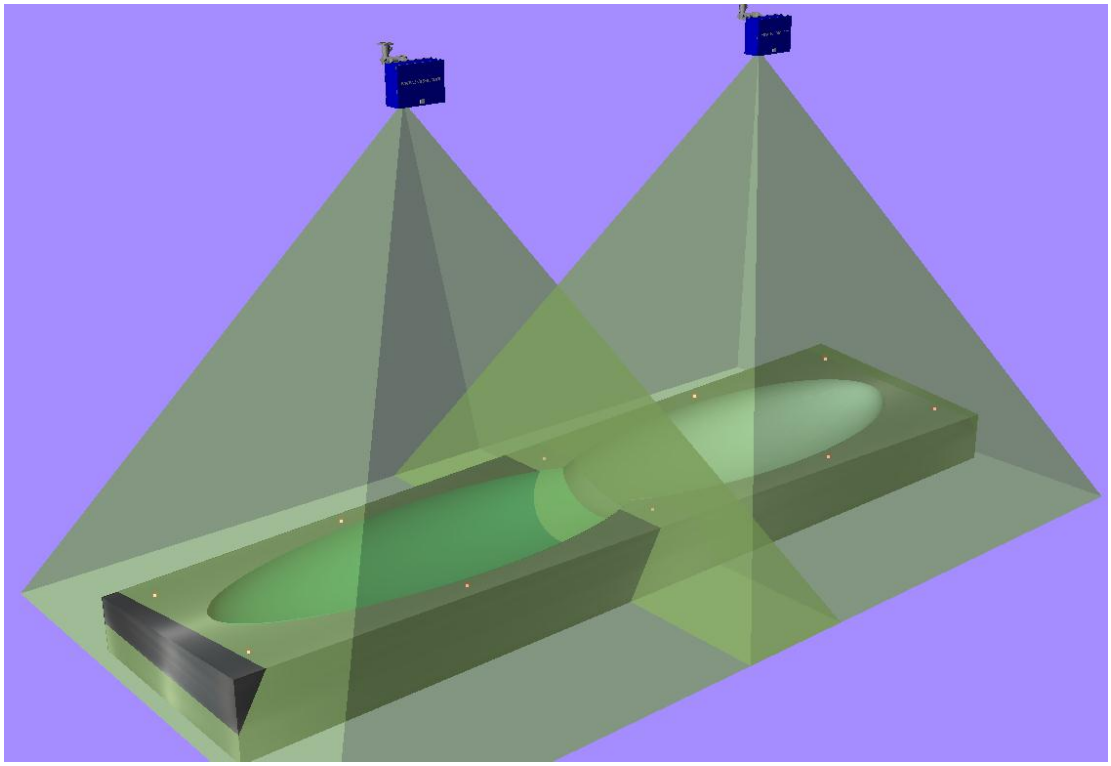
Projector Installation

The following information about projector installation is a guide to be used in conjunction with the information found in the document **LP Manual English**.

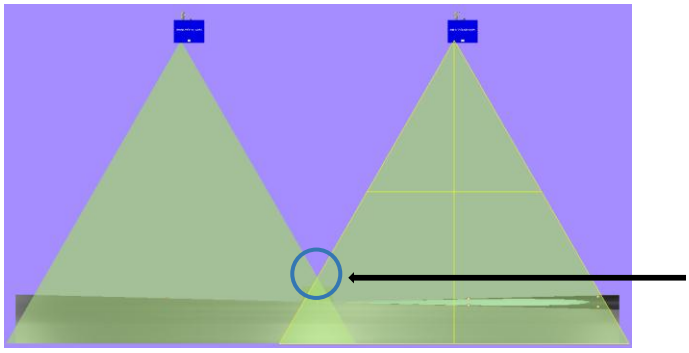
1. Ensure the projectors cover the entire projection area and the calibration points.

   Go to the 3D calibration window, go to the field border page and select **Max. Field**

   Position the projector but take care to avoid looking directly into the laser light



2. Ensure where there are 2 projectors or more that the maximum field borders overlap at the highest point of the projection.

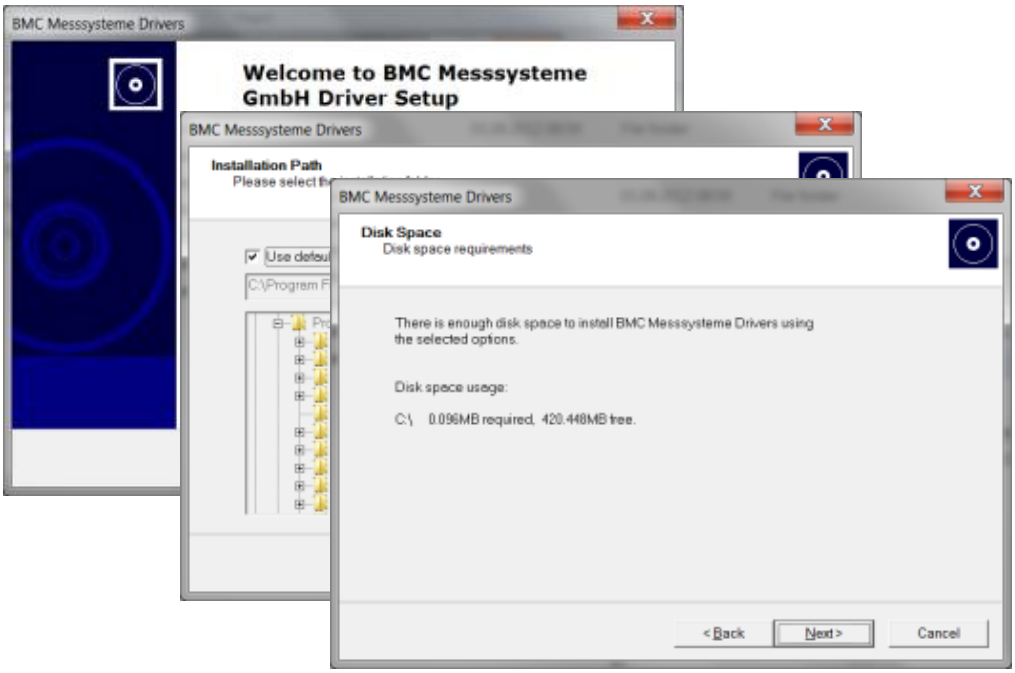


Overlap of the maximum field border above the height of the highest projection point

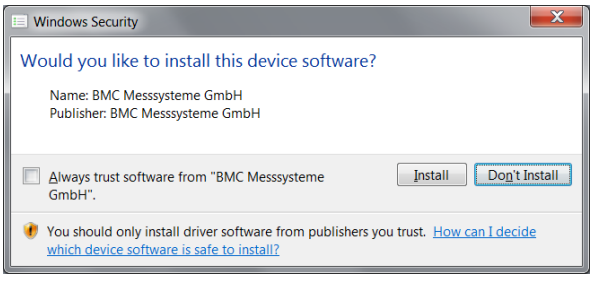
MEM-PIO Driver Installation

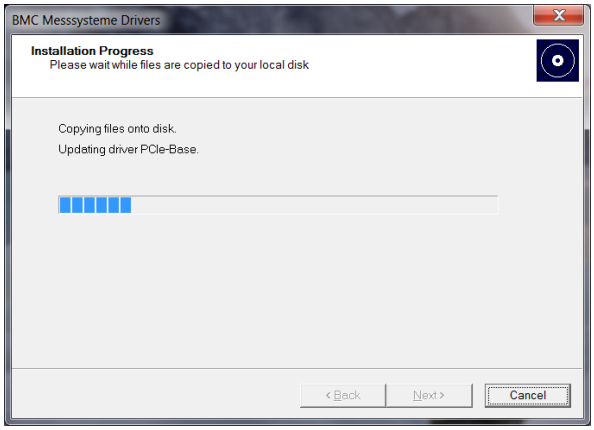
If you require remote control of the projection system or some form of output from the control computer, then you will have to connect the MEM-PIO cable to the computer. The MEM-PIO is a red cable with a USB Type A connector on one end and a DB-15 socket on the other end. It can be attached to a remote control system or the OR8 I/O module. The drivers for it must be installed before it is connected to the control computer.

Open the folder \LPM 8.1\DRIVERS. Select the file **install-drivers-4.5.486.exe**. The following welcome page will be displayed. Select **Next** and the menu will ask where you would like to install the drivers. Ensure that the **Use default folder to install BMC Messsysteme Drivers** is selected. Select Next. The next menu will display the amount of disk space you have remaining. If it is too small, consider installing a new hard drive in the computer. Select **Next**.

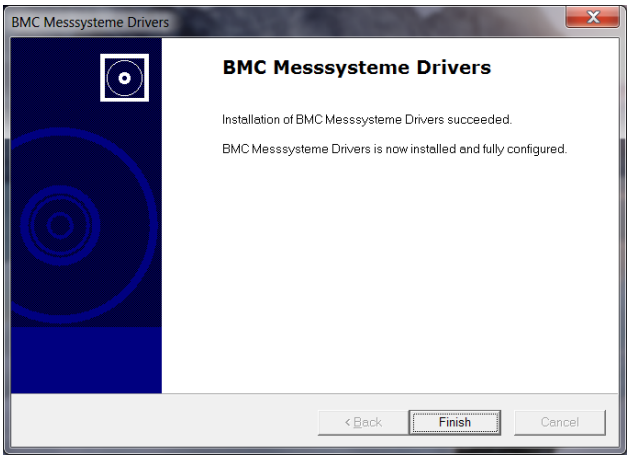


The drivers will begin to load. If you get the following warning, select **Install**. Wait for the drivers to install.

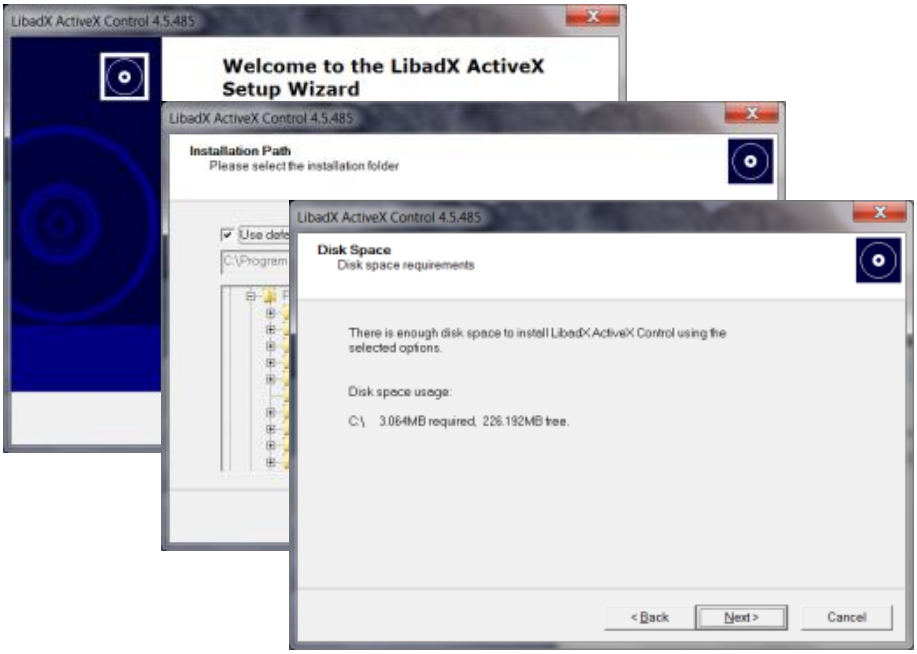




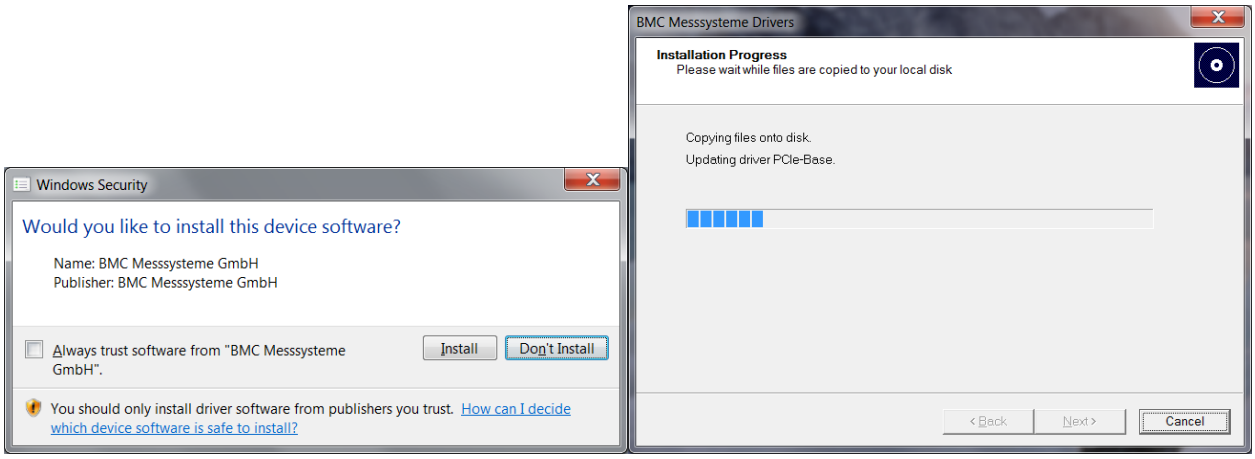
The next menu will inform that the installation is complete. Select **Finish**. If a window pops up asking you if the program installed correctly, select **The program installed correctly** button.



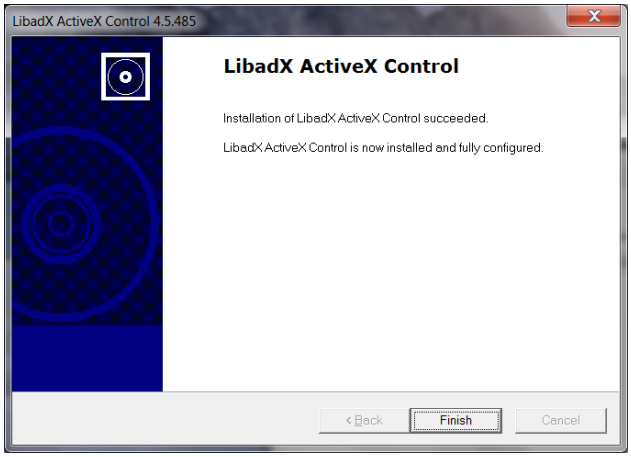
Next, you will have to install the ActiveX files. Select and double click on the file **libad-actx-4.5.485.exe**. The following welcome page will be displayed. Select **Next** and the menu will ask where you would like to install the drivers. Ensure that the **Use default folder to install BMC Messsysteme Drivers** is selected. Select Next. The next menu will display the amount of disk space you have remaining. If it is too small, consider installing a new hard drive in the computer. Select **Next**.



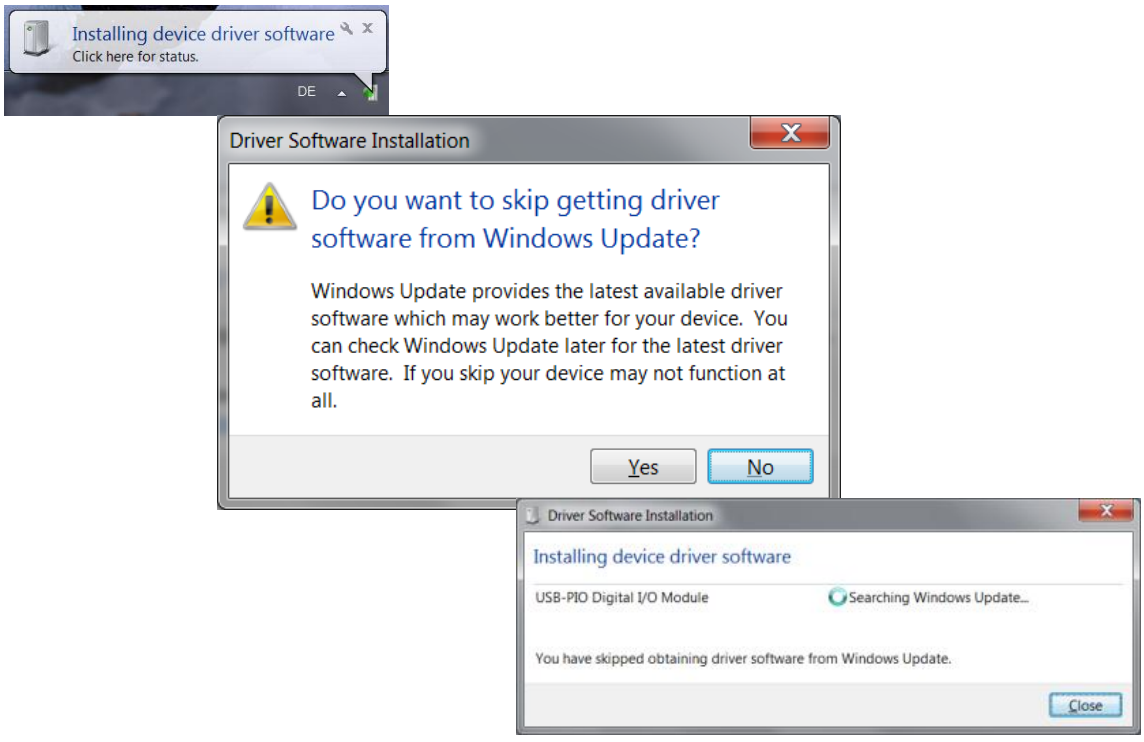
The drivers will begin to load. Wait for the drivers to install.



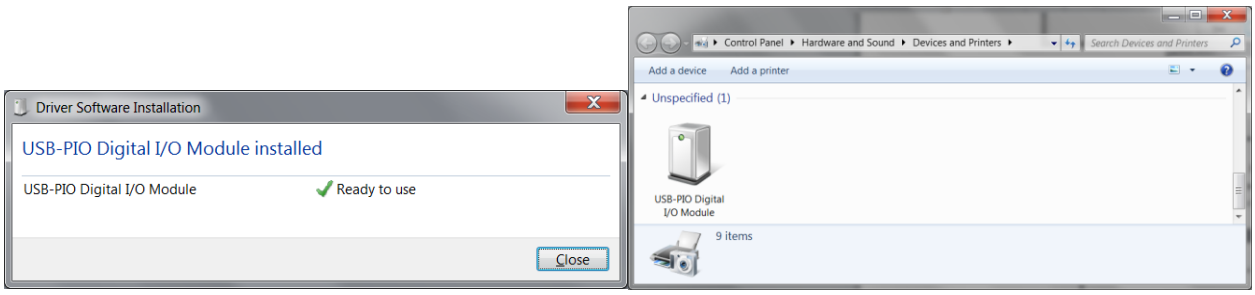
The next menu will inform that the installation is complete. Select **Finish**. If a window pops up asking you if the program installed correctly, select **The program installed correctly** button.



Insert the MEM-PIO cable into the USB input you wish to use. A pop-up window will be seen in the bottom left hand window of the screen displaying that the driver is loading. If it takes too long, it may be trying to find the drivers from Windows Update. Cancel by pressing **Cancel**. Select **Yes** to skip getting the driver from Windows Update.



The computer will then check its folders and find the drivers. If it finds the drivers, it will show you that the device is **Ready to Use**. Select **Close**. To check if it is connected correctly, go to the Windows **Start** Icon in the bottom left of your screen and select **Devices and Printers** from the Start menu. The USB-PIO will be displayed under **Unspecified**.



Your MEM-PIO (USB-PIO) is now installed and ready for use. Ensure it is connected correctly to your remote control receiver or your OR8 I/O device.

Table of Figures

Figure 1: Distance Tolerance and Maximum Angle....13
Figure 2: Difference in Angle Tolerance....14
Figure 3: The Example Mold....25
Figure 4: The 3d Calibration Menu....32
Figure 5: The 3D Field Border Menu....36
Figure 6: The Step/Object Work Flow....39
Figure 7: The Projection Step/Object Work Flow....40
Figure 8: Projecting the First Object....53
Figure 9: Projecting the Second Object....54
Figure 10: Projecting the Third Object....55
Figure 11: The Rhino User Interface....56
Figure 12: The Layer Dialog Window....57
Figure 13: Types of Curves....61
Figure 14: Difference between a Solid and a Surface....62
Figure 15: The Object Properties Dialog Window....66